Grothendieck rings of general fusion categories of rank four

Abstract

Exactly fifteen based rings occur as Grothendieck rings of complex fusion categories of rank four. Every rank-four fusion category admits a spherical structure, and each of the fifteen rings has a unitary realization. No multiplicity bound or pivotal hypothesis is assumed. A uniform theoretical argument using sphericalization, central induction, dimension-field descent and indicator estimates first proves $\mathrm{FPdim}(\mathcal{C}) < 1000$ and $N_{XY}^{Z} \leq 31$. An exhaustive exact census and categorical obstruction certificates then identify all surviving rings. The resulting sharp bounds are $24+12\sqrt{3}$ for the global Frobenius–Perron dimension and 6 for the fusion multiplicities, with equality precisely for the C_3 near-group ring of parameter six.

AI-generated research · independent verification pending.

The results, proofs, code and data have not yet been independently verified by human experts or formalized in Lean. Treat them with caution; automated checks establish only their stated computational scope.

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1 Introduction

This article establishes the rank-four part of the classification program without a restriction on the fusion multiplicities and without assuming a pivotal structure. Its conclusion concerns both the based rings that occur and the sphericality of every category realizing them. The proof uses one common framework—sphericalization, central induction and exact character arithmetic—to reduce arbitrary rank-four categories to a finite, explicitly bounded collection. The exact certificate stage then identifies the fifteen realizable rings.

The companion finite census article, *Rank-four Grothendieck rings through multiplicity 128*, gives a larger independent finite verification. The present proof includes its own smaller census and does not require the larger census as an input.

Theorem 1.1 (Classification)**.** *There are exactly fifteen Grothendieck rings of complex fusion categories of rank four, up to based isomorphism. They are listed in Table 1. Every complex categorification of each of these rings admits a spherical structure. Moreover,*

$$\mathrm{FPdim}(\mathcal{C}) \le 24 + 12\sqrt{3}, \qquad \max_{X,Y,Z} N_{XY}^{Z} \le 6.$$

Both bounds are sharp, and equality in either holds precisely for the near-group ring $\mathrm{NG}(C_3, 6)$.

μ	A unitary realization	D	Reference
1	$\mathrm{Vec}_{C_2\times C_2}$	4	[10, §3.4]
1	$\mathrm{Fib} \boxtimes \mathrm{Vec}_{C_2}$	$5+\sqrt{5}$	[10, §3.4]
1	$\mathrm{Rep}(D_{10})$	10	[10, §3.4]
1	$\mathrm{Ad}(\mathrm{SU}(2)_6)$	$8+4\sqrt{2}$	[5, Theorem 3.1]
1	$\mathrm{Fib} \boxtimes \mathrm{Fib}$	$(15+5\sqrt{5})/2$	[10, §3.4]
1	$\mathrm{Ad}(\mathrm{SU}(2)_7)$	δ_7	[10, §3.4]
1	Vec_{C_4}	4	[10, §3.4]
1	Tambara–Yamagami on C_3	6	[24]
1	Principal even part of S'	$8+4\sqrt{2}$	[23, Theorem 1]
2	Principal even part of $2D2$	$20+8\sqrt{5}$	[5, Theorem 3.1]
2	Haagerup category $\mathcal{H}_1$	$(39+9\sqrt{13})/2$	[8, Table 2]
2	$\mathrm{Rep}(A_4)$	12	[11, Theorem 1.1]
3	$\mathrm{Ad}(E_8)$	δ_E	[22, 21]
3	Near-group type $(C_3, 3)$	$(21+3\sqrt{21})/2$	[9, Example 9.4]
6	Near-group type $(C_3, 6)$	$24+12\sqrt{3}$	[9, Theorem 10.19]

Table 1: One unitary realization of each surviving based ring, with its multiplicity μ and global Frobenius–Perron dimension D. Here D_{10} has order ten, S' has index $3+2\sqrt{2}$, and the near-group category of type $(C_3, 3)$ is the Izumi–Xu category.

The numbers δ_7 and δ_E are the largest real roots of $x^3-27x^2+162x-243$ and $x^4-45x^3+375x^2-1125x+1125$, respectively. The supplementary model check verifies these polynomials, the positive dimension characters, and all tensor identifications exactly.

The classification and sphericality are proved in Corollary 14.6, and sharpness in Corollary 14.7. The classification is computer assisted. The complete local census below $D < 1000$

contains 1 692 fusion rings: 1 642 self-dual rings and 50 rings with a dual pair. The supplementary package supplies its own enumeration and classification programs, every full multiplication tensor, and the exact certificate ledger. Table 2 records the successive reductions to the fifteen models. The finite verification is specified in Section 14.1, and all its mathematical exclusion criteria are established in Section 13. In particular, positive-dimensional indicator estimates are invoked only after a proved pseudo-unitary reduction. Other branches are tested through the sphericalization, whose signed quotient and dimension character are both exhaustively retained.

The bound that makes this finite verification exhaustive is independent of the computation.

Theorem 1.2 (Uniform bound)**.** *For every complex fusion category $\mathcal{C}$ of rank four,*

$$\mathrm{FPdim}(\mathcal{C}) < 1000, \qquad N_{XY}^{Z} \le 31 \quad (X, Y, Z \in \mathrm{Irr}\,\mathcal{C}).$$

The bound is independent of the fusion-ring census. Its constants come from the dimension and character estimates proved below; no sharpness is asserted for this theoretical reduction. Its proof is completed in Section 12. Central induction in the sphericalization gives positive integral Galois columns and a descent theorem for the dimension field. Quadratic arithmetic and averaged indicator identities then give $D < 1000$ in the nonspherical quadratic case and $D < 950$ in the nonspherical quartic case. Pivotal categories satisfy the independent bound $D < 1000$.

The structural ingredients also prove sphericality when the category has a non-self-dual simple, a nontrivial invertible simple, or a cubic Frobenius–Perron character field. The two Fibonacci normal forms needed in the proof are treated directly. A short quadratic argument treats the Haagerup ring. These results establish sphericality of every categorification of each surviving ring; existence of one spherical realization alone would not suffice for that conclusion.

A character field and its integral based order are distinguished throughout. In particular, a quadratic character component may have a constant rational formal codegree, and this possibility is retained in the signed reductions. The arithmetic charts used for the uniform bound are bounded integer calculations derived within the proof, rather than assumptions taken from a fusion-ring census.

Notation and proof structure

The ordinary and signed dimension characters are denoted by d_i, x_i, with global dimensions $D = \sum d_i^2$ and $E = \sum x_i^2$. The ordinary and signed formal codegrees are f_i, g_i; $u = \sum f_i^{-2}$ and $v = \sum g_i^{-2}$. A character-value field is not silently identified with the field of its global codegree: that identification is supplied where needed by Section 7. The proof separates as follows.

Case	Reduction used in the main proof
Non-self-dual rings	Automatic sphericality; a theoretical candidate classification and exclusion (Sections 4, 14).
Self-dual, quadratic codegree	Dimension-field descent and quadratic arithmetic (Section 8).
Cubic character field	Positive central measure and periodic moments (Section 9).
Quartic global codegree	Central ratios and a bounded arithmetic chart (Section 11).
Spherical case	Independent pivotal bound (Section 10).
Bounded tables	Complete ordinary census, signed-cover verification, and exact certificates (Section 14).

All categorical reductions needed for the classification are proved in this article. The exact enumeration and certificates are supplied in the `unrestricted` package, with their interfaces

described in the verification guide. Existence of the named unitary categories is supplied by the construction references in Table 1.

2 Sphericalization, central induction, and rigidity

Throughout, a fusion category is finite, semisimple, rigid and $\mathbb{C}$-linear, with simple tensor unit. Its rank is the number of isomorphism classes of simple objects. A fusion ring is a based ring with nonnegative integral structure constants and the usual coefficient-of-unit pairing. A *Grothendieck ring* is a fusion ring realized by such a category. A pivotal structure is a monoidal isomorphism from the identity to the double dual; it is spherical when its left and right traces agree. We impose no pivotal hypothesis; the general existence question is [6, Conjecture 2.8].

For a self-dual basis $1, X, Y, Z$ we use the convention

$$\begin{aligned} X^2 &= 1 + aX + bY + cZ, \quad XY = bX + dY + eZ, \quad XZ = cX + eY + fZ, \\ Y^2 &= 1 + dX + gY + hZ, \quad YZ = eX + hY + iZ, \quad Z^2 = 1 + fX + iY + jZ. \end{aligned} \tag{2.1}$$

Thus a table is denoted by $(a, b, c, d, e, f, g, h, i, j)$. The same convention applies to signed tables. All subsequent changes of labels or signs are explicitly local to the argument concerned.

Lemma 2.1 (Rank-four algebras and coefficient bounds)**.** *A semisimple complex algebra of dimension four admitting a one-dimensional representation is commutative. In particular every rank-four fusion ring, and every rank-four signed sphericalization quotient, has commutative complexification. For an ordinary fusion ring, write $d_i = \mathrm{FPdim}(b_i)$ and $D = \sum_j d_j^2$. Then*

$$N_{ij}^k \le d_i < \sqrt{D}.$$

Proof. The only noncommutative semisimple complex algebra of dimension four is $M_2(\mathbb{C})$, which has no one-dimensional representation. The regular based $*$-algebras in question are semisimple; their Frobenius–Perron or signed dimension characters supply the required representation. In the ordinary ring, reciprocity gives $N_i^* = N_{i^*}$, so commutativity makes N_i normal. Its operator norm is its spectral radius d_i. Every matrix entry is bounded by that norm, and $d_i^2 < D = \sum_j d_j^2$. □

For a commutative self-dual table its characters have real values, and its formal codegrees are $f_\chi = \sum_i \chi(X_i)^2$. Orthogonality gives

$$\sum_\chi \frac{\chi(X_i)\chi(X_j)}{f_\chi} = \delta_{ij}, \qquad \sum_\chi f_\chi^{-1} = 1. \tag{2.2}$$

All codegrees are positive. The d-number and cyclotomicity conditions are collected in Proposition 2.8. In degree two, the d-number criterion says that the norm divides the square of the trace. A real character field of a categorification is totally real and abelian.

We write $R = K_0(\mathcal{C})$, $d_i = \mathrm{FPdim}(X_i)$ and $D = \sum_i d_i^2$. In the self-dual case, let $\mathcal{S}$ denote the sphericalization and ε its canonical central sign object. Its central lift is the nontrivial object of the canonical $\mathrm{Rep}(C_2)$ in the equivariantization construction of sphericalization [1, Section 5.3]; its spherical dimension and ribbon twist are both -1. Write $T = K_0(\mathcal{S})/(\varepsilon + 1)$ and $x_i = \dim(X_i^+)$, so $E = \sum_i x_i^2$ and $\dim \mathcal{S} = 2E$. The lifts are self-dual: duality preserves spherical dimensions, and the two lifts have opposite nonzero dimensions. The notation $\beta_i = x_i$ is used when the lift signs are chosen to make the embedded spherical dimensions positive at the selected real embedding. A Galois change of embedding is made explicitly;

neither sphericality nor a unitary realization alone is used to assert pseudo-unitarity. The ordinary and signed codegrees are denoted by f_i, g_i, and

$$u = \sum_i f_i^{-2}, \qquad v = \sum_i g_i^{-2}.$$

Lemma 2.2 (Sphericalization and induction)**.** *Let $\mathcal{C}$ be a complex fusion category with a self-dual rank-four ring R, FP dimensions d_i, and $D = \sum_i d_i^2$. Its sphericalization has a commutative signed quotient T of rank four. Choose positive embedded spherical dimensions β_i for its lifts and put $E = \sum_i \beta_i^2$. Then:*

1. *$|T_{ij}^k| \le N_{ij}^k$ and $T_{ij}^k \equiv N_{ij}^k \pmod 2$;*
2. *all ordinary and signed codegrees lie in $L = \mathbb{Q}(\beta_i)$, and E/f is an algebraic integer for every such codegree f;*
3. *every signed codegree is at most D and $E \le D$;*
4. *if $E = D$, the chosen lifts give a spherical section of $\mathcal{C}$.*

These statements remain valid after real embeddings of L and changes of lift signs. In particular, in seeking a nonspherical categorification one may assume $E < D$ and select the larger conjugate when $[L : \mathbb{Q}] = 2$.

Proof. The sphericalization $\mathcal{S}$ has two self-dual lifts of each simple, exchanged by its canonical central sign object ε, whose dimension is -1 and whose square is one; see [1, Section 5.3] and [16, Section 5.1]. Taking the quotient by $\varepsilon + 1$ gives signed coefficients $T_{ij}^k = N_{ij}^{k,+} - N_{ij}^{k,-}$, while $N_{ij}^k = N_{ij}^{k,+} + N_{ij}^{k,-}$. Its real regular matrices are symmetric. Semisimplicity follows from the positive coefficient-of-unit form; the dimension character excludes $M_2(\mathbb{C})$, the sole noncommutative semisimple algebra of dimension four. Thus T and $K_0(\mathcal{S})$ are commutative. The latter has codegrees $2f_j, 2g_j$ and spherical dimension $2E$.

Central induction into $Z(\mathcal{S})$ has one simple constituent of $I(1)$ per character, of dimension E/f_j or E/g_j and twist one [16, Theorems 2.5 and 2.13]. These dimensions are algebraic integers; codegree-field containment is [16, Corollary 2.15]. The signed g_j are d-numbers because $2g_j$ are codegrees of $\mathcal{S}$. Spectral comparison $|T_i| \le N_i$ bounds every signed character value in absolute value by d_i. Thus its codegree is at most D. If $E = D$, then $\beta_i = d_i$ throughout, and

$$\sum_k (N_{ij}^k - T_{ij}^k) d_k = 0$$

forces $T_{ij}^k = N_{ij}^k$. The positive lifts are tensor closed and form the required spherical section. The same algebraic arguments hold after each real embedding, followed by choosing the signs of the lifts to make their values positive. □

The central quotients $E/f_i, E/g_i$ belong to the signed order itself, not merely to its field's ring of integers: the forgetful image is an integral signed combination of the chosen lifts. For a field algebra, the coefficient-of-unit functional satisfies

$$\tau_T(a) = \mathrm{Tr}_{L/\mathbb{Q}}(a/E). \tag{2.3}$$

Indeed $\mathrm{Tr}(a) = \sum_i \tau_T(x_i a x_i) = \tau_T(aE)$. In particular $\mathrm{Tr}(x_i/E) = \delta_{i0}$. When both ordinary and signed algebras are the same quartic field,

$$\mathrm{disc}(R) = N(D), \quad \mathrm{disc}(T) = N(E), \quad N(E/D) \in \mathbb{Z}_{>0}^2. \tag{2.4}$$

The first two identities follow by comparing the trace pairing with the orthonormal based pairing. The last follows because the ratio of two order discriminants is a rational square and E/D is integral.

2.1 Spectral and trace saturation

A self-dual based ring has a basis $b_0 = 1, b_1, b_2, b_3$ and a symmetric integral multiplication tensor, with coefficient-of-unit form $\langle b_i, b_j \rangle = \delta_{ij}$. A signed table below has the same unit and form; its nonunit structure constants need not be positive. A gauge change replaces each b_i by $s_i b_i$, where $s_0 = 1$ and $s_i \in \{1, -1\}$.

Lemma 2.3. *Let $N = (N_{ijk})$ be the structure tensor of a self-dual rank-four fusion ring, and let $T = (T_{ijk})$ be a symmetric integral structure tensor defining an associative unital algebra on the same orthonormal basis $1, X, Y, Z$, with $T_{0ij} = \delta_{ij}$. If*

$$|T_{ijk}| = N_{ijk} \qquad (0 \le i, j, k \le 3),$$

then there are signs $s_0 = 1$, $s_1, s_2, s_3 \in \{1, -1\}$ such that

$$T_{ijk} = s_i s_j s_k N_{ijk}.$$

Proof. Associativity and symmetry give, for every i, j,

$$\sum_k T_{iik} T_{jjk} = \sum_k T_{ijk}^2.$$

The same identity holds for N, and the right-hand sides agree. Each term $N_{iik}N_{jjk} - T_{iik}T_{jjk}$ is nonnegative; therefore each is zero. For a fixed k, all nonzero coefficients T_{iik} consequently have the same sign.

Use the usual ten-entry convention

$$(a, b, c, d, e, f, g, h, i, j) = (N_{111}, N_{112}, N_{113}, N_{122}, N_{123}, N_{133}, N_{222}, N_{223}, N_{233}, N_{333}).$$

The three groups of diagonal coefficients are

$$(a, d, f), \qquad (b, g, i), \qquad (c, h, j).$$

Independent sign changes of X, Y, Z make every signed coefficient in these groups nonnegative. Hence the two tables now agree except possibly at e. If they agree there too, we are done. Otherwise $e > 0$ and the signed table has coefficient $-e$.

Compare the associativity equations for the two choices of sign of e. Their linear and constant parts give

$$\begin{gathered} a = d + f, \qquad g = b + i, \qquad j = c + h, \\ bh + ci = bc, \qquad dh = cd + fh, \qquad bf + di = fi, \\ e^2 = 1 + df + bi + ch. \end{gathered}$$

We claim that one of the three displayed diagonal groups is zero. If $d > f$, then $bf = i(f - d)$ forces $i = 0$ and $bf = 0$. For $f > 0$ this gives $b = g = i = 0$. For $f = 0$, the equation $dh = cd$ gives $h = c$; then $e^2 = 1 + c^2$ forces $c = h = j = 0$ by integrality. If $d = f > 0$, the equations give $b = c = 0$ and then $e^2 = 1 + d^2$, an impossibility. If $d = f = 0$, then $a = d = f = 0$. Finally, if $d < f$, the identity $cd = h(d - f)$ forces $h = 0$ and $cd = 0$. For $d > 0$ this gives $c = h = j = 0$. For $d = 0$, the equation $bf = fi$ gives $b = i$; then $e^2 = 1 + b^2$ forces $b = g = i = 0$. This proves the claim in every case.

Flip the basis sign belonging to a zero group. It preserves all nine diagonal coefficients and reverses the sign of e. The two tensors now agree, proving the lemma. □

Corollary 2.4. *Under the same hypotheses, suppose only that $|T_{ijk}| \le N_{ijk}$. Let C_N and C_T be multiplication by the respective Casimir elements $\sum_i b_i^2$. If*

$$\operatorname{Tr}(C_T) = \operatorname{Tr}(C_N),$$

then T and N are gauge equivalent.

Proof. In the regular representation the multiplication matrices are symmetric, and

$$\operatorname{Tr}(C_N) = \sum_{i,j,k} N_{ijk}^2, \qquad \operatorname{Tr}(C_T) = \sum_{i,j,k} T_{ijk}^2.$$

Equality, together with termwise domination, forces equality of every absolute value. Apply Lemma 2.3. □

Corollary 2.5. *Equality of the ordinary and signed Casimir traces in a rank-four self-dual sphericalization implies that the original category is spherical.*

Proof. Corollary 2.4 supplies a basis sign change for which the signed and ordinary tensors agree. The chosen lifts are tensor closed, so the spherical structure restricts to a tensor subcategory equivalent to the original category. □

Lemma 2.6 (Galois transport)**.** *If a spherical dimension character belongs to the Galois orbit of the Frobenius–Perron character, the category has a pseudo-unitary Galois conjugate with the same fusion rules. Its central identities can therefore be evaluated with positive Frobenius–Perron dimensions.*

Proof. Extend the required embedding of the dimension field to an automorphism of $\mathbb{C}$. Apply it to skeletal tensor, duality and spherical structure constants. All their identities and the fusion rules persist, while the dimension character becomes the Frobenius–Perron character. This requires no algebraic field of definition for the associativity coefficients. Equivalently, apply the automorphism to the central identities; restriction multiplicities are fixed and twists remain roots of unity. □

Corollary 2.7 (Multiplicity-one sphericality)**.** *Every complex categorification of a self-dual rank-four fusion ring of multiplicity one admits a spherical structure.*

Proof. For its signed quotient, domination and parity give $|T_{ij}^k| \le N_{ij}^k$ and $T_{ij}^k \equiv N_{ij}^k \pmod 2$. Since $N_{ij}^k \in \{0,1\}$, $|T_{ij}^k| = N_{ij}^k$ for every coefficient. Lemma 2.3 makes the two tables gauge equivalent. The corresponding tensor-closed lifts form a spherical subcategory equivalent to the original category. □

Proposition 2.8 (Established inputs)**.**

1. *In any complex categorification every formal codegree is a d-number. For its monic minimal polynomial $t^e + a_1 t^{e-1} + \cdots + a_e$, this means $a_e^j \mid a_j^e$ for $1 \le j \le e$ [17].*
2. *If $S \subseteq R$ is a fusion subring, $D/\operatorname{FPdim}(S)$ is an algebraic integer [4, Theorem 7.17.6].*
3. *Every character value of a complex-categorifiable fusion ring is cyclotomic [6, Theorem 8.51 and Corollary 8.53].*
4. *In a spherical categorification with dimension character χ, every $\chi(b_i)$ is nonzero, $\chi(b_i) = \chi(b_{i^*})$, and its global dimension $\Delta = f_\chi$ satisfies $\Delta/f \in \overline{\mathbb{Z}}$ for every formal codegree f. Its dimension field contains all formal codegrees [16, Theorem 2.13 and Corollaries 2.14–2.15].*

5. *A weakly integral fusion category is pseudo-unitary [6, Proposition 8.24]. A categorifiable fusion ring with precisely two distinct simple Frobenius–Perron dimensions has a pseudo-unitary Galois conjugate [20, Lemma 3.1].*

For reference, for a spherical category of global dimension H, central induction and higher indicators satisfy

$$\sum_{A\in \operatorname{Irr} Z(\mathcal{A})} [I(V) : A] \dim(A)\theta_A^n = H\nu_n(V). \tag{2.5}$$

Here the twists are roots of unity, and $\nu_n(V) \in \mathbb{Q}(\zeta_n)$; see [13, Theorem 4.1] and [14, Theorem 5.1]. Induction of the unit has one multiplicity-one constituent per character, of dimension H/f_χ and twist one [16, Theorems 2.5 and 2.13].

Lemma 2.9 (The central sign and its translate twists)**.** *Let $\mathcal{S}$ be a rank-four self-dual sphericalization, with central sign ε, and let $A_\chi \subset I(1)$ be the multiplicity-one constituent corresponding to a character χ. Its canonical central sign translate satisfies*

$$\theta_{\widehat{\varepsilon}\otimes A_\chi} = \chi(\varepsilon).$$

Thus ordinary constituents have translate twist $+1$ and signed constituents have translate twist -1. The constituents of $I(1)$ and their sign translates are disjoint. Moreover $\nu_n(\varepsilon) = 0$ for odd n and -1 for even n.

Proof. Central induction identifies modular characters by

$$\frac{S_{A_\chi,W}}{S_{A_\chi,0}} = \chi(F(W))$$

[16, Example 2.9 and the paragraph preceding Theorem 2.13]. At $W = \widehat{\varepsilon}$ the ratio is $\chi(\varepsilon)$; it is also $\dim(\varepsilon)$ times the monodromy scalar. Since $\dim(\widehat{\varepsilon}) = \theta_{\widehat{\varepsilon}} = -1$ and $\theta_{A_\chi} = 1$, balancing gives the asserted twist. The constituents have dimensions $E/f_i, E/g_i > 0$, whereas their translates have the opposite dimensions, proving disjointness. For the indicator, the invariant space is zero for odd tensor powers of ε. For even powers it is one-dimensional, and the rotation is multiplied by the inverse pivotal map -1. □

Lemma 2.10. *In a pivotal fusion category, a self-dual simple with second indicator -1 has even multiplicity in every adjoint product $Y^* \otimes Y$.*

Proof. Let j be the pivotal structure and choose $f : X \to X^*$ with transpose $f^t = f^* j_X = -f$. On $W = Y \otimes X \otimes Y^*$, the map $\alpha = j_Y \otimes f \otimes \mathrm{id}_{Y^*} : W \to W^*$ satisfies $\alpha^t = -\alpha$, by monoidality and $j_Y^* j_{Y^*} = 1$. The pairing $B(v, w) = v^*\alpha w$ on $\mathrm{Hom}(1, W)$ is nondegenerate by semisimplicity and alternating by this transpose identity. Its dimension is even. Rigidity identifies that dimension with the desired multiplicity. □

3 Indicator inequalities

Lemma 3.1 (Positive-measure inequalities)**.** *Let μ be a finite positive measure on the unit circle, with mass M and real first two moments A, B. Then*

$$0 \le M + B \le 2M, \qquad 2A^2 \le M(M + B).$$

If its first m moments all equal the real number c, then

$$\mu(\{1\}) \le \frac{M + mc}{m + 1}.$$

These statements also apply after subtracting any known positive atomic submeasure.

Proof. Writing $z = e^{it}$, one has $\int \cos^2 t \, d\mu = (M + B)/2$; Cauchy–Schwarz gives the first inequalities. The nonnegative Fejér polynomial $F_m(z) = |1 + z + \cdots + z^m|^2/(m+1)$ has value $m + 1$ at one and integral $M + mc$, proving the atom bound. □

Lemma 3.2 (Second moment)**.** *Let u, v be the ordinary and signed inverse-square codegree sums. For any signed dimension character,*

$$E(u + v - 1) \le \sum_{i=0}^{3} |\beta_i| \le 1 + \sqrt{3(E-1)}.$$

Proof. The eight constituents of $I(1)$ have positive dimensions E/f and squared mass $H = E^2(u+v)$. Their translates by the canonical central sign object have opposite dimensions, are disjoint from them, and have twists ± 1. The center's total squared dimension is $4E^2$. The central indicator formula [13, Theorem 4.1] gives

$$\sum_A (\dim A)^2 \theta_A^2 = 4Es, \qquad s = 1 + \sum_{i=1}^{3} \nu_2(X_i^+)\beta_i.$$

Here each $\nu_2(X_i^+) = \pm 1$ and the two lifts have opposite dimensions and opposite second indicators. Removing the sixteen known objects, the weighted sum of squared real parts of the remaining twists is

$$\tfrac{1}{2}((4E^2 - 2H) + (4Es - 2H)) = 2E^2 + 2Es - 2H \ge 0.$$

This gives the first bound, and Cauchy's inequality gives the second. □

Lemma 3.3 (Quartic trace and norm)**.** *If L has degree four, the signed rational algebra is L itself, and its four character codegrees are the conjugates of E, with multiplicity. Moreover*

$$\mathrm{Tr}_{L/\mathbb{Q}}(E) \le \sum_j f_j, \qquad \mathrm{Norm}_{L/\mathbb{Q}}(E/f) \in \mathbb{Z}_{>0}$$

for every ordinary codegree f. If v is the signed inverse-square sum, then

$$v \ge \frac{1}{2} - \frac{9}{2} \frac{\mathrm{Tr}_{L/\mathbb{Q}}(E)}{\mathrm{Norm}_{L/\mathbb{Q}}(E)}.$$

Proof. The dimension character is a surjection from a rational algebra of dimension four onto L, hence an isomorphism. The Casimir's regular trace is the sum of the codegrees. Its trace is also $\sum_i \mathrm{Tr}(T_i^2) = \sum_{i,j,k} (T_{ij}^k)^2$, at most the corresponding ordinary sum. Norm integrality is Lemma 2.2. For $q_i = 1/g_i$ the reciprocal identity gives $\sum q_i = 1$, and the elementary four-variable Schur inequality $9e_3(q) + 2\sum q_i^2 \ge 1$ gives the result, since $e_3(q) = \mathrm{Tr}(E)/\mathrm{Norm}(E)$. □

3.1 Galois averages in a quartic signed field

Assume here that the signed dimension field L has degree four. It identifies the signed rational algebra with L. We use the normalized trace identity (2.3).

Proposition 3.4. *Suppose that, for every integer $1 \le n \le m$,*

$$L \cap \mathbb{Q}(\zeta_n) = \mathbb{Q}. \tag{3.1}$$

Then

$$2u + v \le \frac{3}{4} + \frac{3}{m+1}. \tag{3.2}$$

Proof. Choose a finite Galois extension $\Omega/\mathbb{Q}$ containing L, all twists of simple objects of $Z(\mathcal{S})$, and $\mathbb{Q}(\zeta_n)$ for $1 \le n \le m$. Put $G = \mathrm{Gal}(\Omega/\mathbb{Q})$. On the unit circle define

$$\mu = \frac{4}{|G|} \sum_{\sigma \in G} \sum_{A \in \mathrm{Irr}\, Z(\mathcal{S})} \frac{\sigma(\dim A)^2}{4\sigma(E)^2} \, \mathrm{Dirac}_{\sigma(\theta_A)}.$$

Every central dimension lies in L, since forgetting expresses it as an integral linear combination of the x_i. All its conjugates are real, so this measure is positive. The global-dimension identity for the center shows that its total mass is four.

The central-induction indicator formula [13, Theorem 4.1] is $\mathrm{tr}(\theta^n|_{I(V)}) = 2E\nu_n(V)$. Multiplying by $\dim V$ and summing, adjunction gives

$$\sum_A \dim(A)^2 \theta_A^n = 2E \sum_{V \in \mathrm{Irr}\, \mathcal{S}} \dim(V)\nu_n(V).$$

Consequently, for $1 \le n \le m$,

$$\int z^n \, d\mu(z) = \frac{2}{|G|} \sum_{\sigma \in G} \sum_{V \in \mathrm{Irr}\, \mathcal{S}} \sigma\!\left(\frac{\dim V}{E}\right) \sigma(\nu_n(V)). \tag{3.3}$$

The cyclic operator defining $\nu_n(V)$ has order dividing n; hence $\nu_n(V) \in \mathbb{Q}(\zeta_n)$ [14, Theorem 5.1]. By (3.1), restriction of G to L and $\mathbb{Q}(\zeta_n)$ is onto the product of their Galois groups. The average of each product in (3.3) therefore factors into the two separate averages. Equation (2.3) annihilates all nonunit terms. The two remaining objects are $\mathbf{1}$ and ε. The unit has every indicator equal to one. The sign has indicator -1 at even orders and zero at odd orders. Their normalized dimension traces are respectively 1 and -1. It follows that

$$\int z^n \, d\mu(z) = \begin{cases} 1, & n \text{ even}, \\ \frac{1}{2}, & n \text{ odd}. \end{cases} \tag{3.4}$$

Lemma 2.9 gives the indicator signs, the eight constituents of $I(1)$ of dimensions $E/f_i, E/g_i$, and their eight disjoint sign translates. Thus translates of the R constituents have twist $+1$, while translates of the T constituents have twist -1. These statements are preserved by every $\sigma \in G$. Summing the squared weights yields the two estimates

$$\mu(\{1\}) \ge 2u + v, \qquad \mu(\{-1\}) \ge v. \tag{3.5}$$

By Cauchy–Schwarz, $v \ge 1/4$. Consequently $\nu = \mu - \frac{1}{4}\,\mathrm{Dirac}_{-1}$ is a positive measure. Its mass is $15/4$, and its first m moments are all $3/4$. Apply Lemma 3.1 to obtain $(m+1)\mu(\{1\}) \le (15 + 3m)/4$. Together with (3.5), this proves the claim. □

Proposition 3.5 (Squared moments)**.** *If $L \cap \mathbb{Q}(\zeta_{2j}) = \mathbb{Q}$ for $1 \le j \le a$, then*

$$u + v \le \frac{1}{2} + \frac{3}{2(a+1)}.$$

In particular $u + v \le 7/8$ always, and $u + v \le 4/5$ if $\sqrt{2} \notin L$.

Proof. Push the preceding averaged measure forward by $z \mapsto z^2$. Its total mass is four, its first a moments are one, and its atom at one is at least $2(u + v)$. Apply Lemma 3.1. The first three intersections are trivial because L is totally real. The fourth can be nontrivial only if $\mathbb{Q}(\sqrt{2}) \subset L$. □

Corollary 3.6. *If the signed dimension field L is quartic, then*

$$\sqrt{2} \notin L \implies u+v \le \frac{4}{5}, \qquad \sqrt{5} \notin L \implies 2u+v \le \frac{9}{8}.$$

Proof. The first assertion is Proposition 3.5. For the second, the only nontrivial totally real intersection of L with a cyclotomic field of order at most seven can be $\mathbb{Q}(\sqrt{5})$. Apply Proposition 3.4 with $m = 7$. □

Lemma 3.7. *Suppose the signed dimension field is $K = \mathbb{Q}(\sqrt{2}, \sqrt{5})$. With the preceding notation, one has*

$$5u + 4v \le \frac{15}{4}.$$

Proof. Use the unsquared averaged central measure μ above. The nontrivial quadratic subfields of K have conductors $5, 8, 40$. Hence $K \cap \mathbb{Q}(\zeta_n) = \mathbb{Q}$ whenever $5 \nmid n$ and $8 \nmid n$. The same indicator calculation gives

$$\int z^n \, d\mu = \begin{cases} 1/2 & n \text{ odd}, \\ 1 & n \text{ even}, \end{cases} \qquad 5 \nmid n, \quad 8 \nmid n.$$

These identities hold at every such order: extending the averaging field leaves the measure unchanged.

Restrict μ to the fortieth roots of unity. Averaging the functions z^{40j} over j gives their indicator function, since the measure has finite root-of-unity support. Thus, for each residue n divisible by neither 5 nor 8, the restricted measure retains the displayed nth Fourier coefficient. Subtract $\nu = \frac{3}{4}\,\mathrm{Dirac}_1 + \frac{1}{4}\,\mathrm{Dirac}_{-1}$. The resulting forty weights have Fourier support only at multiples of 5 or 8. By the discrete Fourier transform they are therefore

$$r_{ij} = A_i + B_j, \qquad i \in \mathbb{Z}/8, \quad j \in \mathbb{Z}/5,$$

where a fortieth root with exponent l corresponds to $(i, j) = (l \bmod 8, l \bmod 5)$. All cells other than $(0, 0)$ and $(4, 0)$ are nonnegative. The forced atoms give

$$r_{00} \ge 2u + v - \frac{3}{4}, \qquad r_{40} \ge v - \frac{1}{4}.$$

The total of the forty weights is at most $4 - 1 = 3$. For an additive array there is the exact identity

$$\sum_{i,j} r_{ij} - 5r_{00} - 3r_{40} = \frac{1}{2} \sum_{j \ne 0} r_{4j} + \frac{5}{4} \sum_{\substack{i \ne 0,4 \\ j \ne 0}} r_{ij} \ge 0.$$

Consequently $3 \ge 5(2u + v - 3/4) + 3(v - 1/4)$, proving the assertion. □

3.2 Pivotal positivity and second-indicator parity

Lemma 3.8 (A pivotal Fejér bound)**.** *For a pseudo-unitary self-dual rank-four category,*

$$(9u - 3)D \le 6 + 2\sqrt{3(D-1)}.$$

The same holds for the FP data whenever a spherical dimension character is Galois conjugate to the FP character.

Proof. Let A_j be the constituents of $I(1)$, with dimensions D/f_j. Their known twist-one mass L_X in $I(X)$ satisfies

$$\sum_{X\neq 1} d_X L_X = \sum_j (D/f_j)(D/f_j - 1) = D^2 u - D.$$

Apply $|1 + z + z^2|^2$ to the positive twist weights of $I(X)$. The first indicator vanishes, the second is ± 1, and the polynomial has value nine at one. Thus $9L_X \leq D(3d_X + 2)$. Multiplying by d_X and summing gives $9(Du - 1) \leq 3(D - 1) + 2\sum_{X\neq 1} d_X$; Cauchy's inequality finishes. For a conjugate dimension character, apply the embedding to the induction and indicator identities. Forgetful multiplicities are nonnegative integers, so all embedded central dimensions are positive; twists remain roots of unity and second indicators remain signs. The same proof applies. □

We will also use the unsimplified form of the same three-term estimate:

$$9(D^2 u - D) \leq D(3(D - 1) + 2S), \qquad S = \sum_{X\neq 1} d_X. \tag{3.6}$$

It follows directly from the proof of Lemma 3.8 before applying Cauchy's inequality.

3.3 Quadratic comparison and determinant concentration

Lemma 3.9 (Quadratic comparison)**.** *Let D, D' be the ordinary FP orbit and $E \geq E'$ a proper signed dimension orbit in the same quadratic field. Set $P = D^{-1} + D'^{-1}$, $q = E^{-1} + E'^{-1}$, $N = DD'$ and $k = EE'/N$. Then k is a positive integer,*

$$kq < P + (k - 1)/D, \qquad q \leq 2P/3. \tag{3.7}$$

Proof. Norm divisibility gives $k \in \mathbb{Z}_{>0}$. Since $D > E \geq \sqrt{kN}$, compare $x + kN/x$ at E and D to obtain the strict inequality. If $k \geq 2$, then $D/D' > k$ and $1/D < P/(k + 1)$, whence $q < 2P/(k + 1) \leq 2P/3$. If $k = 1$, $s = D/E > 1$ is a totally positive unit, so its integral trace is at least three. Using $E \geq E'$ gives $q/P \leq 2/(s + s^{-1}) \leq 2/3$. □

4 Non-self-dual categories

4.1 Statement and proof

The non-self-dual sphericality proof uses no classification theorem, fusion-ring census, or multiplicity bound.

Theorem 4.1. *Every complex fusion category of rank four with a non-self-dual simple object admits a spherical structure.*

We prove the theorem using the sphericalization and formal codegrees. The construction and its section criterion are recalled in [16, Section 5.1]; see also [1, Section 5.3]. Write the simple objects as $1, X, Y, Z$, where $X^* = Z$ and $Y^* = Y$. Let $R = K_0(\mathcal{C})$, let $\mathcal{S}$ be the sphericalization, and let ε be its central sign object. Its two quotients are

$$R = K_0(\mathcal{S})/(\varepsilon - 1), \qquad T = K_0(\mathcal{S})/(\varepsilon + 1).$$

Choose dual-compatible simple lifts. If N_{ij}^k are the coefficients of R and t_{ij}^k those of T, then

$$|t_{ij}^k| \leq N_{ij}^k, \qquad t_{ij}^k \equiv N_{ij}^k \pmod 2. \tag{4.1}$$

These are respectively the sum and difference of the two lift multiplicities. Changing a lift changes its basis sign in T.

Lemma 4.2. *Suppose that $\mathcal{C}$ admits no spherical structure. There is a real quadratic field K and imaginary quadratic fields L, M such that*

$$R \otimes \mathbb{Q} \cong K \times L, \qquad T \otimes \mathbb{Q} \cong K \times M.$$

The formal codegrees are, respectively,

$$D, D', f, f \qquad \text{and} \qquad E, E', g, g,$$

where $D = \mathrm{FPdim}(\mathcal{C})$ is irrational, $f, g \in \mathbb{Z}_{\geq 3}$, and primes denote the nontrivial embedding of K. We can choose the lifts and a real character β of T so that all its values are positive, its codegree is E, and

$$E \geq E', \qquad E < D, \qquad E/D \text{ and } E/g \text{ are algebraic integers.} \tag{4.2}$$

Every displayed codegree is greater than one, and $f \leq D$.

Proof. Lemma 2.1 applies to both regular based $*$-algebras, so both quotients and their direct sum $K_0(\mathcal{S}) \otimes \mathbb{C}$ are commutative.

The complex-linear based duality has trace two, since it fixes $1, Y$ and exchanges X, Z. On primitive idempotents it permutes the characters. Exactly two characters are therefore fixed by duality. The positive Hermitian pairing gives $\chi(b^*) = \overline{\chi(b)}$, so these are precisely the two real characters. In particular, the Galois orbit of any real character has size at most two: the equations $\chi(b) = \chi(b^*)$ persist under Galois automorphisms.

If D were rational, $\mathcal{C}$ would be weakly integral and hence pseudo-unitary and spherical by [6, Proposition 8.24]. Otherwise the Frobenius–Perron character field is quadratic and equals $K = \mathbb{Q}(D)$. The dimension-field theorem [16, Corollary 2.15], applied to $\mathcal{S}$, puts K inside its spherical dimension field. That field has degree at most two by the preceding character argument, so it too is K. The remaining two-dimensional factors have two nonreal characters and are imaginary quadratic fields. Their conjugate characters have the same real formal codegree, hence a rational algebraic integer; these are f and g.

For a commutative algebra with the above orthonormal based pairing, the formal codegrees are positive and satisfy $\sum_\chi f_\chi^{-1} = 1$. Positivity follows also from $f_\chi = \sum_i |\chi(b_i)|^2$. As there are four characters, each codegree is strictly greater than one. The repeated codegrees therefore satisfy $f, g > 2$. Character spectral domination gives $|\chi(b_i)| \leq \mathrm{FPdim}(b_i)$ for R, whence $f \leq D$.

Initially write δ for the codegree of the spherical dimension character of T, so $\dim(\mathcal{S}) = 2\delta$. The formal codegrees of $\mathcal{S}$ are twice those of R and T. The induction theorem [16, Theorem 2.13] implies that δ/D, δ/D' and δ/g are algebraic integers. Conjugate only the signed dimension character, if necessary, to choose its codegree $E \geq E'$. The three divisibility statements show that E/D and E/g remain integral after either embedding. Every value of the conjugated character is real and nonzero. Choose lift signs to make these values positive; dual compatibility is preserved because the character agrees on dual basis elements. This does not change the category or its sphericalization.

By (4.1), spectral domination bounds each positive $\beta(b_i)$ by $d_i = \mathrm{FPdim}(b_i)$. Thus $E = \sum_i \beta(b_i)^2 \leq \sum_i d_i^2 = D$. If equality holds, every $\beta(b_i) = d_i$, and the two dimension equations give

$$\sum_k (N_{ij}^k - t_{ij}^k) d_k = 0.$$

All summands are nonnegative, so $t_{ij}^k = N_{ij}^k$ for every triple. The unselected lift consequently has multiplicity zero in every product of selected lifts. They form a fusion subcategory of $\mathcal{S}$, on which the forgetful functor is an equivalence with the given category $\mathcal{C}$. Restriction transports a spherical structure to $\mathcal{C}$, a contradiction. Hence $E < D$. □

Lemma 4.3. *In the situation of Lemma 4.2, put*

$$n = DD', \qquad k = \frac{EE'}{DD'}.$$

Then $k \in \{1, 2\}$ and $g \in \{3, 4\}$, with $k = 2$ possible only when $g = 3$.

Proof. The norm of the totally positive algebraic integer E/D makes k a positive integer. Set

$$p = 1 - \frac{2}{f} = \frac{1}{D} + \frac{1}{D'}, \qquad q = 1 - \frac{2}{g} = \frac{1}{E} + \frac{1}{E'}.$$

Since $E \geq \sqrt{kn}$ and $E < D$, strict monotonicity of $t + kn/t$ above its minimum gives

$$kq < p + \frac{k-1}{D} \leq 1 + \frac{k-3}{D}. \tag{4.3}$$

Here we used $f \leq D$. If $k \geq 3$, then $q \geq 1/3$ and $D \geq 4$ give the contradiction

$$\frac{k}{3} < 1 + \frac{k-3}{D} \leq \frac{k+1}{4} \leq \frac{k}{3}.$$

Thus $k \leq 2$. If $k = 2$ and $g \geq 4$, the left side of (4.3) is at least one, whereas its right side is $1 - 1/D < 1$.

If $k = 1$, put $r = D/E$. It is the inverse of an algebraic-integer unit of norm one, hence is itself integral. It is totally positive, and $r > 1$, so $r + r^{-1}$ is an integer at least three. Furthermore $E' = rD'$ and $E \geq E'$ imply $D \geq r^2 D'$. Therefore

$$\frac{q}{p} = \frac{D/r + rD'}{D + D'} \leq \frac{2}{r + r^{-1}} \leq \frac{2}{3}.$$

The first inequality is equivalent to $(r^2 - 1)(r^2 D' - D) \leq 0$. Since $p < 1$, we get $q < 2/3$ and $g < 6$.

A nonreal character of T has values $(1, \alpha, t, \overline{\alpha})$, with $t \in \mathbb{Z}$ and α an imaginary quadratic algebraic integer. Put $a = \mathrm{Tr}(\alpha)$ and $b = \mathrm{N}(\alpha)$. Orthogonality to its conjugate character gives

$$2b = 1 + a^2 + t^2, \qquad g = 1 + t^2 + 2b = 2 + a^2 + 2t^2. \tag{4.4}$$

For $g = 5$, the latter identity forces $a^2 = t^2 = 1$, making $2b = 3$, impossible. Hence g is three or four. □

Proof of Theorem 4.1. Suppose that $\mathcal{C}$ admits no spherical structure, and use the preceding lemmas. Unit, duality and Frobenius reciprocity give the signed table

$$\begin{aligned} X^2 &= PX + LY + CZ, & XY &= QX + JY + LZ, \\ Y^2 &= 1 + JX + HY + JZ, & YZ &= LX + JY + QZ, \\ Z^2 &= CX + LY + PZ, & XZ &= 1 + PX + QY + PZ. \end{aligned} \tag{4.5}$$

All six parameters are integers. This is the signed version of the standard non-self-dual table in [11, Proposition 2.1].

If $g = 3$, (4.4) gives $t = 0$, $a = \pm 1$, $b = 1$. Orthogonality to the positive character β gives $1 + a\beta(X) = 0$, hence $a = -1$, $\beta(X) = \beta(Z) = 1$, and α is a primitive cube root of unity. Evaluating X^2, XY and Y^2 in this character gives

$$C = J = 1, \qquad P = L = Q = 0, \qquad H = \mu \in \mathbb{Z}.$$

Writing $v = \beta(Y) > 0$, we obtain

$$v^2 = 3 + \mu v, \quad v' = -3/v, \quad E = 3 + v^2, \quad E' = 3 + 9/v^2. \tag{4.6}$$

The ordering $E \geq E'$ implies $v \geq \sqrt{3}$, so $\mu \geq 0$. In particular,

$$E + E' = \mu^2 + 12, \qquad EE' = 3(\mu^2 + 12). \tag{4.7}$$

If $g = 4$, (4.4) gives $a = 0$, $t = \pm 1$, $b = 1$. Orthogonality to β gives $t = -1$ and $\beta(Y) = 1$, with $\alpha = \pm i$. Evaluation of (4.5) gives

$$L = 1, \qquad J = Q = H = 0, \qquad P = C = \mu \in \mathbb{Z}.$$

Writing $u = \beta(X) = \beta(Z) > 0$, we have

$$u^2 = 1 + 2\mu u, \quad u' = -1/u, \quad E = 2 + 2u^2, \quad E' = 2 + 2/u^2.$$

Again $E \geq E'$ gives $\mu \geq 0$. The case $\mu = 0$ would make β rational, contrary to its field being K, so $\mu \geq 1$. Thus $u = \mu + \sqrt{\mu^2 + 1} > 2$ and $E' < 5/2$. Here $k = 1$, so the totally positive unit $r = D/E > 1$ has integral trace at least three:

$$r \geq \frac{3 + \sqrt{5}}{2} > \frac{5}{2} > E'.$$

This gives $D' = E'/r < 1$, contrary to Lemma 4.2.

It remains to exclude $g = 3$. First suppose $\mu > 0$. Integrality of $E/3$ and (4.7) give $(\mu^2 + 12)/3 \in \mathbb{Z}$, hence $3 \mid \mu$ and $\mu \geq 3$. Moreover

$$K = \mathbb{Q}(\sqrt{\mu^2 + 12}), \qquad \mu^2 + 12 = 3(3(\mu/3)^2 + 4).$$

The radicand has 3-adic valuation one, so K is neither $\mathbb{Q}(\sqrt{5})$ nor $\mathbb{Q}(\sqrt{2})$. From (4.6), $v > 7/2$ and $v^2 = 3 + \mu v > 27/2$, whence

$$E' < \frac{11}{3} < 2 + \sqrt{3}. \tag{4.8}$$

For $k = 1$, the unit $r = D/E > 1$ cannot have trace three, since that would generate $\mathbb{Q}(\sqrt{5})$. Its trace is at least four, so $r \geq 2 + \sqrt{3} > E'$, again giving $D' < 1$.

For $k = 2$, put $a_0 = E/D$ and $s = 2D/E$. The known algebraic integer a_0 has norm two, so $s = 2/a_0 = a_0'$ is an algebraic integer of norm two. It is totally positive and $s > 2$. Its integral trace $s + 2/s$ is therefore greater than three. Trace four would generate $\mathbb{Q}(\sqrt{2})$, which is excluded; thus the trace is at least five and $s > 4 > E'$. Since $D' = E'/s$, this too contradicts $D' > 1$.

Finally suppose $\mu = 0$. Then $E = E' = 6$ and $n = DD'$ is either 36 or 18. Denote the unsigned parameters corresponding to (C, H, J, L, P, Q) by $(c, e, k_0, \ell, p_0, q_0)$. Their parities are

$$c, k_0 \text{ odd}, \qquad e, \ell, p_0, q_0 \text{ even}.$$

For a nonreal character $(1, \alpha, t, \overline{\alpha})$ of R, the equations for $Y(X - Z)$ and $X^2 - Z^2$ give $t = q_0 - \ell$ and $\alpha + \overline{\alpha} = p_0 - c$. Applying (4.4) to R therefore gives

$$f = 2 + (p_0 - c)^2 + 2(q_0 - \ell)^2 \equiv 3 \pmod 8.$$

But $\mathrm{Tr}(D) = n(1 - 2/f)$ is an integer, so $f \mid 2n$. Since f is odd and $n \in \{18, 36\}$, it divides nine; the congruence and $f \geq 3$ force $f = 3$. For $n = 36$ the trace of D is 12, giving $D = D' = 6$, contrary to irrationality. For $n = 18$ the trace is 6, giving negative discriminant $36 - 72$, contrary to reality. Every possibility has now been excluded. □

Corollary 4.4. *Every non-self-dual rank-four complex fusion category has a pseudo-unitary Galois conjugate.*

Proof. It is spherical by Theorem 4.1. Duality has trace two on the based algebra and permutes primitive idempotents, so exactly two characters are real. This condition is Galois stable. If D is rational, weak integrality gives pseudo-unitarity. Otherwise the Frobenius–Perron orbit consists of both real characters: its degree is two. The spherical dimension character is real and hence belongs to this orbit. Apply Lemma 2.6. □

5 Categories with a nontrivial invertible object

5.1 A nontrivial invertible object forces sphericality

Theorem 5.1. *Every self-dual rank-four complex fusion category with a nontrivial invertible simple object admits a spherical structure.*

Use the sphericalization notation of Section 2, choosing positive signed dimensions β_i and a maximal conjugate E. Lemma 2.2 shows that, in the absence of a spherical section,

$$E < D. \tag{5.1}$$

If D is rational, the category is weakly integral and already spherical by [6, Propositions 8.23–8.24]. We therefore restrict attention to irrational D.

Lemma 5.2 (The fixed action, quadratic case)**.** *Suppose the nontrivial invertible simple G fixes the other two simples. A proper signed quotient cannot have quadratic dimension field.*

Proof. The positive lift G^+ has dimension one and squares to the positive unit. It fixes the positive lifts of both other simples: a sign translate would have the wrong nonzero dimension. Both quotients therefore have the rational character $(1,-1,0,0)$ on $1, G, X, Y$, of codegree two. Field containment makes the ordinary FP field quadratic as well. Its other rational codegree h_0 gives

$$p = \frac{1}{D} + \frac{1}{D'} = \frac{1}{2} - \frac{1}{h_0} < \frac{1}{2}.$$

The signed rational algebra is $\mathbb{Q}^2 \times K$. Write its other rational character as $(1,1,r,t)$, its codegree as $h = 2 + r^2 + t^2$, and its positive dimension character as $(1,1,u,v)$. Thus

$$2 + ru + tv = 0, \qquad q = \frac{1}{E} + \frac{1}{E'} = \frac{1}{2} - \frac{1}{h}.$$

Lemma 3.9 gives $q < 1/3$, so $2 < h < 6$. The case $h = 5$ is impossible because $r^2 + t^2 = 3$.

Write the signed multiplication as

$$X^2 = 1 + G + aX + bY, \qquad XY = bX + cY, \qquad Y^2 = 1 + G + cX + eY,$$

where a, b, c, e are integers. If $h = 4$, then $|r| = |t| = 1$. Both positive signs contradict $2 + ru + tv = 0$. Both negative signs give $u + v = 2$, so $E < 6$, whereas $q = 1/4$ and $E \geq E'$ give $E \geq 8$. After interchanging X, Y, the remaining case is $(r,t) = (1,-1)$, with $v = u + 2$. Evaluation of the products gives

$$(a,b,c,e) = (b-1, b, b+1, b+2), \qquad u^2 = (2b-1)u + 2b + 2.$$

Put $\Delta = 4b^2 + 4b + 9$, an odd integer. Substitution gives

$$E = 4b + 10 + (4b+2)u, \qquad EE' = 8\Delta.$$

But $E/4$ is an algebraic integer, whose norm would be $\Delta/2$, a contradiction.

It remains that $h = 3$. After interchanging X, Y, positivity forces $(r, t) = (0, -1)$ and $v = 2$. The table becomes

$$X^2 = 1 + G + \mu X + 2Y, \qquad XY = 2X, \qquad Y^2 = 1 + G + Y.$$

Consequently

$$u^2 = 6 + \mu u, \quad E = 6 + u^2, \quad E' = 6 + 36/u^2, \quad E + E' = \mu^2 + 24.$$

Our ordering makes $\mu \geq 0$. Integrality of $E/2$ and $E/3$ forces $6 \mid \mu$; write $\mu = 6j$. Put $k = EE'/(DD')$ and $s = kD/E = (E/D)'$. Then

$$k < s < E', \qquad s' = k/s < 1.$$

If $j = 0$, then $E = E' = 12$ and $K = \mathbb{Q}(\sqrt{6})$. Write $s = c + d\sqrt{6}$ with integers c, d. Since $s > s' > 0$, $d \geq 1$ and $c > d\sqrt{6}$, so $c \geq 3$. Therefore

$$p = \frac{s + k/s}{12} = \frac{c}{6} \geq \frac{1}{2},$$

a contradiction.

Suppose $j > 0$. Then $u > 6j$, $E > 42$, and $E' < 7$. Equation (3.7), with $q = 1/6$ and $p < 1/2$, forces $k \leq 3$: if $k \geq 4$, then

$$\frac{k}{6} < \frac{1}{2} + \frac{k-1}{42} \leq \frac{k}{6}.$$

The integral trace $T = s + k/s$ satisfies $k + 2 \leq T \leq 7$. The field $K = \mathbb{Q}(\sqrt{3(3j^2+2)})$ is ramified at three, so $T^2 - 4k$ has odd three-adic valuation. The only possibilities are

$$(k, T) = (1, 4), (1, 5), (3, 6).$$

Indeed $k = 2$ is impossible modulo three; for $k = 1$ the traces $3, 6, 7$ have respectively discriminants $5, 32, 45$, none ramified at three; and $k = 3$ forces $T = 6$. The first two possibilities would give $K = \mathbb{Q}(\sqrt{3})$ or $\mathbb{Q}(\sqrt{21})$. They would force respectively $3j^2 + 2 = a^2$ or $3j^2 + 2 = 7a^2$ for an integer a, impossible modulo three. Thus $k = 3$ and $s = 3 + \sqrt{6}$. But

$$p = sq - \frac{s - k/s}{E} = \frac{1}{2} + \sqrt{6}\left(\frac{1}{6} - \frac{2}{E}\right) > \frac{1}{2},$$

again a contradiction. □

Proposition 5.3. *Let $\mathcal{C}$ be a self-dual rank-four complex fusion category. Suppose that both its ordinary rational Grothendieck algebra and the signed rational algebra of its sphericalization are the product of $\mathbb{Q}$ with a totally real cubic field, and that the Frobenius–Perron and spherical dimension characters belong to the respective cubic factors. Then the ordinary rational character does not have formal codegree two.*

Proof. Suppose its codegree is two. After relabeling, the rational character is $(1, -1, 0, 0)$ on the basis $\mathbf{1}, G, X, Y$: orthogonality to the positive Frobenius–Perron character fixes the negative sign and gives $\mathrm{FPdim}(G) = 1$. Thus G is invertible. Its values on the three cubic characters are all one, so $G^2 = \mathbf{1}$, $GX = X$, and $GY = Y$.

Let $\mathcal{S}$ be the sphericalization, with central sign ε. Choose positive lifts G^+, X^+, Y^+ and their sign translates. The lift G^+ has dimension one, satisfies $(G^+)^2 = \mathbf{1}$, and fixes X^+, Y^+ on both sides: the alternative sign translates have the wrong nonzero dimension. In the

signed quotient write G, X, Y for their classes. Its rational character is again $(1, -1, 0, 0)$ and has codegree two. Its multiplication has the form

$$X^2 = 1 + G + aX + bY, \quad XY = bX + cY,$$
$$Y^2 = 1 + G + cX + eY,$$

with integers a, b, c, e, which may be negative. The signed Casimir is therefore

$$1 + G^2 + X^2 + Y^2 = 4 + 2G + (a + c)X + (b + e)Y.$$

Put $x = \dim X^+$, $y = \dim Y^+$, and $E = 2 + x^2 + y^2$. Since the dimension character is faithful on the cubic quotient by $G - 1$, the numbers $1, x, y$ are linearly independent over $\mathbb{Q}$. Writing $p = a + c$ and $q = b + e$ gives

$$E = 6 + px + qy, \qquad E/2 = 3 + (p/2)x + (q/2)y. \tag{5.2}$$

Let $F : Z(\mathcal{S}) \to \mathcal{S}$ be forgetful and I its right adjoint. The ordinary and signed rational characters give two distinct one-dimensional characters of $K_0(\mathcal{S})$, with ε acting by $+1$ and -1, respectively. Both have formal codegree four. The central-induction theorem therefore supplies distinct constituents B_+, B_- of $I(\mathbf{1})$, each occurring once and satisfying

$$\dim B_+ = \dim B_- = \frac{\dim \mathcal{S}}{4} = \frac{E}{2}, \qquad \theta_{B_+} = \theta_{B_-} = 1.$$

Moreover $I(\mathbf{1})$ and $I(\varepsilon)$ have no common constituent: by adjunction and the induction formula their inner product is the trace of multiplication by ε, which is zero since it interchanges every pair of lifts. Hence each $F(B_\pm)$ contains the unit once and contains no ε.

Write its signed multiplicity at G as

$$r_G = [F(B_\pm) : G^+] - [F(B_\pm) : \varepsilon G^+].$$

Comparing dimensions with (5.2), using the independence of $1, x, y$, gives $1 + r_G = 3$. Thus $[F(B_\pm) : G^+] \geq 2$, so each $B_\pm$ occurs in $I(G^+)$ at least twice.

On the other hand,

$$\dim \operatorname{End}(I(G^+)) = \sum_{V \in \operatorname{Irr}(\mathcal{S})} \dim \operatorname{Hom}(VG^+, G^+V) = 8.$$

Here G^+ is invertible and commutes with every simple class; each summand equals one. The two multiplicities already contribute $2^2 + 2^2 = 8$. Consequently

$$I(G^+) = 2B_+ \oplus 2B_-.$$

Its twist is the identity, and its twist trace is $2E = \dim \mathcal{S}$. The center formula of [13, Theorem 4.1] for the first Frobenius–Schur indicator would give $\nu_1(G^+) = 1$. This contradicts $\nu_1(G^+) = 0$, since G^+ is a nontrivial simple object. □

Proof of Theorem 5.1. Assume there is no spherical section of the sphericalization. We have already excluded rational D. Since every simple is self-dual, the group of invertible simples has exponent two. If it has four elements the category is pointed and spherical. Thus it has exactly two elements $1, G$, and G either fixes or interchanges the other simples X, Y.

First suppose $GX = Y$. Then $G^2 = 1$ and Frobenius reciprocity gives

$$X^2 = Y^2 = 1 + aX + bY, \qquad XY = G + bX + aY$$

for nonnegative integers a, b. Put $m = a + b > 0$. Their common FP dimension satisfies $d^2 = 1 + md$, and

$$D = 2 + 2d^2, \qquad DD' = 4(m^2 + 4).$$

Choose positive lifts G^+, X^+ and put $Y^+ = G^+X^+$. Both latter lifts have the same dimension $x > 0$, and the signed table has $X^2 = 1 + a'X + b'Y$, with $|a'| \le a$, $|b'| \le b$. Thus

$$x^2 = 1 + \mu x, \qquad E = 2 + 2x^2, \qquad \mu = a' + b', \quad |\mu| \le m.$$

The case $\mu = 0$ would give a rational dimension field, contradicting field containment of the irrational D. Otherwise the signed field is quadratic and contains the ordinary quadratic field, so they coincide. Induction divisibility gives the positive integer

$$\frac{EE'}{DD'} = \frac{\mu^2 + 4}{m^2 + 4} \le 1.$$

Hence $\mu^2 = m^2$. The larger conjugate of E equals D, contradicting (5.1) after the permissible Galois conjugation. This proves the assertion for the interchanged action.

Now suppose $GX = X$ and $GY = Y$. The signed quotient has the rational character $(1, -1, 0, 0)$, so its dimension field has degree at most three. A rational dimension field would contain the irrational D, which is impossible. A quadratic dimension field is excluded by Lemma 5.2. Finally, if the signed field is cubic, it contains the irrational D, so both ordinary and signed rational algebras are products of $\mathbb{Q}$ and a cubic field, with the dimension characters in the cubic factors. Their rational characters have codegree two. This contradicts Proposition 5.3. Every case gives a spherical section, completing the proof. □

6 The Fibonacci normal form and two sphericality tests

Lemma 6.1 (Fibonacci normal form)**.** *A rank-four fusion ring containing f with $f^2 = 1 + f$ is self-dual and has a basis $1, f, X, Y$ with*

$$\begin{aligned} fX = Y, \qquad fY = X + Y, \qquad X^2 = 1 + aX + bY, \\ XY = f + bX + (a + b)Y, \qquad Y^2 = 1 + f + (a + b)X + (a + 2b)Y, \end{aligned} \tag{6.1}$$

where $a, b \ge 0$ are integers. Its positive character is $(1, \phi, x, \phi x)$, where $\phi = (1 + \sqrt{5})/2$ and $x^2 = 1 + (a + b\phi)x$. In a normalized signed sphericalization quotient, the same formulas hold with integers α, β satisfying $|\alpha| \le a$, $|\beta| \le b$, $\alpha \equiv a$ and $\beta \equiv b \pmod 2$.

Proof. On the two simples outside the Fibonacci subring, left and right multiplication by f give commuting symmetric nonnegative integral matrices satisfying $M^2 = I + M$. Their trace is one and determinant is -1, so each is a reordering of $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. The two reorderings do not commute; hence the left and right actions coincide. Duality commutes with this action and cannot exchange the two outside simples. All four simples are therefore self-dual. Frobenius reciprocity and associativity give the displayed products. For a signed quotient, first change the sign of f so that $f^2 = 1 + f$. A sign change of an outside basis element normalizes its nonzero off-diagonal action coefficient to one. The same calculation then gives the signed formulas; domination and parity give the stated restrictions on α, β. □

Proposition 6.2. *Every complex categorification of either Fibonacci normal form $(a, b) = (1, 0)$ or $(a, b) = (1, 1)$ admits a spherical structure.*

Proof. The first table has multiplicity one, so Corollary 2.7 applies. For $(a, b) = (1, 1)$, the signed parameters α, β belong to $\{1, -1\}$. Reversing the signs of both outside basis elements changes both parameters' signs. Equal signs therefore recover the ordinary table up to gauge and give a spherical section.

For opposite signs it suffices to take $t = \alpha + \beta\phi = 1 - \phi$. A character evaluating f at ϕ and X at a root of $x^2 - tx - 1$ has character field

$$L = \mathbb{Q}(\sqrt{5}, \sqrt{6 - \phi}), \qquad N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(6 - \phi) = 29.$$

The radicand is not a square in $\mathbb{Q}(\sqrt{5})$, since its norm is not a rational square. If $L/\mathbb{Q}$ were normal, its two conjugate quadratic extensions of $\mathbb{Q}(\sqrt{5})$ would agree. Their radicands would have square ratio, forcing 29 to be a square in $\mathbb{Q}(\sqrt{5})$. This is impossible. Thus L is not normal, contrary to cyclotomicity of signed character values from Lemma 2.2. Only the ordinary gauge remains. □

7 Central columns and descent of the dimension field

Let $\mathcal{C}$ be a self-dual rank-four complex fusion category, $\mathcal{S}$ its sphericalization, and ε its sign object. Write $R = K_0(\mathcal{C})$ and $T = K_0(\mathcal{S})/(\varepsilon + 1)$, and put $D = \mathrm{FPdim}(\mathcal{C})$ and $2E = \dim(\mathcal{S})$. The ordinary and signed rational algebras are commutative, as explained above. Let L be the field generated by the signed spherical dimensions. It is totally real and abelian. Central induction gives $E/D \in L$, so $D \in L$.

We use the canonical normalized modular matrix $S = s/(2E)$ of $Z(\mathcal{S})$. For a one-dimensional character χ of $K_0(\mathcal{S})$, let A_χ be the corresponding simple constituent of $I(\mathbf{1})$. The central-induction theorem and modular-character formula give

$$[I(\mathbf{1}) : A_\chi] = 1, \qquad \dim A_\chi = \frac{2E}{f_\chi}, \qquad \frac{S_{A_\chi, V}}{S_{A_\chi, 0}} = \chi(F(V)). \tag{7.1}$$

Here $F : Z(\mathcal{S}) \to \mathcal{S}$ is forgetful; see [16, Example 2.9 and Theorem 2.13].

Proposition 7.1 (Simultaneous Galois columns)**.** *Let L be the signed dimension field and $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$. Let B_σ be the induction constituent associated with the signed dimension character followed by σ. If χ is any ordinary or signed character and f_χ its codegree in R or T, then*

$$\chi(F(B_\sigma)) = \frac{f_\chi}{\sigma(f_\chi)}.$$

Consequently the codegrees determine both its folded and signed forgetful classes, even if the rational algebras are products of fields. Their coefficient vectors w_σ, z_σ satisfy

$$w_{\sigma,0} = z_{\sigma,0} = 1, \qquad w_{\sigma,i} \geq |z_{\sigma,i}|, \qquad w_{\sigma,i} \equiv z_{\sigma,i} \pmod 2.$$

Proof. Let A_χ be the induction constituent for χ. Its dimension is E/f_χ and $S_{A_\chi,0} = 1/(2f_\chi)$. The modular-character formula gives

$$S_{B_\sigma, A_\chi} = \frac{\sigma(E/f_\chi)}{2\sigma(E)} = \frac{1}{2\sigma(f_\chi)}.$$

Symmetry of S gives the asserted character value. All characters together separate the commutative semisimple algebra, so these values determine each class. If the two lift multiplicities at X_i are m_i^+, m_i^-, the two coefficients are their sum and difference. Adjunction gives unit multiplicity one. The sign multiplicity is zero: $I(\varepsilon) = \widehat{\varepsilon} \otimes I(1)$ has negative-dimensional constituents, whereas the constituents of $I(1)$ have positive dimensions. □

Proposition 7.2 (Positive Galois columns)**.** *Suppose* $[\mathbb{Q}(D) : \mathbb{Q}] = 4$*. Identify* R *with its Frobenius–Perron image and* T *with its spherical-dimension image in the common field* $K = L = \mathbb{Q}(D)$*. For every* $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$*, the element*

$$u_\sigma = \frac{D}{\sigma(D)}$$

belongs to R *and has nonnegative integral coefficients in its distinguished basis. Write* w_σ *for its coefficient vector and* z_σ *for the signed coefficient vector of* $v_\sigma = E/\sigma(E) \in T$*. Then*

$$w_{\sigma,0} = z_{\sigma,0} = 1, \qquad w_{\sigma,i} \geq |z_{\sigma,i}|, \qquad w_{\sigma,i} \equiv z_{\sigma,i} \pmod 2.$$

Proof. Apply Proposition 7.1 to all ordinary and signed characters. In a quartic field the dimension maps are injective, so the folded and signed elements are respectively $D/\sigma(D)$ and $E/\sigma(E)$. The coefficient assertions are those of the proposition. □

Proposition 7.3 (Descent of the dimension field)**.** *If* $L \neq \mathbb{Q}(D)$*, then* $\mathcal{C}$ *admits a nontrivial faithful grading by a finite cyclic group.*

Proof. Choose a nonidentity $\sigma \in \mathrm{Gal}(L/\mathbb{Q}(D))$. Define B_σ as in the preceding proof. The same calculation applies without any injectivity assumption on the dimension maps and gives

$$\mathrm{FPdim}(F(B_\sigma)) = D/\sigma(D) = 1.$$

Adjunction gives $[F(B_\sigma) : \mathbf{1}] = 1$. Consequently $F(B_\sigma) = \mathbf{1}$ in $\mathcal{S}$. The character defining B_σ differs from the dimension character, whose constituent is the tensor unit. Thus $B_\sigma \neq \mathbf{1}$ in $Z(\mathcal{S})$.

A central structure on the tensor unit is a monoidal natural automorphism γ of $\mathrm{Id}_{\mathcal{S}}$, and here γ is nonidentity. Its value at ε is one. Indeed the modular-character formula evaluated at the canonical central lift $\widehat{\varepsilon}$ gives -1; on the other hand this value is $\dim(\varepsilon)$ times the monodromy scalar, namely $-\gamma_\varepsilon$. Hence $\gamma_\varepsilon = 1$.

The scalar values of γ therefore agree on the two lifts of every simple of $\mathcal{C}$. They descend to a nonidentity monoidal natural automorphism of $\mathrm{Id}_{\mathcal{C}}$: if an ordinary fusion coefficient is nonzero, one of its lifted coefficients is nonzero and supplies the required product relation. The finitely many scalar values form a subgroup of $\mathbb{C}^\times$, since tensor products give products and duals give inverses. This finite group is cyclic and gives the asserted faithful grading. □

Lemma 7.4. *A rank-four complex fusion category with a nontrivial faithful grading is either weakly integral or has a nontrivial invertible simple object.*

Proof. The neutral component has rank one, two, or three. In rank one, $X \otimes X^*$ is a sum of units for every simple X, and its unit multiplicity is one. Thus every simple is invertible.

In rank two, the neutral component is pointed or has Fibonacci fusion rules by [18]. In the pointed case its global dimension is two, and equality of the dimensions of homogeneous components makes the whole category weakly integral. In the Fibonacci case write $f^2 = 1 + f$ and $\phi = (1 + \sqrt{5})/2$. A component of rank one is impossible: the integer multiplicity for multiplication by f would satisfy $m^2 = 1 + m$. Thus there is just one other component, of rank two. On its simples X, Y, multiplication by the self-dual object f has a symmetric nonnegative integer matrix M satisfying $M^2 = I + M$. Its eigenvalues are the two roots of $t^2 - t - 1$, so its trace is one and determinant is minus one. Up to exchanging X, Y,

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathrm{FPdim}(Y) = \phi\,\mathrm{FPdim}(X).$$

Equality of component dimensions now gives $(1+\phi^2)\,\mathrm{FPdim}(X)^2 = 1+\phi^2$. Hence X is a nontrivial invertible object.

In rank three, there is a single nonneutral simple X, and the grading group has order two. Each neutral simple Y satisfies $Y \otimes X = m_Y X$, so $\mathrm{FPdim}(Y) = m_Y$ is integral. The two homogeneous components have equal Frobenius–Perron dimension; hence the total dimension is an integer. See [6, Proposition 8.20] for equality of component dimensions. □

Corollary 7.5. *If the rational Grothendieck algebra of a rank-four complex fusion category is a quartic field, its Frobenius–Perron global dimension has degree four.*

Proof. Its Frobenius–Perron character identifies the rational algebra with a real abelian quartic field K. Since this character is injective and $\mathrm{FPdim}(X^*) = \mathrm{FPdim}(X)$, every simple object is self-dual. Write d_i for its simple dimensions. If $D = \sum_i d_i^2$ is rational, all four formal codegrees are D, so the reciprocal-codegree identity gives $D = 4$. All four d_i are then one, contradicting injectivity of the character.

Otherwise, if $\mathbb{Q}(D)$ is a proper subfield of K, choose a nonidentity σ fixing D. Character spectral domination gives $|\sigma(d_i)| \le d_i$. Equality of their square sums forces $\sigma(d_i) = s_i d_i$, with $s_i \in \{\pm 1\}$. Applying σ to the dimension equations and comparing with the positive equations shows that $s_i s_j = s_k$ whenever $N_{ij}^k > 0$. Since the d_i generate K, some s_i is negative. Thus these signs define a nontrivial grading. Lemma 7.4 gives weak integrality or a nontrivial invertible object. Weak integrality would make D rational. A nontrivial invertible is also impossible in a totally real field algebra: it has finite order and thus equals 1 or -1, while its positive dimension is one. Both alternatives are contradictory. □

Corollary 7.6. *Every nonspherical self-dual rank-four complex fusion category satisfies* $L = \mathbb{Q}(D)$.

Proof. Otherwise the preceding proposition and lemma apply. Weakly integral fusion categories are spherical by [6, Proposition 8.24]. Theorem 5.1 gives a spherical structure in the remaining case, a contradiction. □

7.1 The Haagerup ring

Theorem 7.7. *Every complex fusion categorification of the rank-four Haagerup ring admits a spherical structure.*

Proof. In the tuple convention (2.1), this ring is

$$R_1 = (0,0,1,1,1,1,1,1,2,2).$$

Swapping Y, Z gives the tuple $(0,1,0,1,1,1,2,2,1,1)$. In the original order, the characters of R_1 are $(1,0,1,-1)$, $(1,1,-1,0)$, and $(1, z-1, z, z+1)$ with $z^2 = 3z+1$. Consequently, writing $\eta = (3+\sqrt{13})/2$, its FP codegree is

$$D = 3\sqrt{13}\eta = \frac{39+9\sqrt{13}}{2}, \qquad DD' = 117.$$

Suppose the category has no spherical structure. Corollary 7.6 makes the signed dimension field $K = \mathbb{Q}(\sqrt{13})$. Select its larger dimension codegree $E \ge E'$. Then $E < D$, while central induction makes $EE'/117$ a positive integer.

The fundamental unit of K is η, of norm -1. Indeed a positive unit of norm -1 has the form $(a+b\sqrt{13})/2$, with $a, b > 0$ and $a^2 + 4 = 13b^2$; therefore $a \ge 3$, with equality at η. A smaller positive unit of norm one would produce a smaller norm-minus-one unit by division into η. Because E is a d-number, E/E' is a totally positive unit, so $E/E' = \eta^{2h}$

for an integer $h \geq 0$. Set $y = E\eta^{-h}$. Since $\eta' = -\eta^{-1}$, one has $y' = (-1)^h y$. Thus $y \in \mathbb{Q}$ if h is even and $y \in \mathbb{Q}\sqrt{13}$ if h is odd. The element y is integral, because η is an integral unit. As $\mathcal{O}_K = \mathbb{Z}[(1+\sqrt{13})/2]$, the two intersections are respectively $\mathbb{Z}$ and $\mathbb{Z}\sqrt{13}$. Hence the divisibility $117 \mid EE'$ gives precisely

$$E = 3s\sqrt{13}\eta^h \quad (h \geq 1 \text{ odd}), \qquad \text{or} \qquad E = 39s\eta^h \quad (h \geq 0 \text{ even}), \qquad s \in \mathbb{Z}_{>0}.$$

For the first form, its norm is 13 times the square of its rational coefficient, forcing that coefficient to be divisible by three. For the second, the rational coefficient has square divisible by $117 = 9 \cdot 13$, forcing divisibility by 39. The first form satisfies $E \geq D$; the second satisfies $E \geq 39 > D$. Both contradict properness. □

8 Quadratic codegree obstructions

8.1 A quadratic signed dimension field with codegree two

Theorem 8.1. *Let $\mathcal{C}$ be a self-dual rank-four complex fusion category. If the signed quotient of its sphericalization has a quadratic spherical dimension field and a rational formal codegree two, then $\mathcal{C}$ admits a spherical structure.*

We prove that a proper signed quotient is impossible. Choose a field embedding and lifts with positive dimension values, and let $E \geq E'$ be the two dimension codegrees of the signed quotient T. If the ordinary Frobenius–Perron dimension D is rational, the category is weakly integral and hence pseudo-unitary [6]. Otherwise D is quadratic in the signed dimension field K. A proper quotient has $E < D$: equality forces equality in the spectral comparison with the ordinary nonnegative table, giving a spherical section. We do not assume that the ordinary Frobenius–Perron *character* field equals its codegree field; that distinction will be addressed below.

8.1.1 The signed table and its norm restrictions

The rational codegree-two character has row $(1, -1, 0, 0)$ after positive lift choices. Orthogonality makes every other character value on X equal to one. Thus multiplication by X is a symmetric integral involution. Its negative eigenspace is already spanned by $1 - X$, so it fixes Y, Z and exchanges $1, X$. Consequently

$$\begin{aligned} &X^2 = 1, && XY = Y, && XZ = Z, \\ &Y^2 = 1 + X + aY + bZ, && YZ = bY + cZ, && Z^2 = 1 + X + cY + dZ, \end{aligned} \tag{8.1}$$

with integral coefficients and $b^2 + c^2 = 2 + ac + bd$. The other rational character is $(1, 1, r, t)$, with

$$h = 2 + r^2 + t^2 > 2, \qquad 2 + ru + tv = 0,$$

where $(1, 1, u, v)$ is the positive dimension character.

Put

$$p = D^{-1} + (D')^{-1}, \quad q = E^{-1} + (E')^{-1} = \frac{1}{2} - \frac{1}{h}, \quad k = \frac{EE'}{DD'} \in \mathbb{Z}_{>0}.$$

Central induction gives integrality of $E/D, E/2, E/h$. Each of the other two ordinary codegrees is at most D, so $p \leq 1 - 2/D$, whether or not those characters are rational. Comparison on the upper-root interval yields

$$kq < p + \frac{k-1}{D} \leq 1 + \frac{k-3}{D}. \tag{8.2}$$

Moreover

$$s_0 = \frac{kD}{E} = (E/D)' \quad \text{is integral,} \qquad \mathrm{Norm}(s_0) = k, \qquad k < s_0 < E'. \tag{8.3}$$

The last inequality follows from $D' = E'/s_0 > 1$. For $k = 1$, Lemma 8.2 gives $s_0 \geq (3+\sqrt{5})/2$; for $k = 2$, the integral trace $s_0 + 2/s_0$ is at least four (since $s_0 > 2$), so $s_0 \geq 2 + \sqrt{2}$. If $h \geq 6$, then $q \geq 1/3$ and (8.2), with $D \geq 4$, forces $k \leq 2$: for $k \geq 3$ it would give $k/3 < (k+1)/4 \leq k/3$.

For $h \geq 11$, norm divisibility gives $EE' \geq h^2$, whence

$$E' \leq \frac{4h}{h - 2 + \sqrt{(h-2)^2 - 16}} < \frac{4h}{2h-5} \leq \frac{44}{17} < \frac{3+\sqrt{5}}{2}.$$

This contradicts (8.3). The values $h = 5, 8, 9$ are impossible because $h - 2 = 3, 6, 7$ is not a sum of two squares. We next treat $h = 4, 6, 7, 10$ explicitly.

For $h = 4$, the entries r, t have absolute value one. Both positive are incompatible with $2 + ru + tv = 0$. Both negative give $u + v = 2$ and $E < 6$, whereas $E \geq 2/q = 8$. Thus take $(r, t) = (1, -1)$. Evaluation of (8.1) gives

$$v = u + 2, \quad (a, b, c, d) = (b - 1, b, b + 1, b + 2), \quad u^2 = (2b - 1)u + 2b + 2.$$

For $\Delta = 4b^2+4b+9$, the codegrees have trace 2Δ and norm 8Δ. Since Δ is odd, $\mathrm{Norm}(E/4) = \Delta/2$ is not an integer, a contradiction.

For $h = 6$, take $(r, t) = (0, -2)$. Then

$$v = 1, \quad (a, b, c, d) = (\mu, 1, 0, -1), \quad u^2 = 3 + \mu u, \quad E = 3 + u^2, \quad E' = 3 + 9/u^2.$$

The ordering gives $\mu \geq 0$. The trace and norm of E are $\mu^2 + 12$ and $3(\mu^2 + 12)$; integrality of $E/6$ forces $6 \mid \mu$. If $\mu > 0$, then $E' < 13/4$ and $K = \mathbb{Q}(\sqrt{\mu^2 + 12})$ is ramified at three. Hence K is neither $\mathbb{Q}(\sqrt{5})$ nor $\mathbb{Q}(\sqrt{2})$. For $k = 1$ the trace of s_0 is at least four, so $s_0 \geq 2+\sqrt{3} > E'$. For $k = 2$ its trace is at least five, so $s_0 > 4 > E'$. Both are impossible. If $\mu = 0$, then $E = E' = 6$ and $K = \mathbb{Q}(\sqrt{3})$. Norm two is impossible in $\mathbb{Z}[\sqrt{3}]$ modulo three, so $k = 1$. The norm-one unit $1 < s_0 < 6$ has even trace four or six; trace six gives the wrong field. Thus this branch leaves only

$$D = 12 + 6\sqrt{3}. \tag{8.4}$$

For $h = 7$, integrality of $E/7$ gives $EE' = 49A$ and $\mathrm{Tr}(E/7) = 5A/2$, so A is even and $EE' \geq 98$. Consequently

$$E' \leq (35 - \sqrt{833})/2 < 13/4.$$

This excludes $k = 2$ and forces, for $k = 1$, a unit of trace three, hence $K = \mathbb{Q}(\sqrt{5})$. The absolute values of r, t are one and two. Both negative give $u + 2v = 2$ and $E < 6$, contrary to $E \geq (E + E')/2 \geq 35/2$. The two remaining sign choices yield, for an integer j,

(r, t)	(a, b, c, d)	equation for v
$(1, -2)$	$(8j - 5, 4j - 2, 2j, j - 1)$	$v^2 + (1 - 5j)v + 4j - 2 = 0$
$(-1, 2)$	$(8j + 5, 4j + 2, 2j, j + 1)$	$v^2 - (5j + 1)v - 4j - 2 = 0.$

For $\Delta = 25j^2 \pm 26j + 9$ with the corresponding sign, the codegree trace and norm are $5\Delta, 14\Delta$. Integrality of $E/7$ forces $7 \mid \Delta$, while

$$25\Delta = (25j \pm 13)^2 + 56$$

then gives $v_7(\Delta) = 1$. This field cannot be $\mathbb{Q}(\sqrt{5})$.

For $h = 10$, take $(r, t) = (2, -2)$; the other signs contradict positivity or give $u + v = 1$ and $E < 3$, whereas $E \geq 5$. Evaluation gives $v = u + 1$ and $(a, b, c, d) = (b + 1, b, b + 2, b + 1)$. With $m = b + 1$,

$$u^2 = (2m - 1)u + m + 1, \qquad \mathrm{Tr}(E) = 2(4m^2 + 5), \quad \mathrm{Norm}(E) = 5(4m^2 + 5).$$

The norm is odd, contradicting the integrality of $E/2$.

8.1.2 The codegrees two and three

Only $h = 3$ remains. Its rational row can be taken as $(1, 1, 0, -1)$, so evaluation gives

$$(a, b, c, d) = (6j, 2, 0, 1), \quad v = 2, \quad u^2 = 6 + 6ju, \quad j \geq 0,$$
$$E = 6 + u^2, \qquad E' = 6 + 36/u^2.$$

The multiple of six follows from the integral traces of $E/2$ and $E/3$. Suppose first $j > 0$. Then $E > 42$, $E' < 7$, and (8.2) gives $k \leq 6$. The integral trace $t_0 = s_0 + k/s_0$ lies between $k + 2$ and seven. The field $K = \mathbb{Q}(\sqrt{3(3j^2 + 2)})$ is ramified at three, so $t_0^2 - 4k$ has odd three-adic valuation. The entire resulting list is

$$(k, t_0, K) = (1, 4, \mathbb{Q}(\sqrt{3})),\ (1, 5, \mathbb{Q}(\sqrt{21})),\ (3, 6, \mathbb{Q}(\sqrt{6})),\ (4, 7, \mathbb{Q}(\sqrt{33})).$$

Indeed $1 \leq k \leq 6$ and $k + 2 \leq t_0 \leq 7$ are finite integer intervals; imposing odd three-adic valuation gives exactly these four entries. The first two are impossible modulo three: they would require $3j^2 + 2$ to be a square or seven times a square. For the last, $s_0 = (7 + \sqrt{33})/2 > 25/4$, while $j \geq 2$ gives $E' < 6 + 1/j^2 \leq 25/4$; at $j = 1$ the field is instead $\mathbb{Q}(\sqrt{15})$. The remaining possibility is

$$\begin{gathered} k = 3, \quad K = \mathbb{Q}(\sqrt{6}), \quad 2w^2 - 3j^2 = 2, \quad u = 3j + w\sqrt{6}, \\ DD' = 48w^2, \qquad EE' = 144w^2, \qquad p = \frac{w + j}{2w}. \end{gathered} \tag{8.5}$$

The Pell equation makes j even. Since $j = 2$ would give $w^2 = 7$, one has $j \geq 4$, $w \geq 5$, and

$$(j/w)^2 = \frac{2}{3}(1 - w^{-2}) \geq \frac{16}{25}, \qquad p \geq \frac{9}{10}.$$

Apply Lemma 3.2, with U_2, V_2 the ordinary and signed inverse-square codegree sums:

$$E(U_2 + V_2 - 1) \leq 1 + \sqrt{3(E - 1)}. \tag{8.6}$$

Cauchy's inequality on the other two ordinary codegrees gives

$$U_2 \geq p^2 + \frac{(1 - p)^2}{2} - \frac{2}{DD'} \geq \frac{163}{200} - \frac{1}{600} = \frac{61}{75}.$$

The signed reciprocal identity gives

$$V_2 = \frac{1}{4} + \frac{1}{9} + \frac{1}{36} - \frac{2}{144w^2} \geq \frac{233}{600}.$$

Hence $U_2 + V_2 - 1 \geq 121/600$. But $E \geq (E + E')/2 = EE'/12 \geq 300$, so (8.6) bounds this by less than $1/300 + 1/10 = 31/300 < 121/600$. This excludes the whole positive Pell family without restrictions on the other ordinary characters.

If $j = 0$, then $E = E' = 12$, $K = \mathbb{Q}(\sqrt{6})$. Write $s_0 = c + d\sqrt{6}$. The inequalities $0 < s_0' < 1 < s_0 < 12$ imply $1 \leq c \leq 6$ and $1 \leq d \leq 2$. Then $0 < c - d\sqrt{6} < 1$ leaves exactly $s_0 = 3 + \sqrt{6}$ of norm three or $s_0 = 5 + 2\sqrt{6}$ of norm one. Thus the remaining possibilities are

$$D = 12 + 4\sqrt{6} \quad \text{or} \quad D = 60 + 24\sqrt{6}. \tag{8.7}$$

8.1.3 Excluding the three ordinary dimensions

The ordinary Frobenius–Perron character field is $K = \mathbb{Q}(D)$: a quartic character field with quadratic global codegree is excluded by Corollary 7.5.

Write the three nonunit dimensions as $a_i + b_i\sqrt{d}$. Here $d = 3$ or six, so all a_i, b_i are integers. Put $A = \sum_i a_i^2$, $B = \sum_i b_i^2$, $C = \sum_i a_i b_i$. Every simple dimension must be nonzero. For $D = 12 + 6\sqrt{3}$, we have $A + 3B = 11$, $C = 3$. Cauchy's inequality $AB \geq 9$ forces $B = 2, A = 5$. After simultaneous coordinate signs and a permutation, take $b = (1, 1, 0)$. Then $a_1 + a_2 = 3$ and

$$(a_1 - a_2)^2 + 2a_3^2 = 1,$$

so $a_3 = b_3 = 0$, a contradiction. For $D = 12 + 4\sqrt{6}$, the equations are $A + 6B = 11$, $C = 2$, forcing $B = 1, A = 5$. Take $b = (1, 0, 0)$. Then $a_1 = 2$ and $a_2^2 + a_3^2 = 1$, again giving a zero dimension. For $D = 60 + 24\sqrt{6}$, the equations are $A + 6B = 59$, $C = 12$. The inequality $B(59 - 6B) \geq 144$ gives $9/2 \leq B \leq 16/3$, hence $B = 5, A = 29$. Take $b = (2, 1, 0)$. The relation $2a_1 + a_2 = 12$ yields

$$(5a_1 - 24)^2 + 5a_3^2 = 1,$$

again forcing a zero dimension.

The three contradictions prove Theorem 8.1.

8.2 Short signed quadratic orbits

Lemma 8.2 (The quadratic-unit gap)**.** *Let $0 < t < 1$ be a totally positive quadratic algebraic integer of norm one. Then $t \leq (3 - \sqrt{5})/2$. Unless its field is $\mathbb{Q}(\sqrt{5})$, one has $t \leq 2 - \sqrt{3}$.*

Proof. Its integral trace $t + t^{-1}$ is at least three. Trace three gives $\mathbb{Q}(\sqrt{5})$; otherwise the trace is at least four. Solve the quadratic equation and take its smaller root. □

Theorem 8.3. *Let $\mathcal{C}$ be a self-dual rank-four complex fusion category with irrational quadratic Frobenius–Perron global dimension D. Suppose its signed dimension field is $K = \mathbb{Q}(D)$. If $\mathcal{C}$ is not spherical, its signed rational algebra is $K \times K$. The two formal codegrees in the complementary character orbit are genuinely irrational, and their product is at least* 100*.*

We give all the small-row reductions needed in the proof. In particular, a signed basis involution is not treated as an ordinary invertible object, and a short signed element is never transferred to the ordinary ring.

8.2.1 Common quadratic comparisons

Assume throughout the proof that $\mathcal{C}$ is not spherical. Choose the signed dimension orbit $E \geq E'$ and put

$$N = DD', \qquad H = EE' = kN, \qquad p = D^{-1} + D'^{-1}, \qquad q = E^{-1} + E'^{-1}.$$

Spectral domination and saturation give $E < D$, and central induction gives $k \in \mathbb{Z}_{>0}$. Every ordinary codegree is at most D. The two outside the Frobenius–Perron orbit therefore have reciprocal sum at least $2/D$. Lemma 3.9 gives

$$p \leq 1 - \frac{2}{D}, \qquad kq < p + \frac{k-1}{D} \leq 1 + \frac{k-3}{D}. \tag{8.8}$$

If $q \geq 1/3$, then $k \leq 2$: for $k \geq 3$, use $D \geq 4$ in the last inequality. When $k = 2$ it gives $q < 1/2$. When $k = 1$, the totally positive unit $r = D/E > 1$ satisfies

$$r \geq \frac{3 + \sqrt{5}}{2}, \qquad \frac{q}{p} \leq \frac{2}{r + r^{-1}} \leq \frac{2}{3}.$$

In particular $q < 2/3$. If $q = 1/4$, then $E \geq 8$ and $D > 8$; (8.8) gives $k \leq 4$.

All simple Frobenius–Perron dimensions belong to K by Corollary 7.5; this justifies the ordinary dimension reconstructions below.

We repeatedly use the following two bounds, where u, v are the ordinary and signed reciprocal-square codegree sums:

$$u \geq p^2 + \frac{(1-p)^2}{2} - \frac{2}{N}, \qquad u + v - 1 \leq \frac{1 + \sqrt{3(E-1)}}{E}. \tag{8.9}$$

The first is Cauchy's inequality for the two complementary ordinary reciprocals, and does not assume that their characters are quadratic. The second is Lemma 3.2.

Lemma 8.4 (Small integral character rows)**.** *For a self-dual integral Frobenius table of rank four, a rational character cannot have codegree five, eight, or sixteen. Distinct rational characters cannot have codegree pair* $(3,4)$*,* $(3,12)$*, or* $(9,9)$*. A character orthogonal to a codegree-four character has even codegree.*

Proof. A rational character is an integral row $(1, a, b, c)$ with codegree $1 + a^2 + b^2 + c^2$. At codegree five its nonunit row is $(\pm 2, 0, 0)$ up to permutation; evaluating the corresponding square would require $4 = 1 \pm 2m$. For eight and sixteen, the nonunit squared norm is 7 modulo eight and cannot be a sum of three squares. At codegrees $3, 4, 12, 9$ the absolute nonunit rows are respectively $(1,1,0), (1,1,1), (3,1,1), (2,2,0)$. The three excluded pairs cannot have nonunit inner product -1. Finally a codegree-four row is odd in every coordinate. Orthogonality modulo two makes the sum of the other row's coordinates, hence its squared norm, even. □

8.2.2 Rational complementary signed codegrees

Lemma 8.5. *The two complementary signed codegrees cannot both be rational.*

Proof. A rational signed codegree two already gives sphericality by Theorem 8.1; hence assume all rational codegrees under consideration are at least three.

First suppose the complementary characters form a quadratic orbit, with a repeated rational codegree g. Then $q = 1 - 2/g$. The preceding comparisons leave only $g = 3, 4, 5$, and $k = 1$ for $g = 4, 5$. Write the two character rows as $(t_i \pm s_i\sqrt{d})/2$, with $t_0 = 2, s_0 = 0$ and $d > 1$ squarefree. Integrality, orthogonality and the rational codegree give

$$\sum_i t_i^2 = d \sum_i s_i^2 = 2g, \qquad \sum_i t_i s_i = 0.$$

If $d \not\equiv 1 \pmod 4$, all t_i, s_i are even; otherwise they have the same coordinatewise parity. For $g = 3$, the possibilities $d = 2, 3, 6$ would make a sum of even squares equal to six. For $g = 4$, $d = 2$ and exactly one nonunit s_i is nonzero. Orthogonality of the dimension row to the difference of the two rows then forces a zero simple dimension. Both cases are impossible.

For $g = 5$ the two rows, after signed permutation, are

$$(1, -1, \phi, \phi'), \qquad (1, -1, \phi', \phi), \qquad \phi = \frac{1+\sqrt{5}}{2}, \quad \phi' = 1 - \phi.$$

Evaluation of the symmetric table gives

$$(t, t, t, t-1, t, t-1, t, t-1, t-1, t), \qquad t \in \mathbb{Z}. \tag{8.10}$$

This calculation uses only the displayed character rows and the orthonormal based pairing, so it also applies to ordinary tables. The remaining characters are $(1, z+1, z, z)$, where $z^2 + (2-3t)z - t = 0$. With $\Delta = 9t^2 - 8t + 4$ their codegrees satisfy

$$E + E' = 3\Delta, \qquad EE' = 5\Delta.$$

The value $t = 0$ gives rational selected dimensions. Otherwise $\Delta \geq 5$, and the smaller root is

$$E' = \frac{10}{3 + \sqrt{9 - 20/\Delta}} \leq \frac{15 - 5\sqrt{5}}{2} < \frac{3 + \sqrt{5}}{2} \leq r.$$

This contradicts $D' = E'/r > 1$.

It remains to treat two rational characters, of integral codegrees $3 \leq g \leq h$. The small-row restrictions are Lemma 8.4. Since $q = 1 - 1/g - 1/h \geq 1/3$, we have $k \leq 2$. The conditions $q < 1/2$ for $k = 2$ and $q < 2/3$ for $k = 1$ now give

$$\begin{array}{l|l} k = 2 & (g, h) = (3, 3), \\ k = 1 & (3, 3),\ (3, h)\ (h \geq 6),\ (4, 4),\ (4, 6),\ (4, 10). \end{array}$$

For $(3, h)$ with $h \geq 6$, central integrality gives $H \geq h^2$. As $q = 2/3 - 1/h$, the smaller root satisfies

$$E' \leq \frac{2}{q + \sqrt{q^2 - 4/h^2}} \leq 9 - 3\sqrt{5} < \frac{3 + \sqrt{5}}{2} \leq r.$$

The second bound is attained at $h = 6$: both q and $q^2 - 4/h^2 = 4/9 - 4/(3h) - 3/h^2$ increase thereafter. For $(4, 6)$, integrality of $E/4, E/6$ implies that of $E/12$ by Bezout, so $E \geq 12$ and $E' \leq 2$. For $(4, 10)$ the same argument gives $E \geq 20$ and $E' \leq 5/3$. Again $D' = E'/r < 1$.

For $(4, 4)$ the rational rows are, up to signed permutation, $(1, -1, 1, -1)$ and $(1, -1, -1, 1)$. Their orthogonal complement has selected row $(1, 1, z, z)$. Character evaluation gives

$$X^2 = 1, \quad XY = Z, \quad XZ = Y, \quad Y^2 = Z^2 = 1 + t(Y + Z), \quad YZ = X + t(Y + Z).$$

Here $z^2 = 1 + 2tz$. Select the larger codegree and change both signs if necessary, so $t \geq 1, z > 1$; $t = 0$ would give rational dimensions. Then $E' = 2 + 2/z^2 < 5/2 < r$.

Finally, for $(3, 3)$ the rows are $(1, 0, 1, -1)$ and $(1, 1, -1, 0)$, up to signed permutation. Evaluation gives the tuple

$$(m - 1, m - 1, m, m, m, m, m, m, m + 1, m + 1), \qquad m \in \mathbb{Z},$$

and remaining characters $(1, z - 1, z, z + 1)$ with

$$z^2 = 3mz + 1, \qquad E = 3 + 3z^2, \qquad EE' = 9(9m^2 + 4). \tag{8.11}$$

The value $m = 0$ gives a zero simple dimension. Otherwise the larger codegree has $|z| > 3$ and $E' = 3 + 3/z^2 < 10/3$. The field $K = \mathbb{Q}(\sqrt{9m^2 + 4})$ is neither $\mathbb{Q}(\sqrt{2})$ nor $\mathbb{Q}(\sqrt{5})$, since either equality would require a square congruent to two modulo three. If $k = 1$, the positive unit r cannot have trace three, so $r \geq 2 + \sqrt{3} > E'$. If $k = 2$, put $s = 2D/E > 2$. This is a totally positive integer of norm two. Its trace is at least four, and equality would give $K = \mathbb{Q}(\sqrt{2})$. Thus its trace is at least five, so $s > 4 > E'$. In either case $D' < 1$, a contradiction. □

The selected character gives a quadratic quotient K of the semisimple rational algebra $T_{\mathbb{Q}}$, leaving a complementary algebra of dimension two. If the latter is $\mathbb{Q} \times \mathbb{Q}$, both codegrees are rational, contrary to Lemma 8.5. Otherwise it is a quadratic field K'. Its codegree G belongs to K' by its definition and to K by central induction. If $K' \neq K$, then $G \in K \cap K' = \mathbb{Q}$, giving a repeated rational codegree, again excluded by the lemma. Thus G is genuinely irrational and $K' = K$. This proves the $K \times K$ assertion in Theorem 8.3 without asserting field containment for arbitrary complementary character values.

8.2.3 A signed basis involution

Lemma 8.6. *Under the common quadratic hypotheses of Theorem 8.3, if the signed quotient has a nonunit basis element X with $X^2 = 1$, then the original category is spherical.*

Proof. Multiplication by X is a symmetric integral involution, hence a signed permutation matrix. It interchanges $1, X$. On the other two basis elements it either fixes them or interchanges them. Since every selected simple dimension is nonzero, in the fixed case the two signs equal the selected value of X. Changing its basis sign gives $XY = Y, XZ = Z$; then $(1, -1, 0, 0)$ is a codegree-two character, and Theorem 8.1 applies.

In the interchange case choose signs to obtain

$$X^2 = 1, \quad XY = Z, \quad XZ = Y, \qquad Y^2 = Z^2 = 1 + aY + bZ, \quad YZ = X + bY + aZ.$$

The selected dimensions can be taken as $(1, 1, z, z)$, with $z^2 = 1 + (a + b)z$. If z is rational, it is ± 1; central induction then makes the ordinary global dimension rational, and weak integrality gives sphericality. Otherwise select the larger signed codegree and change both signs so $z > 1$. Thus z is a norm-minus-one quadratic unit and $E = 2 + 2z^2$. Here $q = 1/2$, so (8.8) forces $k = 1$. Properness would give a totally positive unit $r = D/E > 1$.

Let $\eta > 1$ be the fundamental unit of K. The existence of z of norm minus one implies $N(\eta) = -1$. Write $\eta - \eta^{-1} = t \in \mathbb{Z}_{>0}$. If $t \geq 2$, then $z \geq \eta$, and $r \geq \eta^2 \geq 3 + 2\sqrt{2} > 3 > E' = 2 + 2/z^2$, which is impossible. If $t = 1$, then $K = \mathbb{Q}(\sqrt{5})$ and $\eta = \phi$. Write $z = \phi^{2j+1}$, $j \geq 0$. For $j \geq 1$, $E' \leq 2 + 2\phi^{-6} < \phi^2 \leq r$. For $j = 0$, only $r = \phi^2$ keeps $D' = E'/r > 1$; it gives $D = 10 + 4\sqrt{5}$ and $D^{-1} + D'^{-1} = 1$. The other ordinary reciprocal codegrees would then sum to zero. Thus no proper signed dimension exists. □

8.2.4 The complete short character chart

Lemma 8.7. *Consider a pair of conjugate quadratic characters of a self-dual integral Frobenius table of rank four, orthogonal to a character whose four coordinates are nonzero. Suppose their formal codegree G is a d-number and $L = GG' < 100$. Up to simultaneous signed permutation of the three nonunit coordinates and conjugation, their rows are exactly those in the following table. Here*

$$\chi = (1, (t_i + s_i\sqrt{d})/2)_{i=1}^3, \qquad \ell = G^{-1} + G'^{-1}.$$

Signs of structure constants are unrestricted; this chart applies equally to ordinary and signed tables. Entries marked $*$ have rational codegree.

d	t	s	L	ℓ	$\|U\|^2$
5	2 −1 −1	0 1 1	20	1/2	1
5	2 −1 1	0 1 1	25*	2/5	6
5	−3 −1 −1	1 1 1	25	3/5	6
2	2 −2 −2	0 2 2	32	1/2	1
3	0 −4 −2	0 2 2	36	2/3	2
5	−3 −1 1	1 1 1	45	1/3	2
5	−3 −1 −4	1 1 2	45	2/3	2
13	2 −3 −3	0 1 1	52	1/2	1
2	2 −2 2	0 2 2	64*	1/4	3
2	−2 −2 −6	2 2 4	64	3/4	3
2	−4 −2 0	2 2 2	72	1/3	2
5	2 −4 −4	0 2 2	80	1/2	1
3	2 −4 0	0 2 2	96	1/4	3
3	−4 −4 −6	2 2 4	96	3/4	3

The last column is the squared coefficient norm of the primitive integral nonunit part of the rational sector idempotent. In the four rows where it is one, that basis element squares to one.

Proof. Put $J = |s|^2$ and $C = t \cdot s$. Orthogonality of the conjugate rows gives

$$|t|^2 = dJ - 4, \quad G = \frac{dJ + C\sqrt{d}}{2}, \quad L = \frac{d}{4}(4J + |t \times s|^2), \quad \ell = \frac{dJ}{L}. \tag{8.12}$$

Character values are algebraic integers, so t_i, s_i are integers; they are both even if $d \not\equiv 1 \pmod 4$, and have the same coordinatewise parity otherwise. Orthogonality to a nonzero-coordinate character rules out support one for s. The reciprocal identity gives $0 < \ell < 1$, hence $dJ < L < 100$. The d-number condition is $L \mid (dJ)^2$.

Here is a bounded arithmetic check of the chart. Take s nonnegative and ordered, and take $t_i \geq 0$ wherever $s_i = 0$. If $d \not\equiv 1 \pmod 4$, write $s = 2b, t = 2a$. The bounds are

$$d \in \{2, 3, 6, 7, 10, 11\}, \qquad 2 \leq |b|^2 \leq \lfloor 24/d \rfloor,$$
$$|a|^2 = d|b|^2 - 1, \qquad |b|^2 + |a \times b|^2 \leq \lfloor 24/d \rfloor.$$

If $d \equiv 1 \pmod 4$, they are

$$d \in \{5, 13, 17, 21, 29, 33, 37, 41\}, \qquad 2 \leq J \leq \lfloor 99/d \rfloor, \qquad |t|^2 = dJ - 4.$$

For each permitted d, J, take the nonnegative ordered three-square representations of J. For each such s, take

$$|t_1| \leq \sqrt{dJ - 4}, \quad |t_2| \leq \sqrt{dJ - 4 - t_1^2}, \quad t_3 = \pm\sqrt{dJ - 4 - t_1^2 - t_2^2},$$

retaining integral values with the stated parity. Apply (8.12), $dJ < L < 100$ and $L \mid (dJ)^2$. The resulting L lists, before identifying the repeated values, are

$d = 2$	$32, 64, 64, 72$
$d = 3$	$36, 96, 96$
$d = 5$	$20, 25, 25, 45, 45, 80$
$d = 13$	$52,$

and the other displayed radicands give no row. Substitution gives exactly the fourteen t, s pairs above; the remaining identifications are precisely signed permutations and conjugation. This is a finite sum-of-squares check, with every range and divisibility specified independently of any fusion-ring multiplicity.

For completeness, the sector idempotent P has unit coefficient ℓ and nonunit coefficients

$$P_i = \frac{d(Jt_i - Cs_i)}{2L}.$$

Clear their denominators and divide by their common gcd to obtain U. If $U = \lambda(P - \ell)$, its two eigenvalues are $\lambda(1 - \ell)$ and $-\lambda\ell$; this computes the last column. In the norm-one rows $\ell = 1/2$ and $\lambda = 2$, so $U^2 = 1$. □

Apply the chart to the complementary signed orbit. The starred rows are excluded by Lemma 8.5, the norm-one rows by Lemma 8.6, and every row with $q = 1 - \ell \geq 2/3$ by (8.8). Only the following cases remain:

L	25	36	45	64	96
K	$\mathbb{Q}(\sqrt{5})$	$\mathbb{Q}(\sqrt{3})$	$\mathbb{Q}(\sqrt{5})$	$\mathbb{Q}(\sqrt{2})$	$\mathbb{Q}(\sqrt{3})$
q	$2/5$	$1/3$	$1/3$	$1/4$	$1/4.$

(8.13)

8.2.5 Ordinary dimensions needed at the small endpoints

Lemma 8.8. *None of the following can be the ordinary global dimension of the nonspherical category under consideration:*

$$5\phi^2, \quad 12+6\sqrt{3}, \quad 15+6\sqrt{5}, \quad 75+33\sqrt{5}, \quad 8+4\sqrt{2}, \quad 24+16\sqrt{2}, \quad 36+20\sqrt{3}.$$

Proof. The simple dimensions belong to their indicated quadratic field. For $d = 2, 3$, write the three nonunit dimensions as $a_i + b_i\sqrt{d}$. Orthogonality of this character to its conjugate gives $1 + |a|^2 = d|b|^2$. If $D = A + B\sqrt{d}$, then

$$|b|^2 = \frac{A}{2d}, \qquad |a|^2 = \frac{A}{2} - 1, \qquad a \cdot b = \frac{B}{2}.$$

For $d = 5$, use $(t_i + s_i\sqrt{5})/2$ and (8.12). The resulting small integer-vector equations are recorded below. In the middle column the three entries are $(|s|^2, |t|^2, t \cdot s)$ for $d = 5$ and $(|b|^2, |a|^2, a \cdot b)$ otherwise.

D	invariants	dimension row or consequence
$5\phi^2$	$(3, 11, 5)$	$(1, \phi, \phi, \phi^2)$
$12+6\sqrt{3}$	$(2, 5, 3)$	one dimension is zero
$15+6\sqrt{5}$	$(6, 26, 12)$	$(1, \phi, \phi^2, \phi^3)$
$75+33\sqrt{5}$	$(30, 146, 66)$	$(1, \phi^2, \phi^3, \phi^5)$
$8+4\sqrt{2}$	$(2, 3, 2)$	$(1, 1, \eta, \eta)$
$24+16\sqrt{2}$	$(6, 11, 8)$	$(1, \eta, \eta, \eta^2)$
$36+20\sqrt{3}$	$(6, 17, 10)$	$(1, \epsilon, \epsilon, 2\epsilon - 1)$.

Here $\eta = 1 + \sqrt{2}$ and $\epsilon = 2 + \sqrt{3}$. We justify uniqueness in this small table. For $d = 2, 3$, a vector of squared norm two is a signed permutation of $(1, 1, 0)$, and one of squared norm six is a signed permutation of $(1, 1, 2)$. The indicated dot products and companion squared norms then force respectively

$$a = (1, 2, 0), \quad (1, 1, \pm 1), \quad (1, 1, 3), \quad (2, 2, 3),$$

up to the corresponding signed permutations. For $d = 5$, squared norms three, six and thirty force $s = (1, 1, 1), (1, 1, 2), (1, 2, 5)$. Parity and the dot products force $t = (1, 1, 3), (1, 3, 4), (3, 4, 11)$. For the last case, the particularly short check is $|t - (11/5)s|^2 = 4/5$. Restoring the signs required by positive Frobenius–Perron dimensions gives the displayed rows.

An object of dimension one is invertible, so the fifth row gives sphericality by Theorem 5.1. The second row is impossible. In either of the first and third rows, an object X of dimension ϕ satisfies $X^2 = 1 + Y$ for some object Y of dimension ϕ. If $Y \neq X$, reciprocity puts X in XY; the remaining dimension is one but $[XY : 1] = 0$, an impossibility. Thus $Y = X$. Apply the normal form in Lemma 6.1. In the first row its remaining parameter is $x = \phi$, so $a + b\phi = 1$, giving $(a, b) = (1, 0)$; Corollary 2.7 gives sphericality. In the third row $x = \phi^2$, so $a + b\phi = \phi^2 - \phi^{-2} = \sqrt{5} = -1 + 2\phi$. This forces $a = -1$, impossible already for a fusion ring.

For the fourth row, label the dimensions of X, Y, Z by ϕ^2, ϕ^3, ϕ^5. Evaluation of $X^2 = 1 + aX + bY + cZ$ gives $a + b + 3c = 1$ and $a + 2b + 5c = 3$, hence $a + c = -1$. This contradicts nonnegativity.

For the sixth row, $d_X = d_Y = \eta$ and $d_Z = \eta^2 = 1 + 2\eta$. Evaluation of X^2, Y^2 gives $c = h = 0$, $a + b = d + g = 2$. Evaluation of $XY = bX + dY + eZ$ then gives $e = 1$, $b = d = 0$. Consequently $X^2 = 1 + 2X$, excluded by the rank-two classification [18]. For the last row, $\epsilon^2 = 4\epsilon - 1$ and $d_Z = 2\epsilon - 1$. Evaluation of the two squares forces $X^2 = Y^2 = 1 + 2Z$, so $b = d = 0$. Evaluation of $XY = eZ$ would simultaneously require $e = 1$ and $2e = 4$, a contradiction. □

8.2.6 Excluding the five signed rows

For a totally positive quadratic d-number in $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$ or $\mathbb{Q}(\sqrt{5})$, its norm has square class in, respectively,

$$\{1,2\}, \qquad \{1,6\}, \qquad \{1,5\}. \tag{8.14}$$

Indeed its conjugate quotient is a totally positive unit. The least such units are $3+2\sqrt{2}$, $2+\sqrt{3}$ and ϕ^2. Separate the even and odd powers of that quotient: the original number is a rational multiple of a unit or of one plus that unit. The norms of the latter three elements have square classes two, six and five. This proves (8.14), including the restrictions it imposes on the integral ratio $k = H/N$.

The quadratic unit descriptions used below are elementary Pell reductions. In $\mathbb{Z}[\sqrt{3}]$ the prime over three is generated by $\sqrt{3}$ and the prime over two by $1+\sqrt{3}$; in $\mathbb{Z}[\sqrt{2}]$ the prime over two is generated by $\sqrt{2}$. Divide a generator by powers of the least positive unit until it lies in the fundamental interval. The norm equation in that interval leaves precisely the generators just stated. In $\mathbb{Q}(\sqrt{5})$ the units are $\pm\phi^n$. These assertions can also be checked directly by writing an algebraic integer and its conjugate as $z, N(z)/z$.

Complementary norm 25. The row in the chart gives, by its two orthogonality equations, selected dimensions $(1, 1, z, -1-z)$. Orthogonality to the selected conjugate gives $\mathrm{Tr}(z) + 2N(z) = -3$. Put $h = N(z)+1 \in \mathbb{Z}$ and $A = 4h^2+5$. Direct substitution gives

$$E = A + 2|h|\sqrt{A}, \qquad H = 5A.$$

The selected dimension field is $\mathbb{Q}(\sqrt{5})$ precisely when $A = 5w^2$. This implies $5 \mid h$. If $h \neq 0$, then $|h| \geq 10$, since $h = 5v$ gives $w^2 = 20v^2+1$ and $v = \pm 1$ is impossible. Thus $E \geq 405$. Here $q = 2/5$ gives $k \leq 2$, and (8.14) excludes $k = 2$. Hence $r = D/E \geq \phi^2$. Since $D' = E'/r > 1$, we have $E' > \phi^2$, so

$$E < \frac{1}{2/5 - \phi^{-2}} < 56,$$

a contradiction. If $h = 0$, then $E = 5$ and $r = \phi^2$ is the only positive unit with $E'/r > 1$. It gives $D = 5\phi^2$, excluded by Lemma 8.8.

Complementary norm 36. The short row gives $(1, x, 1, -1)$ with $N(x) = -3$. With $\epsilon = 2+\sqrt{3}$, select the larger square to write

$$x = \sqrt{3}\epsilon^j, \qquad E = 3+3\epsilon^{2j}, \qquad j \geq 0.$$

As $q = 1/3$, the norm classes force $k = 1$. Then $r = D/E \geq \epsilon$. For $j \geq 1$, $E' \leq 3+3\epsilon^{-2} < \epsilon$, impossible. For $j = 0$, $E = 6$, and only $r = \epsilon$ keeps $D' > 1$. It gives $D = 12+6\sqrt{3}$, excluded by Lemma 8.8.

Complementary norm 45. The selected dimensions are $(1, 1-z, -1-z, z)$, with $N(z) = -1$. Thus

$$|z| = \phi^{2j+1}, \qquad E = 3+3\phi^{4j+2}, \qquad j \geq 0.$$

Again $q = 1/3$ and the norm classes give $k = 1$. Since $E' \leq 3+3\phi^{-2} < \phi^4$, properness forces $r = \phi^2$. It follows that

$$p = \frac{\phi^2}{3} - \frac{\sqrt{5}}{E}.$$

The cases $j = 0, 1$ give respectively $D = 15 + 6\sqrt{5}$ and $D = 75 + 33\sqrt{5}$, already excluded. For $j \geq 2$, $E > 300$, $N \geq 1125$, and $p \geq 5/6$. Here $v = 23/45 - 2/N$, so (8.9) gives

$$u + v - 1 \geq \frac{79}{360} - \frac{4}{N} > \frac{1}{5}, \qquad u + v - 1 < \frac{1}{300} + \sqrt{\frac{3}{300}} < \frac{1}{5},$$

a contradiction.

Complementary norm 64. Orthogonality gives, after signs, the selected row $(1, 1 - z, 1 + z, 1)$ with $N(z) = -2$. Put $\epsilon = 3 + 2\sqrt{2}$. Then

$$z = \sqrt{2}\epsilon^j, \quad E = 4 + 4\epsilon^{2j}, \quad H = 64w^2, \qquad j \geq 0,$$

where w is the rational coefficient of ϵ^j. Here $q = 1/4$, so $k \leq 4$; the norm classes leave $k = 1, 2, 4$. For $k = 1$, $r \geq \epsilon$. If $j \geq 1$ then $E' < \epsilon$. If $j = 0$, $E = 8$ and only $r = \epsilon$ is possible; this gives $D = 24 + 16\sqrt{2}$, already excluded.

For $k = 4$, the totally positive integer $s = 4D/E > 4$ has norm four. Its ideal is (2), so $s = 2\epsilon^a$ with $a \geq 1$. But $s \geq 2\epsilon > 8 \geq E'$, contradicting $D' = E'/s > 1$. For $k = 2$, the corresponding integer $s = 2D/E > 2$ has norm two, so $s = (2 + \sqrt{2})\epsilon^a$. Since $E' \leq 8$, only $a = 0$ is possible. Consequently

$$p = \frac{2 + \sqrt{2}}{4} - \frac{2\sqrt{2}}{E}.$$

At $j = 0$, this gives $D = 8 + 4\sqrt{2}$, already excluded. For $j \geq 1$, $E > 100$, $H \geq 576$, $p \geq 5/6$ and $N = H/2$. Since $v = 19/32 - 2/H$, (8.9) gives

$$u + v - 1 \geq \frac{29}{96} - \frac{6}{H} \geq \frac{7}{24}, \qquad u + v - 1 < \frac{1}{100} + \sqrt{\frac{3}{100}} < \frac{37}{200},$$

again impossible.

Complementary norm 96. The selected row is $(1, 1 - z, 1 + z, -1)$, so $E = 4 + 2z^2$ and $N(z) = -2$. Put $\epsilon = 2 + \sqrt{3}$. Selecting the larger square gives

$$z = (1 + \sqrt{3})\epsilon^j, \quad E = 4 + 4\epsilon^{2j+1}, \quad H = 96w^2, \qquad j \geq 0,$$

where w is the $\sqrt{3}$ coefficient of z. The norm classes and $k \leq 4$ leave $k = 1, 4$. For $k = 4$, the integer $s = 4D/E > 4$ has norm four and equals $2\epsilon^a$, $a \geq 1$. This exceeds $E' \leq 12 - 4\sqrt{3}$, a contradiction. Thus $k = 1$. The same bound for E' is less than ϵ^2, so $r = \epsilon$, and

$$p = \frac{\epsilon}{4} - \frac{2\sqrt{3}}{E}.$$

At $j = 0$, $D = 36 + 20\sqrt{3}$, already excluded. For $j \geq 1$, $E > 200$, $N \geq 864$, $p \geq 11/12$, and $v = 29/48 - 2/N$. Hence

$$u + v - 1 \geq \frac{43}{96} - \frac{4}{N} > \frac{2}{5}, \qquad u + v - 1 < \frac{1}{200} + \sqrt{\frac{3}{200}} < \frac{2}{5}.$$

This excludes the last row of (8.13) and proves Theorem 8.3.

8.3 An unconditional quadratic dimension bound

Throughout, $D \geq D'$ and $E \geq E'$ are the ordinary and signed quadratic formal-codegree pairs, $E < D$,

$$N = DD', \quad EE' = hN, \quad p = D^{-1} + D'^{-1}, \quad q = E^{-1} + E'^{-1}, \quad t = E/D.$$

Central integrality gives $h \in \mathbb{Z}_{>0}$ and makes t a totally positive algebraic integer of norm h. Write $L \geq 100$ for the signed complementary codegree product. Every formal codegree is greater than one. Let u, v be the ordinary and signed reciprocal-square sums. We use the already established second-moment inequality

$$E(u + v - 1) \leq 1 + \sqrt{3(E-1)}.$$

The exact identities and bounds are

$$N = \frac{D}{p - 1/D}, \qquad q = \frac{t}{h}(p - 1/D) + \frac{1}{tD}, \qquad E^2 \geq hN,$$

$$u \geq p^2 + \frac{(1-p)^2}{2} - \frac{2}{N}, \qquad v \geq F(q) - \frac{2}{hN} - \frac{1}{50}, \qquad F(z) = z^2 + (1-z)^2.$$

The proof branches first on h, then on the ordinary complementary codegree product M. For $h = 1$, Lemma 8.2 leaves the exceptional unit $(3 - \sqrt{5})/2$ or $t \leq 2 - \sqrt{3}$:

Branch	Closing estimates
$h \geq 2$	Lemma 8.9
$h = 1$, $M \geq 100$	Lemma 8.10
$h = 1$, $M < 100$, quadratic complement	Lemma 8.12
$h = 1$, $M < 100$, rational complement	Lemmas 8.11, 8.15

The repeated quadratic minimizations below use the exact identities

$$F(p) + F(cp) - 1 = 2(1 + c^2)\left(p - \frac{1+c}{2(1+c^2)}\right)^2 + \frac{(1-c)^2}{2(1+c^2)}, \tag{8.15}$$

$$p^2 + \frac{(1-p)^2}{2} + F(cp) - 1 = \frac{3 + 4c^2}{2}\left(p - \frac{1+2c}{3+4c^2}\right)^2 + \frac{1-2c}{3+4c^2}. \tag{8.16}$$

Thus the bounds follow by dropping a square; interval restrictions and endpoint positivity remain explicit in each application.

Lemma 8.9 (All norm ratios greater than one)**.** *If $h \geq 2$, then $D < 1000$, without any assumption on the ordinary complementary codegrees.*

Proof. Suppose $D \geq 1000$. Since $0 < t < 1$, its integral trace satisfies $t + h/t \geq h + 2$. Consequently

$$\frac{t}{h} \leq \frac{2}{h + 2 + \sqrt{h^2 + 4}} \leq 1 - \frac{1}{\sqrt{2}} < \frac{3}{10}.$$

Also $E > \sqrt{2D} > 44$. If $E < 60$, then $p > 2D/E^2 > 5/9$, and

$$q \leq \frac{E}{2D} + \frac{1}{E} \leq \frac{7}{150}.$$

For the last inequality, the function $E/2000 + 1/E$ increases on $[\sqrt{2000}, 60]$ and has terminal value $7/150$. Thus

$$u + v - 1 \geq \frac{11}{27} + F(7/150) - 1 - \frac{23}{1000} > \frac{29}{100} > \frac{1}{44} + \sqrt{\frac{3}{44}} > \frac{1 + \sqrt{3(E-1)}}{E},$$

a contradiction.

Assume now $E \ge 60$ and put $a = \min(E/(2D), 3/10)$. Since $q \le ap + 1/E$ and $ap < 1/2$, minimizing in p gives

$$C(a) - \frac{1}{50} - \frac{3}{D} \le \frac{3}{E} + \sqrt{\frac{3}{E}}, \qquad C(a) = \frac{1-2a}{3+4a^2}.$$

The function C decreases on $[0, 3/10]$. The following elementary comparisons contradict this inequality:

E range	left side exceeds	right side is less than
$[60, 100]$	$90/301 - 23/1000 > 11/40$	$11/40$
$[100, 200]$	$5/19 - 23/1000 > 21/100$	$21/100$
$[200, 400]$	$15/79 - 23/1000 > 7/50$	$7/50$
$[400, \infty)$	$5/42 - 23/1000 > 19/200$	$19/200$.

In each row the bound on a follows from $D \ge 1000$; the last uses $a \le 3/10$. The right column follows by evaluating at the left endpoint. This completes the proof. □

Lemma 8.10. *If $h = 1$ and the ordinary complementary codegree product M is at least 100, then $D < 1000$.*

Proof. Put $g(c) = (1-c)^2/(2(1+c^2))$ and use (8.15). Since $h = 1$, t is a proper totally positive unit and $t \le (3-\sqrt{5})/2 < 2/5$. Using $q = tp + (1/t - t)/D$ and $M, L \ge 100$ gives

$$u + v - 1 \ge g(t) - \frac{1}{25} - \frac{4}{D} - \frac{2}{tD} + \frac{2t}{D}. \tag{8.17}$$

Suppose $D \ge 1000$. If $t < 1/20$, use $q \le p/20 + 1/E$ to obtain

$$E(u+v-1) \ge \left(\frac{361}{802} - \frac{44}{1000}\right) E - 2 > \frac{2}{5}E - 2 > 1 + \sqrt{3(E-1)}.$$

The last inequality holds for $E \ge 63/2$, while here $E > \sqrt{1000} > 63/2$. After squaring it reduces to positivity of $4E^2 - 135E + 300$, which is positive at $63/2$ and increases thereafter.

For $1/20 \le t \le 2/5$, the lower bound (8.17) increases with D, whereas the second-moment upper bound after division by E decreases. It suffices to compare them at $D = 1000$. Put

$$B(t) = 1000t(g(t) - 1/25) - 4t + 2t^2 - 3.$$

Then

$$B''(t) = \frac{2000(3t^2-1)}{(1+t^2)^3} + 4 < 0.$$

Moreover $B(1/20) = 1388361/80200 > 33/2$ and $B(2/5) = 30297/725 > 41$. Concavity therefore gives $B(t) \ge 13 + 70t$. Finally

$$(13+70t)^2 - 3(1000t - 1) = \left(70t - \frac{59}{7}\right)^2 + \frac{4947}{49} > 0,$$

contradicting the second-moment inequality. □

Lemma 8.11 (The largest unit in the exceptional field)**.** *Suppose the two ordinary complementary characters are rational, with codegrees g, j, where $gj < 100$. If $h = 1$, the common quadratic field is $\mathbb{Q}(\sqrt{5})$, and $t = (3-\sqrt{5})/2$, then $D < 1000$, unless the ordinary ring has a nontrivial invertible basis element.*

Proof. We may suppose $g, j \geq 3$: an ordinary rational character of codegree two forces an invertible basis element. The ordinary rational algebra is $\mathbb{Q}^2 \times \mathbb{Q}(\sqrt{5})$. Its order discriminant is Ngj, so Ngj has square class five. A totally positive quadratic d-number in $\mathbb{Q}(\sqrt{5})$ has norm of square class one or five; hence gj is a square or five times a square.

Assume $3 \leq g \leq j$ and $gj < 100$. Apply the rational-row exclusions of Lemma 8.4. Checking the squares and five times squares below 100 therefore leaves

$$(3,3),\ (3,15),\ (3,27),\ (4,4),\ (4,20),\ (6,6),\ (7,7).$$

The pair $(4,4)$ forces one nonunit Frobenius–Perron dimension to be one, by adding its two orthogonal sign rows. For $(3,3)$ use (8.11) with $m \neq 0$; $m = 0$ has a zero coordinate. Its norm is $9(9m^2+4)$ and its field is $\mathbb{Q}(\sqrt{9m^2+4})$, which is not $\mathbb{Q}(\sqrt{5})$ by reduction modulo three.

For the five remaining pairs put $p = 1 - 1/g - 1/j$. Direct substitution at $t = (3-\sqrt{5})/2$ gives

$$A := p^2 + g^{-2} + j^{-2} + F(tp) - 1 - 1/50 > 1/10.$$

For a rational check, list the pairs as $(3,15)$, $(3,27)$, $(4,20)$, $(6,6)$, $(7,7)$. The values of A are

$$\frac{529}{450} - \frac{12\sqrt{5}}{25},\quad \frac{5569}{4050} - \frac{136\sqrt{5}}{243},\quad \frac{373}{200} - \frac{77\sqrt{5}}{100},\quad \frac{358}{225} - \frac{2\sqrt{5}}{3},\quad \frac{4801}{2450} - \frac{40\sqrt{5}}{49}.$$

Each exceeds $1/10$ on using $\sqrt{5} < 161/72$. Since $q = tp + (1/t - t)/D$, the derivative bound $|F'| \leq 2$ gives

$$E(u+v-1) > \frac{tD}{10} - 4t - 2 + 2t^2 = \frac{tD}{10} + 2t - 4.$$

For fixed t, comparison after division by $E = tD$ is strongest at $D = 1000$. At that endpoint it is enough to show

$$102t - 5 > \sqrt{3000t - 3}.$$

Both sides are positive, and the difference of their squares is

$$30412 - 13596\sqrt{5} > 30412 - 13596\frac{161}{72} = \frac{59}{6} > 0.$$

The second-moment inequality is contradicted. □

Lemma 8.12 (Short ordinary quadratic orbits)**.** *Suppose the ordinary complementary characters form a quadratic orbit whose codegree product M is less than* 100*. Then $D < 1000$, or the ordinary ring has a nontrivial invertible basis element.*

Proof. Lemma 8.9 handles $h \geq 2$, so let $h = 1$. A proper totally positive unit gives $0 < t \leq (3-\sqrt{5})/2 < 2/5$. Apply Lemma 8.7 (its proof uses only integrality, orthogonality, and the Frobenius pairing, so it applies to the ordinary algebra as well). The row with $M = 20$ gives a basis involution. Except for the constant-codegree-five row, the remaining rows have the following reciprocal trace and lower bound for $U := u + 2/N$:

p	1/4	1/3	2/5	1/2	2/3	3/4
$U \geq$	19/32	1/2	11/25	7/16	23/45	19/32.

Substitution shows that, for $0 \leq t \leq 2/5$,

$$U + F(tp) - 1 - 1/50 \geq f(t) := \frac{167}{400} - t + \frac{t^2}{2}.$$

For example the only potentially small difference is the $p = 2/3$ row: its difference from f is $53/720 - t/3 + 7t^2/18$, which decreases on this interval and equals $1/400$ at $t = 2/5$. All other rows follow immediately by substitution. Moreover $p \le 3/4$, whence $E^2 \ge N > D/p \ge 4D/3$.

Suppose $D \ge 1000$. Then $E > 36$, and the exact formula for q gives

$$E(u + v - 1) \ge Ef(t) - 4t - 2 + 2t^2.$$

If $t < 1/25$, the right side exceeds $3E/8 - 11/5$, which is greater than $1 + \sqrt{3(E-1)}$ for $E \ge 36$. This last assertion follows by squaring: at 36 the difference of the squares is $109/100$, and its derivative is positive thereafter.

For fixed t, comparison after division by E becomes stronger as D increases. It remains to consider $D = 1000$ and $1/25 \le t \le 2/5$. The lower bound for $E(u + v - 1) - 1$ is

$$B(t) = \frac{827}{2}t - 998t^2 + 500t^3 - 3.$$

It is concave on this interval. Its values at $1/25, 3/25, 2/5$ are respectively greater than $119/10$, greater than $331/10$, and equal to $868/25$. Thus concavity gives

$$B(t) \ge \begin{cases} 13/10 + 265t, & 1/25 \le t \le 3/25, \\ 162/5 + 29t/5, & 3/25 \le t \le 2/5. \end{cases}$$

The square of the first affine function minus $3000t - 3$ increases on its interval and is positive at $1/25$. The analogous difference for the second decreases and is positive at $2/5$, where it equals $5299/625$. This contradicts the second-moment inequality.

It remains to treat the constant-codegree-five row. Here $p = 3/5$ and $U = 11/25$. If $t \le 3/10$, put

$$f_5(t) = \frac{21}{50} - \frac{6}{5}t + \frac{18}{25}t^2.$$

The same argument gives $E(u + v - 1) \ge Ef_5(t) - 4t - 2 + 2t^2$, and $E > 40$. For $t < 1/25$, use $f_5(t) > 37/100$ to obtain $E(u + v - 1) > 37E/100 - 11/5 > 1 + \sqrt{3(E-1)}$ for $E \ge 40$. For $1/25 \le t \le 3/10$ it again suffices to put $D = 1000$. The lower bound for $E(u + v - 1) - 1$ is now

$$B_5(t) = 416t - 1198t^2 + 720t^3 - 3,$$

which is concave. On $[1/25, 3/25]$ its endpoint values exceed those of $11/5 + 238t$; the square of this affine function minus $3000t - 3$ is increasing and positive at $1/25$. On $[3/25, 3/10]$ both endpoint values of B_5 exceed 30, whereas $\sqrt{3000t - 3} \le \sqrt{897} < 30$. This is again a contradiction.

The only remaining unit is $t = (3 - \sqrt{5})/2$ in $K = \mathbb{Q}(\sqrt{5})$: every other proper totally positive quadratic unit has integral trace at least four, hence is at most $2 - \sqrt{3} < 3/10$. At this exceptional value,

$$f_5(t) = \frac{24t}{25} - \frac{3}{10} > \frac{1}{15},$$

using $\sqrt{5} < 161/72$. The second-moment inequality first gives $D < 3000$. Indeed at $D = 3000$ its violation reduces to $202t - 5 > \sqrt{9000t - 3}$; the difference of squares is $111392t - 40776 > 996$, since $t > 3/8$. Monotonicity treats larger D.

Nonnegativity in (8.10) forces the three nonunit basis signs to agree: a mixed choice requires incompatible signs for the repeated coefficients t and $t - 1$. Thus only a common sign change remains. A cyclic permutation and this common sign change, with $t = sJ$, give

$$(a, b, c, d, e, f, g, h, i, j) = (J, J - s, J - s, J - s, J, J, J, J - s, J, J),$$

where $s = 1, J \geq 1$, or $s = -1, J \geq 0$. The character rows become $(1, s\phi, s\phi', -s)$. Their orthogonal positive row is $(1, z, z, z + s)$, where $z^2 = (3J - 2s)z + sJ$. Put

$$\Delta = 9J^2 - 8sJ + 4, \quad N = 5\Delta, \quad D + D' = 3\Delta.$$

The quadratic d-number criterion gives $5 \mid \Delta$, hence $J \equiv s \pmod 5$. If $1000 \leq D < 3000$, then $D' < 2$ and $\Delta < 1001$. The possible (s, J) are

$$(1, 1), (1, 6), (-1, 4), (-1, 9).$$

Their discriminants are respectively $5, 280, 180, 805$. Only the first and third give field $\mathbb{Q}(\sqrt{5})$; their larger codegrees are $(15 + 5\sqrt{5})/2$ and $270 + 120\sqrt{5}$, both less than 1000. This final contradiction completes the proof. □

Lemma 8.13. *Suppose $h = 1$, $t \leq 2 - \sqrt{3}$, and $D \geq 1000$. Then $E < 64$ and E is rational.*

Proof. The ordinary complementary reciprocals give $u \geq p^2 + (1 - p)^2/2 - 2/N$. Minimizing this expression together with $F(tp)$, and using $q \leq tp + 1/E$, gives

$$C(t) - \frac{1}{50} - \frac{4}{D} \leq \frac{3}{E} + \sqrt{\frac{3}{E}}, \qquad C(t) = \frac{1 - 2t}{3 + 4t^2}. \tag{8.18}$$

Suppose first that $E \geq 64$. For fixed D define

$$B_D(t) = C(t) - \frac{1}{50} - \frac{4}{D} - \frac{3}{Dt} - \sqrt{\frac{3}{Dt}}.$$

This function has no interior minimum on $[64/D, 2 - \sqrt{3}]$. Indeed, with $A(t) = -t^{3/2}C'(t)$,

$$A'(t) = \frac{\sqrt{t}\,(16t^4 - 48t^3 - 144t^2 + 60t + 27)}{(4t^2 + 3)^3} > 0 \quad (0 < t \leq 2/5),$$

and $t^{3/2}B_D'(t) = -A(t) + 3/(D\sqrt{t}) + \sqrt{3}/(2\sqrt{D})$ is strictly decreasing. Both endpoints are positive. At the first,

$$C(64/D) - \frac{1}{50} - \frac{4}{D} \geq C(8/125) - \frac{3}{125} > \frac{53}{200} > \frac{3}{64} + \sqrt{\frac{3}{64}}.$$

At the second, use $2679/10000 < 2 - \sqrt{3} < 67/250$ and

$$C(67/250) > \frac{1411}{10000}, \qquad \frac{30}{2679} + \sqrt{\frac{30}{2679}} < \frac{1171}{10000}.$$

These contradict (8.18), so $E < 64$.

If E is irrational, the d-number property makes E/E' a nontrivial totally positive quadratic unit. Hence $E/E' \geq (3 + \sqrt{5})/2 > 13/5$. Since $EE' = N > D$, this gives $E > 50$ and

$$p = \frac{1}{D} + \frac{D}{E^2}\frac{E}{E'} > \frac{325}{512} > \frac{5}{8}, \qquad q < \frac{64}{1000} + \frac{1}{50} = \frac{21}{250}.$$

Consequently

$$u + v - 1 > \frac{59}{128} + F(21/250) - 1 - \frac{3}{125} > \frac{7}{25},$$

whereas the signed second-moment inequality gives

$$u + v - 1 \leq \frac{1 + \sqrt{3(E - 1)}}{E} < \frac{1}{50} + \sqrt{\frac{3}{50}} < \frac{27}{100}.$$

This final contradiction proves that E is rational. □

Lemma 8.14. *Suppose $h = 1$ and $D \geq 1000$. If E is rational, then neither of these conditions is possible:*

1. *$p \geq 7/12$;*
2. *the ordinary complementary characters are rational of codegrees $3, j$, with $j \geq 7$.*

Proof. Now $N = E^2$, $q = 2/E$, and $E > \sqrt{D} > 30$. Put $A_0 = p^2 + (1-p)^2/2$ in the first case and $A_0 = p^2 + 1/9 + 1/j^2$ in the second. Then

$$u + v - 1 \geq A_0 - \frac{4}{E} + \frac{4}{E^2} - \frac{1}{50}. \tag{8.19}$$

The last three terms increase with $E > 2$.

In the first case, split at $E = 36, 40$. If $E \leq 36$, then $p = D/E^2 + 1/D > 3/4$, so $A_0 > 19/32$; using $E > 30$ in (8.19) gives a bound greater than $2/5$, whereas the signed bound is less than $7/20$. If $36 \leq E \leq 40$, then $p > 5/8$ and $A_0 > 59/128$. The lower bound at $E = 36$ is greater than $33/100$, whereas the signed bound is less than $8/25$. Finally, if $E \geq 40$, then $A_0 \geq 41/96$ and the lower bound at $E = 40$ equals $743/2400 > 3/10$, whereas the signed bound is less than $3/10$.

In the second case, $11/21 \leq p < 2/3$ and

$$A_0 = 2p^2 - \frac{4}{3}p + \frac{5}{9},$$

which increases throughout this interval. Moreover $E > \sqrt{1500} > 38$. If $E \leq 42$, then $p > 1000/42^2 > 9/16$, hence $A_0 > 505/1152$. The lower bound (8.19) at $E = 38$ is greater than $31/100$, whereas the signed bound is less than $31/100$. If $E \geq 42$, then $A_0 \geq 179/441$, and the lower bound at $E = 42$ is

$$\frac{46}{147} - \frac{1}{50} > \frac{29}{100} > \frac{1 + \sqrt{123}}{42}.$$

Every case contradicts the signed second-moment inequality. □

Lemma 8.15. *Suppose the ordinary complementary characters are rational. If $h = 1$ and $t \leq 2 - \sqrt{3}$, then $D < 1000$, unless the ordinary ring has a nontrivial invertible basis element.*

Proof. Assume $D \geq 1000$, and write $3 \leq g \leq j$ for the complementary codegrees; a rational codegree two already forces an invertible basis element. A rational codegree five is impossible, the pair $(3, 4)$ is excluded by orthogonality, and a pair $(4, j)$ has j even. The pair $(4, 4)$ forces a nonunit Frobenius–Perron dimension to be one. Apart from $(3, 3)$ and $(3, 6)$, every remaining pair therefore either has the form $(3, j)$ with $j \geq 7$ or satisfies $p \geq 7/12$. Lemmas 8.13 and 8.14 exclude them.

For $(3, 3)$ the integral character-row calculation gives $N = 9(9m^2 + 4)$, $m \in \mathbb{Z} \setminus \{0\}$, which is not a square. For completeness, in the $(3, 6)$ case the rows can be put, by a signed permutation, in the form

$$(1, 1, 1, 0), \qquad (1, -1, 0, 2).$$

The alternative row $(1, -2, 1, 0)$ forces a zero coordinate in the selected character and is impossible. Evaluating these two rows in the symmetric multiplication table gives

$$\begin{aligned}(a, b, c, d, e) &= (8j - 8, 8 - 8j, 4j - 4, 8j - 7, 4 - 4j),\\ (f, g, h, i, j) &= (2j - 3, 7 - 8j, 4j - 4, 2 - 2j, j), \qquad j \in \mathbb{Z}.\end{aligned}$$

Its two remaining codegrees have reciprocal sum $1/2$ and product

$$N = 9(162j^2 - 312j + 152).$$

Since $N/9 \equiv 2 \pmod 3$, this also is not a square. Thus $E^2 = N$ is impossible in both remaining cases, contradicting Lemma 8.13. □

Theorem 8.16. *A nonspherical self-dual rank-four complex fusion category with irrational quadratic Frobenius–Perron global dimension satisfies*

$$D < 1000.$$

In particular every fusion coefficient is at most 31.

Proof. Corollary 7.6 identifies the signed dimension field with $\mathbb{Q}(D)$, and Theorem 8.3 gives two quadratic signed character orbits with complementary product at least 100. Thus the setup of this subsection applies. A nontrivial ordinary invertible object would give sphericality by Theorem 5.1.

Lemma 8.9 handles $h \geq 2$. If $h = 1$ and the ordinary complementary codegree product is at least 100, apply Lemma 8.10. Otherwise the complementary rational algebra is either a quadratic field or $\mathbb{Q}^2$. The quadratic-field case is Lemma 8.12. In the rational case, a proper totally positive unit either satisfies $t \leq 2 - \sqrt{3}$, or is $(3 - \sqrt{5})/2$ in $\mathbb{Q}(\sqrt{5})$. These alternatives are covered by Lemmas 8.15 and 8.11, respectively.

Finally each ordinary multiplication matrix is symmetric and has operator norm equal to the Frobenius–Perron dimension d_i of its basis element. Hence every entry is at most $d_i \leq \sqrt{D} < 32$, and is therefore at most 31. □

9 The cubic character field

The Frobenius–Perron dimension field means the field generated by all simple Frobenius–Perron dimensions. Rank-four fusion rings are commutative by Lemma 2.1. Suppose a complex fusion categorification exists and its dimension field K has degree three. The ordinary rational algebra is then $\mathbb{Q} \times K$, and character cyclotomicity makes K cyclic and totally real [6, Corollary 8.53]. Every character is consequently real. The based involution acts by complex conjugation on characters, so it is the identity: self-duality follows automatically. Its Frobenius–Perron codegree D is also cubic. Indeed, if D were rational, the reciprocal-codegree identity would give $(f - 1)D = 3f$ for the rational codegree f, hence $(f, D) = (2, 6)$ or $(4, 4)$. The corresponding order discriminants are 432 and 256. The first is not a square, whereas a cyclic cubic field has square discriminant; the second cannot be divisible by the square of a nontrivial odd cubic conductor. Both are impossible.

Central induction places D in the field generated by the signed spherical dimensions. That field has degree at most four, so it has degree three and equals K. Thus the signed rational algebra is also $\mathbb{Q} \times K$, with its dimension character in the cubic factor. The setup below introduces no additional pivotality or equality-of-fields assumption.

9.1 The cubic signed dimension field

Let $\mathcal{C}$ be a self-dual complex fusion category of rank four, and let $\widetilde{\mathcal{C}}$ be its sphericalization, with central sign ε of dimension -1; see [16, Section 5.1] and [1, Section 5.3]. Its signed rational algebra is

$$T_{\mathbb{Q}} = (K_0(\widetilde{\mathcal{C}})/(\varepsilon + 1)) \otimes \mathbb{Q}.$$

Assume that $T_{\mathbb{Q}} \cong \mathbb{Q} \times K$, where K is a totally real cyclic cubic field, and that the spherical dimension character belongs to the cubic factor. Choose dual-compatible lifts X_i^+ of the four simples, with $X_0^+ = \mathbf{1}$, and write $X_i^- = \varepsilon \otimes X_i^+$, $x_i = \dim X_i^+$, and $E = \sum_i x_i^2$. Let ρ denote the rational character of $T_{\mathbb{Q}}$ and let $g = \sum_i \rho(X_i^+)^2$ be its codegree. Both rank-four algebras are commutative by Lemma 2.1. The sphericalization algebra, their direct sum, is therefore commutative as well. The Frobenius trace identity is

$$\operatorname{Tr}_{K/\mathbb{Q}}(x_i/E) = \delta_{i0} - \frac{\rho(X_i^+)}{g}. \tag{9.1}$$

To see this identity, let τ be the coefficient of the unit. In the orthonormal signed basis, the regular trace of multiplication by a equals $\tau(aC_T)$, where C_T is the Casimir. The rational and cubic components of C_T are g and E. Evaluating this formula on $X_i^+ C_T^{-1}$ gives the displayed identity.

Let $f_1, \dots, f_4$ be the formal codegrees of $K_0(\mathcal{C})$, and let g, E_1, E_2, E_3 be those of the signed algebra. Put

$$u = \sum_{i=1}^{4} f_i^{-2}, \qquad v = g^{-2} + \sum_{i=1}^{3} E_i^{-2}.$$

As usual, these numbers are rational. The dimension-field theorem [16, Corollary 2.15] places all the displayed codegrees in K. Throughout, codegrees are counted with their character multiplicities. The two reciprocal-codegree sums are one, and $f, g \geq 2$ whenever the ordinary rational character f is introduced below.

9.2 A positive central measure

Proposition 9.1. *There is a positive, finitely supported measure ξ on the unit circle, of mass three, such that*

$$\int z^n \, d\xi(z) = \begin{cases} 1/2, & n \text{ odd}, \\ 1, & n \text{ even}, \end{cases} \qquad K \cap \mathbb{Q}(\zeta_n) = \mathbb{Q}.$$

If the ordinary algebra is $\mathbb{Q} \times K$, with rational codegree f, its known atoms satisfy

$$\xi(\{1\}) \geq \frac{3}{4}(2u + v) + \frac{f-2}{f^2} + \frac{g-2}{2g^2}, \tag{9.2}$$

$$\xi(\{-1\}) \geq \frac{3}{4}v + \frac{g-2}{2g^2} \geq \frac{1}{4}. \tag{9.3}$$

Proof. Let $F : Z(\widetilde{\mathcal{C}}) \to \widetilde{\mathcal{C}}$ be forgetful and I its right adjoint. Use the canonical normalized center matrix $S = s/(2E)$. The rational character ρ corresponds to a self-dual constituent $B \neq 1$ of $I(1)$ by [16, Example 2.9 and Theorem 2.13]. Put

$$a_A = S_{0A} = \frac{\dim A}{2E} \in K, \qquad b_A = S_{BA} = \frac{\rho(F(A))}{2g} \in \mathbb{Q}.$$

Orthogonality gives

$$\sum_A a_A^2 = \sum_A b_A^2 = 1, \qquad \sum_A a_A b_A = 0. \tag{9.4}$$

In a common cyclotomic Galois field Ω, write $G = \operatorname{Gal}(\Omega/\mathbb{Q})$. Its signed permutations $\widehat{\sigma}$ satisfy

$$\sigma(a_A) = \epsilon_\sigma(A) a_{\widehat{\sigma}A}, \quad b_A = \epsilon_\sigma(A) b_{\widehat{\sigma}A}, \quad \theta_{\widehat{\sigma}A} = \sigma^2(\theta_A). \tag{9.5}$$

The last identity has no scalar anomaly for the canonical center normalization [3, Lemma 4.2 and Section 4.1]. Define

$$\xi = \frac{1}{|G|} \sum_{\sigma \in G} \sum_{A} (3\sigma^2(a_A)^2 + \sigma^2(a_A) b_A) \operatorname{Dirac}_{\sigma^2(\theta_A)} .$$

Here σ^2 means composition. Equation (9.4) gives mass three.

For an allowed n, the restrictions of G to K and $\mathbb{Q}(\zeta_n)$ are independent. Squaring permutes $\mathrm{Gal}(K/\mathbb{Q}) \cong C_3$; no corresponding surjectivity is required on the indicator field. Apply (2.5) and (9.1). The moment of the first term of ξ is

$$\frac{1 - \nu_n(\varepsilon)}{2} - \frac{1}{2g} \operatorname{Av}_{\sigma \in G} \sigma^2 \left(\sum_i \rho(X_i^+)[\nu_n(X_i^+) - \nu_n(X_i^-)] \right) .$$

The sum in parentheses is $2g \sum_A a_A b_A \theta_A^n$ by the same center identity. The second term of ξ cancels this residual. The sign object has indicator -1 for even n and zero for odd n: its even invariant space is one-dimensional and rotation includes its inverse pivotal map -1, whereas the odd invariant space vanishes. This proves the moment formula.

Reindexing each summand by $W = \widehat{\sigma} A$ fixes the twist θ_W and gives its individual weight

$$w_W = \operatorname{Tr}_{K/\mathbb{Q}}(a_W^2) + \frac{b_W}{3} \operatorname{Tr}_{K/\mathbb{Q}}(a_W). \tag{9.6}$$

In particular the measure does not depend on enlarging Ω. Let $h_W = [F(W) : 1] - [F(W) : \varepsilon]$. The Frobenius trace identity says $\operatorname{Tr}(a_W) = h_W/2 - b_W$. Outside $I(1) \cup I(\varepsilon)$ one has $h_W = 0$, so $w_W = \operatorname{Tr}(a_W^2) - b_W^2/3 \geq 0$. For $W \subset I(1)$ put $t_W = \operatorname{Tr}(a_W) > 0$. All conjugate codegrees are positive, and $b_W = 1/2 - t_W$; hence

$$w_W \geq \frac{t_W^2}{3} + \frac{(1/2 - t_W) t_W}{3} = \frac{t_W}{6} > 0.$$

Its sign translate has the same weight. Thus positivity holds pointwise, without grouping atoms.

For the four ordinary constituents the b-values are $(f-3)/(2f), 1/(2f), 1/(2f), 1/(2f)$; the signed values replace f by g. This follows from $a_A = 1/(2f_\chi)$ and $b_A = 1/2 - \operatorname{Tr}(a_A)$. Summing their corrections in (9.6) gives $(f-2)/(2f^2)$ and $(g-2)/(2g^2)$, respectively. By Lemma 2.9, the ordinary sign translates have twist $+1$ and the signed translates have twist -1. Summing their weights gives (9.2)–(9.3); the final bound uses $v \geq g^{-2} + (1 - g^{-1})^2/3$. □

The pushforward under $z \mapsto z^2$ is the resulting squared-twist measure. If $K \cap \mathbb{Q}(\zeta_{2j}) = \mathbb{Q}$ for $1 \leq j \leq m$, its mass is three, its first m moments are one, and its atom at one has weight at least $3(u+v)/2$. The Fejér bound in Lemma 3.1 therefore gives

$$u + v \leq \frac{2(m+3)}{3(m+1)}, \qquad u + v \leq \frac{6}{7} \quad (m = 6). \tag{9.7}$$

Likewise $\xi - \frac{1}{4} \operatorname{Dirac}_{-1}$ has mass $11/4$ and all available moments equal to $3/4$. Thus

$$\frac{3}{4}(2u + v) + \frac{f-2}{f^2} + \frac{g-2}{2g^2} \leq \frac{3}{4} + \frac{2}{m+1} \tag{9.8}$$

when the first m indicator fields are disjoint from K.

9.3 Periodic moments

Proposition 9.2. *Suppose the ordinary and signed rational algebras are $\mathbb{Q} \times K$ as above. If F is the conductor of K, then*

$$\mathcal{L} := \frac{3}{2}(u+v) + \frac{f-2}{f^2} + \frac{g-2}{g^2} \le 1 + \frac{2}{F} \le \frac{9}{7}. \tag{9.9}$$

Proof. A nontrivial cyclic cubic conductor is odd and at least seven. Since $[K:\mathbb{Q}] = 3$, the field K is disjoint from $\mathbb{Q}(\zeta_n)$ whenever $F \nmid n$. Restrict ξ to the $2F$th roots and call the restriction ξ_0. For $F \nmid r$, average its moments with exponents $2Fj + r$ over $0 \le j < M$ and let $M \to \infty$. The geometric average is zero off the $2F$th roots and one on them, so ξ_0 has the same prescribed rth moment. Its mass is at most three. Subtract $\frac{3}{4}\,\mathrm{Dirac}_1 + \frac{1}{4}\,\mathrm{Dirac}_{-1}$. The residual Fourier transform on this cyclic group is supported on $\{0, F\}$, so its weights are constant on each parity class; call them a, b. Each class has roots other than ± 1, at which these are weights of the positive measure ξ_0. Hence $a, b \ge 0$ and

$$1 + F(a+b) \le 3, \qquad \xi(\{1\}) + \xi(\{-1\}) = 1 + a + b \le 1 + \frac{2}{F}.$$

Now use (9.2)–(9.3). □

9.4 The rational codegrees and cubic norms

The two rational codegrees f, g are integers at least two. The ordinary value $f = 2$ is excluded by Proposition 5.3.

Lemma 9.3. *For either algebra, let r be its rational codegree and H a codegree from its cubic character orbit. There are positive integers A, s such that*

$$\mathrm{Tr}(H) = rA, \qquad \mathrm{Norm}(H) = r^3 s^2,$$

and H/r has minimal polynomial

$$z^3 - Az^2 + (r-1)s^2 z - s^2. \tag{9.10}$$

If $r \ge 3$, the inverse-square sum of the four codegrees satisfies

$$r^{-2} + \sum_{\sigma : K \hookrightarrow \mathbb{R}} \sigma(H)^{-2} \ge r^{-2} + \tfrac{1}{2}(1 - r^{-1})^2. \tag{9.11}$$

Proof. First H has degree three. If it were rational, the reciprocal-codegree identity would give $(r-1)H = 3r$, so $(r, H) = (2, 6)$ or $(4, 4)$. The order discriminant is the product of its four formal codegrees. The first case gives 432, which is not a square, contrary to the cyclic cubic field discriminant. The second gives 256; its square root would have to be divisible by the conductor F_K, whereas a nontrivial cyclic cubic conductor is odd and cannot divide sixteen.

Write the minimal polynomial of H as

$$z^3 - az^2 + (r-1)hz - rh.$$

The reciprocal identity gives this form with positive integers a, h. The d-number criterion gives $rh \mid a^3$ and $(rh)^2 \mid ((r-1)h)^3$. Thus $r^2 \mid h$ and $r \mid a$. The order discriminant $r\,\mathrm{Norm}(H) = r^2 h$ is a square, so $h = r^2 s^2$ and $a = rA$ for positive integers A, s. This proves (9.10). For the signed algebra the same argument applies: $2H$ is an actual formal codegree of the sphericalization, and the d-number property is unchanged by nonzero rational scaling.

Put $k = r - 1$ and $n = s^2$. The three positive reciprocal roots of (9.10) have sum k, pairwise-product sum A/n, and product $1/n$. Schur's inequality yields

$$4kA \le k^3 n + 9.$$

If $k \ge 3$ and $4A > k^2 n$, the positive integer $4A - k^2 n$ is at least four when k or s is even, and at least three otherwise. This contradicts the displayed inequality except possibly at $k = 3$ with equality. Equality in Schur for positive numbers makes all three roots equal, contradicting their cubic degree. Hence $A \le k^2 n/4$. For $k = 2$, Schur gives $A \le n + 1$. At $A = n + 1$ the discriminant is $n(n-4) = s^2(s^2 - 4)$, which cannot be a nonzero square. Thus $A \le n$ in this case too. Finally the inverse-square sum is

$$\frac{1 + k^2 - 2A/n}{r^2} \ge \frac{1}{r^2} + \frac{k^2}{2r^2},$$

as claimed. □

Theorem 9.4 (Direct cubic reduction)**.** *The only possibilities are*

$$(F, f, g) = (7, 3, 3), \quad (7, 4, 4), \quad (9, 3, 3), \qquad \mathrm{Norm}(E) = \mathrm{Norm}(D).$$

In the conductor-nine case $u = v = 1/3$, the ordinary and signed cubic codegrees agree as multisets, and the category is spherical.

Proof. For a constituent A from the ordinary cubic orbit, $a_A = 1/(2D_i)$ gives $b_A = 1/(2f)$ in this measure. Since $b_A = \rho(F(A))/(2g)$ and ρ is integral-valued, $g/f \in \mathbb{Z}_{>0}$. Put $a = \mathrm{Tr}(D)$, $b = \mathrm{Tr}(E)$, $c = \mathrm{Norm}(D)$ and $k = \mathrm{Norm}(E)/c \in \mathbb{Z}_{>0}$, using central integrality. Casimir trace domination gives $b + g \le a + f$. For

$$H(r) = \frac{1}{2} - \frac{1}{r} + \frac{3}{2r^2}, \qquad J(r) = 1 - \frac{2}{r} + \frac{2}{r^2},$$

Lemma 9.3 and the reciprocal identities give

$$u \ge H(f), \quad u = J(f) - 2a/c, \quad v = J(g) - 2b/(kc), \quad v - u \ge J(g) - J(f).$$

If $g \ge 2f$, substitution in (9.9) yields

$$\mathcal{L} \ge \frac{3}{2} + \frac{1}{f} - \frac{1}{2f^2} - \frac{2}{g} + \frac{1}{g^2} \ge \frac{3}{2} - \frac{1}{4f^2} \ge \frac{53}{36} > \frac{9}{7},$$

a contradiction. Therefore $g = f$. Now

$$\mathcal{L} \ge \frac{3}{2} - \frac{1}{f} + \frac{1}{2f^2}.$$

This increases for $f > 1$ and equals $33/25 > 9/7$ at $f = 5$. Consequently $f = g \in \{3, 4\}$.

Lemma 9.3 makes k an integer square. If $k > 1$, then $k \ge 4$, and $b \le a$ implies $v \ge \frac{3}{4} J(f) + \frac{1}{4} u$. Thus

$$\mathcal{L} \ge \frac{33}{16} - \frac{17}{8f} + \frac{17}{16f^2} \ge \frac{53}{36} > \frac{9}{7},$$

again impossible. Hence $k = 1$. At $f = 3$ the lower bound for $\mathcal{L}$ is $11/9$, forcing $F \le 9$; at $f = 4$ it is $41/32$, forcing $F \le 64/9$. The only cyclic cubic conductors in these ranges are seven and nine, as follows from primitive Dirichlet characters of order three. This gives the stated list.

At $(F, f) = (9, 3)$ equality holds throughout, so $u = v = 1/3$. Equality of norms and inverse-square sums gives $\mathrm{Tr}(E) = \mathrm{Tr}(D)$, and the reciprocal identity identifies the remaining coefficient of the two cubic polynomials. Their roots therefore agree. Choose a signed character with codegree D and apply spectral saturation in Lemma 2.2 to obtain a spherical section. □

9.5 The conductor-nine ring

Theorem 9.5. *Let $\mathcal{C}$ be a self-dual rank-four complex fusion category whose Frobenius–Perron codegree field has conductor nine. Its Grothendieck ring, up to relabeling, has basis $\mathbf{1}, X, Y, Z$ and products*

$$\begin{aligned} X^2 &= \mathbf{1} + Z, & XY &= Y + Z, & XZ &= X + Y, \\ Y^2 &= \mathbf{1} + X + Y + Z, & YZ &= X + Y + Z, & Z^2 &= \mathbf{1} + Y + Z. \end{aligned} \tag{9.12}$$

This ring occurs as the even part of $\mathrm{SU}(2)_7$, and every complex fusion category with this Grothendieck ring admits a spherical structure.

Proof. Theorem 9.4 gives $f = 3$ and $u = 1/3$. Write $n = s^2$ in Lemma 9.3. The inverse-square sum is $(5 - 2A/n)/9$, so $A = n$. Consequently $D/3$ has polynomial

$$z^3 - nz^2 + 2nz - n.$$

Its discriminant is $n^2(4n - 27)$. The cubic field is cyclic, so this is a nonzero square. Thus $r^2 = 4s^2 - 27$ for a positive integer r, and

$$(2s - r)(2s + r) = 27.$$

The positive factor pairs $(1, 27)$ and $(3, 9)$ give $s = 7$ and $s = 3$. At $s = 7$ the polynomial discriminant is $7^4 \cdot 13^2$, whereas K has discriminant 9^2, which must divide the discriminant of any integral power basis. This excludes $s = 7$. Hence $s = 3$, $n = A = 9$, $\mathrm{Tr}(D) = 27$, and

$$D^3 - 27D^2 + 162D - 243 = 0. \tag{9.13}$$

The rational character has codegree three, so its nonunit values are two signs and a zero. Orthogonality to the positive dimension row rules out both signs positive; two negative signs would give two positive simple dimensions summing to one. After relabeling its row is $(1, 1, -1, 0)$. Frobenius reciprocity makes the three-index structure tensor symmetric. Applying this character to each product therefore puts the table in the form

$$\begin{aligned} X^2 &= 1 + aX + aY + cZ, & XY &= aX + (a+1)Y + cZ, \\ XZ &= cX + cY + pZ, & Y^2 &= 1 + (a+1)X + (a+1)Y + cZ, \\ YZ &= cX + cY + (p+1)Z, & Z^2 &= 1 + pX + (p+1)Y + jZ, \end{aligned} \tag{9.14}$$

where a, c, p, j are nonnegative integers. One associativity equation, obtained by comparing the X coefficients of $X(Z^2)$ and $(XZ)Z$, is

$$cj = 2c^2 + p^2 - a(2p + 1) - 1. \tag{9.15}$$

The trace of the Casimir is the sum of the squares of all structure coefficients, and also the sum of the four codegrees. Substituting (9.14) and using $3 + \mathrm{Tr}(D) = 30$ gives

$$17 + 8a(a + 1) + 6p(p + 1) + 12c^2 + j^2 = 30.$$

Thus $a = 0$, and the remaining equality has only the possibilities $(c, p, j) = (1, 0, 1)$ and $(0, 1, 1)$. Indeed, $c, p \le 1$; when both vanish $j^2 = 13$, and when both equal one the left side already exceeds thirty. Both surviving choices satisfy (9.15). In the second table $X^2 = 1$, $XY = Y$, and $XZ = Z$, so there is a second rational character $(1, -1, 0, 0)$. This contradicts the assumed rational algebra $\mathbb{Q} \times K$. The first choice is precisely (9.12).

For existence use the even part of $\mathrm{SU}(2)_7$ [19, Section 5.3.10]. Its simple labels are V_0, V_2, V_4, V_6. The truncated tensor-product rule is

$$V_i V_j = \sum_{\substack{|i-j| \le \ell \le \min(i+j,\,14-i-j) \\ \ell \equiv i+j \pmod 2}} V_\ell .$$

Taking $X = V_6$, $Y = V_4$, and $Z = V_2$ yields the displayed products. Their Frobenius–Perron dimensions lie in $\mathbb{Q}(\zeta_9 + \zeta_9^{-1})$, and (9.13) has degree three, so this is exactly the codegree field. The sphericality assertion follows from Theorem 9.4. □

9.6 An inert-prime obstruction for a rational codegree

Lemma 9.6. *Let R be a commutative self-dual based ring with $R_{\mathbb{Q}} \cong \mathbb{Q} \times K$, and suppose its rational character has formal codegree equal to a prime p. Then $K \otimes \mathbb{Q}_p$ cannot be an unramified field extension of $\mathbb{Q}_p$ of degree greater than one.*

Proof. Suppose that $L = K \otimes \mathbb{Q}_p$ is such an extension. Put $A = R \otimes \mathbb{Z}_p$ and let B be its image in L. Thus B is a finite $\mathbb{Z}_p$-order in the valuation ring $\mathcal{O}_L$. Write ρ_i for the integral values of the rational character, with $\rho_0 = 1$. Its primitive idempotent is

$$e = \frac{1}{p} \sum_i \rho_i b_i = (1, 0) \quad \text{in } \mathbb{Q}_p \times L.$$

Consequently $A \cap (\mathbb{Q}_p \times 0) = p\mathbb{Z}_p \times 0$: the coefficient of b_0 in te is t/p, and all other coefficients are integral once $t \in p\mathbb{Z}_p$. It follows that the first coordinate modulo p defines a well-defined surjective ring map $B \to \mathbb{F}_p$, and

$$A = \mathbb{Z}_p \times_{\mathbb{F}_p} B.$$

Indeed, if $(a_0, b) \in A$ and $a \equiv a_0 \pmod p$, then $(a, b) = (a_0, b) + (a - a_0, 0) \in A$. The order B is local: its integral normalization $\mathcal{O}_L$ has a unique maximal ideal, and lying-over gives the same assertion for B. Its displayed quotient is therefore its residue field.

The unit-coefficient pairing on A is perfect and associative. It gives an A-module isomorphism $A \cong \mathrm{Hom}_{\mathbb{Z}_p}(A, \mathbb{Z}_p)$. As A is finite free over the discrete valuation ring $\mathbb{Z}_p$, this identifies its canonical module with A, so A is Gorenstein.

The fiber-product theorem [15, Fact 2.6 and Corollary 2.7] states that a nontrivial fiber product of noetherian local rings over their common residue field can be Gorenstein only if both factors are regular local rings of dimension one. Thus B is a discrete valuation ring. It is then integrally closed in L, and hence $B = \mathcal{O}_L$. But the latter has residue field $\mathbb{F}_{p^{[L:\mathbb{Q}_p]}}$, contradicting the residue field $\mathbb{F}_p$ of B. □

Corollary 9.7. *There is no commutative self-dual based ring whose rational algebra is $\mathbb{Q} \times \mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ and whose rational formal codegree is three.*

Proof. The polynomial $x^3 + x^2 - 2x - 1$ of $\zeta_7 + \zeta_7^{-1}$ has discriminant 49 and has no root modulo three: its reduction is $x^3 + x^2 + x - 1$, taking nonzero values at all three elements of $\mathbb{F}_3$. Thus three is unramified and inert in this cubic field. Apply Lemma 9.6 with $p = 3$. □

9.7 Sphericality at rational codegree four

Lemma 9.8. *In the remaining conductor-seven case, put $a = \mathrm{Tr}(D)$, $b = \mathrm{Tr}(E)$, and $c = \mathrm{Norm}(D) = \mathrm{Norm}(E)$. Then*

$$a - b \in 64\mathbb{Z}_{\ge 0}.$$

Proof. Let B be the constituent of $I(\mathbf{1})$ corresponding to the rational signed character. Let C_R and C_T be the ordinary and signed Casimir elements. We show that the ordinary folded and signed forgetful coefficient vectors of B are respectively

$$w = C_R/4, \qquad r = C_T/4.$$

The signed cubic component of $F(B)$ is its dimension $E/4$. Its unit coefficient is one, and its sign-object coefficient is zero, because B belongs to $I(\mathbf{1})$ and the supports of $I(\mathbf{1})$ and $I(\varepsilon)$ are disjoint. The Frobenius trace formula then makes its signed rational component one, so its signed class is $C_T/4$. For an ordinary cubic induction constituent A_i, the rational signed row, already computed in Theorem 9.4, has $S_{B,A_i} = 1/8$. Symmetry and the modular character formula give

$$\chi_i(F(B)) = \frac{S_{A_i,B}}{S_{A_i,0}} = \frac{1/8}{1/(2D_i)} = D_i/4.$$

Its ordinary folded unit coefficient is likewise one, so the ordinary rational component is one and its folded class is $C_R/4$.

These are two coefficient vectors of the same actual object. Thus $w_i \geq |r_i|$ and $w_i \equiv r_i \pmod 2$ for every i. In the orthonormal based trace, the squared coefficient norm of a Casimir is the sum of its formal codegrees. Consequently

$$a - b = 16(\|w\|^2 - \|r\|^2) \in 64\mathbb{Z}_{\geq 0}.$$

□

Theorem 9.9. *Every category in the conductor-seven, rational-codegree-four case admits a spherical structure. Its signed and ordinary cubic codegree polynomials coincide.*

Proof. Retain a, b, c from Lemma 9.8. Here

$$u = \frac{5}{8} - \frac{2a}{c}, \qquad v = \frac{5}{8} - \frac{2b}{c}, \qquad u, v \geq \frac{11}{32}.$$

The periodic inequality is $u + v \leq 29/42$. It follows that

$$\frac{187c}{1344} \leq b \leq a \leq \frac{9c}{64}, \qquad a - b \leq \frac{c}{672}. \tag{9.16}$$

If $a > b$, the trace-gap lemma gives $c \geq 43008$.

For either triple of cubic codegrees let H be its largest element and put $x = 1/H$. Since H is at least a third of the trace, (9.16) gives $0 < x < 1/1900$. Let q_1, q_2 be the other two reciprocal codegrees. Their sum is $3/4 - x$. Also $u, v \leq 29/42 - 11/32 = 233/672$, so

$$q_1^2 + q_2^2 + x^2 \leq \frac{191}{672}, \qquad (q_1 - q_2)^2 \leq \frac{1}{168} + \frac{3}{2}x - 3x^2.$$

The last upper bound is less than $(1/12 - x)^2$, since the difference is

$$\frac{1}{1008} - \frac{5}{3}x + 4x^2 > \frac{1}{1008} - \frac{1}{1140} > 0.$$

Therefore $1/3 < q_1, q_2 < 5/12$. The two smaller codegrees of each triple lie in $(12/5, 3)$.

Align the largest signed codegree $E_{\max}$ with the ordinary Frobenius–Perron codegree D_{FP}. Central induction makes $z = E_{\max}/D_{\mathrm{FP}}$ an algebraic integer: it gives integrality against every conjugate of D, and one may then apply a field embedding sending E to $E_{\max}$. Equality of norms makes z a totally positive unit of norm one. Spectral domination gives $z \leq 1$ at the

chosen embedding. Its other two conjugates are ratios of numbers in $(12/5, 3)$ and are less than $5/4$. Hence

$$3 \le \mathrm{Tr}(z) < 1 + 2 \cdot \frac{5}{4} = \frac{7}{2}.$$

The trace is an integer, so it is three; equality in the arithmetic–geometric mean inequality forces every conjugate to be one. Thus $z = 1$, contradicting $a > b$. We conclude $a = b$. Equality of norms and the reciprocal identity then identifies the two cubic polynomials. Spectral saturation in Lemma 2.2 supplies the spherical structure. □

9.8 Excluding conductor seven

The spherical section obtained above need not have positive dimensions at its original embedding. Its construction does give an embedding which sends every simple dimension to its Frobenius–Perron dimension. Apply that embedding to the central indicator identities. Every central dimension becomes positive, because its forgetful class has nonnegative multiplicities, and every twist remains a root of unity. The first and second indicators, respectively zero and ± 1 for a nonunit self-dual simple, are unchanged. Thus the following argument uses the transformed identities and does not assume that the original category is pseudo-unitary.

Theorem 9.10. *The conductor-seven, rational-codegree-four case cannot occur.*

Proof. Since $u \ge 11/32$ and $S \le \sqrt{3(D-1)}$, Lemma 3.8, in the form (3.6), gives

$$\frac{3}{32} D \le 6 + 2\sqrt{3(D-1)}.$$

This forces $D < 1500$. Indeed, for $D \ge 1500$ its divided right side is at most $1/250 + 1/(5\sqrt{5}) < 3/32$.

Write $a = \mathrm{Tr}(D)$ and $n = s^2$ in Lemma 9.3, so that $a = 4A$ and $\mathrm{Norm}(D) = 64n$. Since the codegree triples now agree, $u = v$, and the periodic inequality gives $u \le 29/84$. Together with $u \ge 11/32$ and $u = (10 - 2A/n)/16$, this yields

$$\frac{47n}{21} \le A \le \frac{9n}{4}. \tag{9.17}$$

Moreover the nonnegative integral vector $w = C_R/4$ from Lemma 9.8 has unit coefficient one. Hence

$$a + 4 = 16\|w\|^2, \qquad A = 3 + 4\sum_{i \ne 0} w_i^2 \equiv 3 \pmod 4.$$

We sharpen the trace bound to obtain $s \le 12$. If $D < 100$, all three ordinary cubic codegrees are at most D, so $a < 300$ and $n \le 21a/188 < 34$. Otherwise put $x = 1/D \le 1/100$ and let q_1, q_2 be the reciprocals of the other two cubic codegrees. They satisfy

$$q_1 + q_2 = \frac{3}{4} - x, \qquad (q_1 - q_2)^2 \le \frac{1}{336} + \frac{3}{2}x - 3x^2,$$

using $u \le 29/84$. The last upper bound is strictly less than $(3/20 - x)^2$, because their difference is

$$\frac{9}{400} - \frac{1}{336} - \frac{9}{5}x + 4x^2 \ge \frac{9}{400} - \frac{1}{336} - \frac{9}{500} > 0.$$

Thus $q_1, q_2 > 3/10$; the other two codegrees are less than $10/3$. Consequently

$$a < D + \frac{20}{3} < \frac{4520}{3}, \qquad n \le \frac{21a}{188} < \frac{7910}{47} < 169.$$

In either case $s \le 12$.

Finally the order discriminant is the product of the formal codegrees,

$$\operatorname{disc}(R) = 4\operatorname{Norm}(D) = 256s^2.$$

The maximal order of $\mathbb{Q} \times K$ has discriminant 49. Its index over R is therefore $16s/7$, an integer. Thus $7 \mid s$, and $1 \le s \le 12$ forces $s = 7$, $n = 49$. Inequality (9.17) now reads $109\frac{2}{3} \le A \le 110\frac{1}{4}$, so $A = 110$. This contradicts $A \equiv 3 \pmod 4$. □

Theorem 9.11. *A rank-four fusion ring whose Frobenius–Perron dimension field has degree three admits a complex fusion categorification if and only if it is the ring* (9.12)*, the even part of* $\mathrm{SU}(2)_7$*. Every complex fusion categorification of this ring admits a spherical structure.*

Proof. The periodic reduction and the inert-prime obstruction leave only the conductor-nine case of Theorem 9.5 and the case excluded in Theorem 9.10. □

10 A uniform pivotal bound

Theorem 10.1. *Every rank-four complex fusion category admitting a pivotal structure has Frobenius–Perron dimension less than* 1000*. Every coefficient of its Grothendieck ring is at most* 31*.*

The proof uses only positive central indicator identities, elementary matrix estimates, and the cubic classification already proved. A spherical dimension character in the Galois orbit of the FP character can be carried to it in these identities, as in Lemma 3.8; hence every estimate below also applies in that situation. Equivalently, applying an extension of that field automorphism to skeletal tensor, duality and spherical structure constants gives a pseudo-unitary category with the same fusion rules. No algebraic field of definition is needed for this observation: an automorphism of the algebraic closure of $\mathbb{Q}$ extends to $\mathbb{C}$.

10.1 Character identities

For a self-dual category with positive FP central identities, put

$$d_i = \mathrm{FPdim}(X_i), \quad D = 1 + \sum_{i=1}^{3} d_i^2, \quad S = \sum_{i=1}^{3} d_i, \quad a_i = N_{ii}^{i},$$
$$A = \sum_{i=1}^{3} d_i a_i, \qquad w = \sum_{j=0}^{3} f_j^{-2}, \quad f_0 = D.$$

Besides orthogonality (2.2), the diagonal matrix coefficient of N_{X_i} gives

$$\sum_j f_j^{-1} = 1, \qquad \sum_j \frac{\chi_j(X_i)^2}{f_j} = 1, \qquad a_i = \sum_j \frac{\chi_j(X_i)^3}{f_j}. \tag{10.1}$$

10.2 Centered character moments

We retain the notation d_i, D, S, A, w above and put

$$t = \sqrt{D}, \quad q_j = f_j^{-1}, \quad Q = 1 - D^{-1}, \quad V = \sum_{j=1}^{3} (q_j - Q/3)^2, \quad \ell = Q/3 - \sqrt{2V/3}.$$

The unit–character orthogonality relation supplements these identities:

$$\sum_{j=1}^{3} q_j \chi_j(X_i) = -d_i/D.$$

Lemma 10.2 (Centered cubic estimate)**.** *Suppose $\ell > 0$. Set*

$$c_4 = \frac{D+1}{(D-1)^2}, \qquad C = \frac{Q-2\ell}{\sqrt{Q\ell(Q-\ell)}}, \qquad R = \sum_{i=1}^{3} d_i \left(1 - \frac{d_i^2}{D-1}\right)^{3/2}.$$

Then

$$A \le c_4 \sum_i d_i^4 - 3 + CR. \tag{10.2}$$

For every $a \ge 0$ we also have

$$A \ge 2(a+1)^2(Dw-1) - (1+a^2)(D-1) - 2aS. \tag{10.3}$$

Proof. For three positive weights of sum Q, each at least ℓ, a weighted-centered real triple of weighted squared norm ρ has third moment at most $C\rho^{3/2}$. Indeed, at an extremum with the mean and squared norm fixed, the Lagrange multiplier equation is quadratic, so there are at most two distinct coordinates. If the positive value has total weight b, the normalized third moment is $(Q-2b)/\sqrt{Qb(Q-b)}$, which decreases with b and is bounded by its value at $b = \ell$.

Apply this to $\chi_j(X_i) + d_i/(D-1)$. Its weighted squared norm is $\rho_i = 1 - d_i^2/(D-1)$, and expansion of the cube gives

$$a_i \le \frac{(D+1)d_i^3}{(D-1)^2} - \frac{3d_i}{D-1} + C\rho_i^{3/2}.$$

Multiplication by d_i and summation proves (10.2). For (10.3), use the positive central twist measure of Lemma 3.8. Its known twist-one mass for $I(X_i)$ is L_i, where $\sum_i d_i L_i = D^2 w - D$. Integrate $|1 + az + az^2 + z^3|^2$. Its constant coefficient is $2 + 2a^2$, its coefficient of $z^2 + z^{-2}$ is $2a$, and that of $z^3 + z^{-3}$ is one. The first indicator of a nonunit simple vanishes, the second has absolute value one, and the third has absolute value at most a_i, since it is the trace of an order-three operator [14, Theorem 5.1]. Thus

$$4(a+1)^2 L_i \le D((2+2a^2)d_i + 4a + 2a_i).$$

Multiply by d_i, sum over nonunit simples, and divide by $2D$. □

Lemma 10.3 (Concentration at dimension 1000)**.** *Suppose $D \ge 1000$, and write*

$$x = \max_i d_i, \qquad B = D - x^2.$$

Then $f_j < 5$ for $1 \le j \le 3$, $B < 4\sqrt{D}$, and

condition	*bound*
all V	$B < (18/5)\sqrt{D}$
$V \ge 1/1000$	$B < (10/3)\sqrt{D}$
$V \ge 1/250$	$B < (5/2)\sqrt{D}$.

(10.4)

Proof. The three-term indicator estimate gives

$$V \le \frac{4}{3D} + \frac{2S}{9D} - \frac{4}{3D^2} < \frac{7}{500}.$$

Thus $\eta = \sqrt{6V}/Q < 3/10$ and $\ell > 2331/10000$. In particular $f_j < 5$ and $|\chi_j(X_i)| < 11/6$, the latter using the unit coordinate in the character norm. The elementary spectral upper bound and the $a = 2$ case of (10.3) give

$$B \le 25 + (4 + 11/6)S < 25 + 245t/24 < 11t.$$

Here $t \ge \sqrt{1000} > 600/19$.

The function

$$F(\eta) = \frac{1+2\eta}{\sqrt{(1-\eta)(2+\eta)}}$$

is convex on $[0, 3/10]$, with $F(0) > 7/10$ and $F(3/10) < 32/25$. Its chord and $Q \ge 999/1000$ imply

$$C = Q^{-1/2}F(\eta) \le \frac{71}{100} + 5\sqrt{V}. \tag{10.5}$$

Write the other two nonunit dimensions as y, z, and put $H = B - 1$. Since $(D/(D-1))^{3/2} < 1003/1000$,

$$R \le \sqrt{2B} + \frac{1003B^{3/2}}{1000D} < \frac{9}{5}\sqrt{B} \qquad (B < 11t). \tag{10.6}$$

Completing the square gives $5R\sqrt{V} - 18DV \le 25R^2/(72D) \le 9B/(8D)$. Substitute these bounds into the centered upper estimate and the $a = 2$ lower estimate. Using

$$c_4 \le D^{-1} + \frac{301}{100D^2}, \qquad S \le t - B/(2t) + \sqrt{2B},$$

and discarding positive terms gives

$$g(B) := 2B - \frac{2007B^2}{1000t^2} - 4t - \frac{1251}{50} - \frac{20\sqrt{B}}{3} - \frac{18B^{3/2}}{25t^2} + \frac{2B}{t} \le 0. \tag{10.7}$$

On $4t \le B \le 11t$,

$$g'(B) \ge 2 - \frac{21077}{500t} - \frac{5}{3\sqrt{t}} - \frac{27\sqrt{11}}{25t^{3/2}} > 0.$$

Moreover

$$g(4t) = 4t - \frac{12283}{250} - \frac{40\sqrt{t}}{3} - \frac{144}{25\sqrt{t}} > 0.$$

The last expression increases for $t \ge 158/5$; its positivity at that endpoint follows from $28/5 < \sqrt{158/5} < 45/8$. Hence $B < 4t$.

Now choose $a = 2 - 4/t$, and put $k = 2(a+1)^2 > 16$. The lower estimate is $A \ge L_0 + kDV - 2aS$, where

$$L_0 = t^2 - \frac{91}{3} + \frac{64}{t} - \frac{40}{3t^2} - \frac{64}{t^3} + \frac{128}{3t^4}.$$

For $s = y + z$ one has $y^4 + z^4 = H^2 - (s^2 - H)^2/2$. The relevant smaller-dimension terms are bounded by

$$c_4\big(H^2 - (s^2 - H)^2/2\big) + (2a + 71/100 + 5\sqrt{V})s.$$

Their derivative in $s \le \sqrt{2H}$ is positive: the negative term is at most $2\sqrt{2}c_4H^{3/2} < 41/10$, whereas $2a + 71/100 > 22/5$. We may therefore replace y, z by equal dimensions in this upper bound. Define

$$\begin{aligned} C_{\mathrm{aux}}(B,t) = L_0 + 3 &- c_4\big((D-B)^2 + (B-1)^2/2\big) \\ &- 2a\sqrt{D-B} - (2a + 71/100)\sqrt{2(B-1)} \\ &- \frac{71}{100}\sqrt{D-B}\left(\frac{B-1}{D-1}\right)^{3/2}, \\ R_0(B,t) = \sqrt{2(B-1)} &+ \sqrt{D-B}\left(\frac{B-1}{D-1}\right)^{3/2}. \end{aligned}$$

Necessarily

$$C_{\mathrm{aux}}(B,t) + kDV - 5R_0(B,t)\sqrt{V} \le 0. \tag{10.8}$$

Since $R_0 < (8/5)\sqrt{B}$, completing the square shows that $C_{\mathrm{aux}}(B,t) - B/D \le 0$. If $V \ge v$ for $v = 1/1000$ or $1/250$, the last two terms of (10.8) increase with $V \ge v$; thus also

$$C_{\mathrm{aux}}(B,t) + kDv - 5R_0(B,t)\sqrt{v} \le 0.$$

Each of these three left sides increases with B on $[(5/2)t, 4t]$. For example its derivative is greater than

$$2 - \frac{12}{t} - \frac{503}{100\sqrt{5t-2}} - \frac{103}{100}\frac{3009}{1000t^{3/2}} - \frac{1}{t^2} > 1.$$

The exact endpoint comparisons recorded below show positivity at $B = (18/5)t, (10/3)t, (5/2)t$, respectively, proving (10.4). □

Rational endpoint comparisons. For clarity, the scalar arithmetic in the preceding proof is included. Put $b = 301/100$, and use the following rational numbers, satisfying $r_- < \sqrt{2e} < r_+$ and $e^{3/2} < h_+$:

e	v	j	r_-	r_+	h_+
18/5	0	0	2683/1000	671/250	6831/1000
10/3	1/1000	4/25	2581/1000	1291/500	3043/500
5/2	1/250	8/25	559/250	2237/1000	3953/1000

In the last two rows $j \ge 5\sqrt{v}$. At $B = et$ the relevant left side is bounded below by

$$L(t) = 18vt^2 + (A - 48v)t - C_{\mathrm{aux}}\sqrt{t} + C_0 + \frac{C_1}{\sqrt{t}} + \frac{C_2}{t} - \frac{C_3}{t^2} + \frac{C_4}{t^3} + \frac{C_5}{t^4},$$

where

$$A = 2e - 4, \qquad C_{\mathrm{aux}} = (471/100 + j)r_+,$$
$$C_0 = 11 - 91/3 - b + 2e - 3e^2/2 + 32v,$$
$$C_1 = 8r_- - (71213/100000 + 1003j/1000)h_+,$$
$$C_2 = 64 + (2b-3)e - e\,\mathbf{1}_{v=0},$$
$$C_3 = 40/3 + 1/2 + 3be^2/2, \qquad C_4 = -64 + be, \qquad C_5 = 128/3 - b/2.$$

To obtain this bound, use $c_4 \le D^{-1} + bD^{-2}$, $\sqrt{D-B} \le t - e/2$, $\sqrt{B-1} \le \sqrt{et}$, and $R_0 \le \sqrt{2et} + (1003/1000)e^{3/2}/\sqrt{t}$. All C_1, C_2, C_3, C_5 are positive, and $C_4 < 0$. At $t_0 = 158/5$, using $28/5 < \sqrt{t_0} < 45/8$, rational substitution gives

e	$L(t_0)$ exceeds	$L'(t)$ exceeds for $t \ge t_0$
18/5	1/2	19/10
10/3	5/2	5/2
5/2	12	4

For the derivative column use

$$36vt_0 + A - 48v - \frac{C_{\mathrm{aux}}}{2(28/5)} - \frac{C_1}{2t_0(28/5)} - \frac{C_2}{t_0^2} - \frac{4C_5}{t_0^5};$$

every discarded derivative term is positive, and every retained negative term decreases in magnitude for $t \ge t_0$. This establishes the three comparisons for all $D \ge 1000$ by rational arithmetic.

10.3 The determinant

Lemma 10.4 (A determinant estimate using the character mean)**.** *Let $D \ge 1000$, $t = \sqrt{D}$, and let x be the largest simple Frobenius–Perron dimension. Put $B = D - x^2$, $q_j = f_j^{-1}$ for the three non-Frobenius–Perron characters, and*

$$m = \frac{1 - D^{-1}}{3}, \qquad V = \sum_{j=1}^{3} (q_j - m)^2.$$

Suppose $V < 7/500$ and

$$B \le \begin{cases} 18t/5, & V \le 1/1000, \\ 10t/3, & 1/1000 \le V \le 1/250, \\ 5t/2, & V \ge 1/250. \end{cases}$$

Then the fusion matrix of the simple object of dimension x is singular.

Proof. Write its other eigenvalues as $\lambda_1, \lambda_2, \lambda_3$. Besides the usual square-sum identity, character orthogonality gives

$$\sum_j q_j \lambda_j = -x/D, \qquad \sum_j q_j \lambda_j^2 = B/D. \tag{10.9}$$

Set $r = B/D$, $z_j = \sqrt{q_j}\lambda_j$, and

$$l = m - \sqrt{2V/3}, \qquad p = m^3 - \frac{mV}{2} - \sqrt{\frac{V^3}{54}},$$
$$c = \frac{\sqrt{(D-B)/(DB)} + \sqrt{V}/(2\sqrt{l})}{\sqrt{m}}.$$

The present bounds imply $l > 0$ and $p > 17/500$. Since $q_j \ge l$, replacing $\sqrt{q_j}$ by $\sqrt{m}$ in the first identity in (10.9) gives

$$\left| \sum_j z_j \right| \le c\sqrt{r}, \qquad \sum_j z_j^2 = r.$$

Here we used $\sum_j (\sqrt{q_j} - \sqrt{m})^2 \le V/(4l)$. For a real triple of square sum r whose sum has absolute value at most $c\sqrt{r}$, where $0 \le c \le 1/\sqrt{3}$, the elementary cubic bound is

$$|z_1 z_2 z_3| \le r^{3/2} P(c), \qquad P(c) = \frac{c(9 - 5c^2) + \sqrt{2(3 - c^2)^3}}{54}. \tag{10.10}$$

To verify it, subtract the common mean and use $|u_1 u_2 u_3| \le (\sum u_j^2)^{3/2}/\sqrt{54}$ when $\sum u_j = 0$. Expanding the original product gives (10.10); P is increasing on $[0, 1/\sqrt{3}]$. The same zero-sum product inequality applied to $q_j - m$ yields $q_1 q_2 q_3 \ge p$. Thus

$$|\det N_X|^2 \le F(D, B, V) := \frac{(D - B)(B/D)^3 P(c)^2}{p}. \tag{10.11}$$

We record why endpoint comparisons suffice. The function F increases with V. In the first two variance intervals, for $25 \le B$ below the stated bounds we have

$$B < D/8, \quad l > 281/1000, \quad m \ge 333/1000, \quad c < 46/100 < 1/\sqrt{3}.$$

Moreover $P'/P < 5/4$, since $P' \le 1/6$ and $P \ge 1/\sqrt{54}$. Consequently

$$B\,\partial_B \log F = 3 - \frac{B}{D-B} - \frac{P'}{P}\frac{D}{D-B}\left(c - \frac{\sqrt{V}}{2\sqrt{lm}}\right)$$
$$> 3 - \frac{1}{7} - \frac{5}{4}\frac{8}{7}\frac{7}{20} = \frac{33}{14} > 0.$$

We may therefore put $B = e\sqrt{D}$ at its upper endpoint. Then

$$(D-B)(B/D)^3 = \frac{e^3(1-e/t)}{t}, \qquad \frac{D-B}{DB} = \frac{t-e}{et^2}.$$

Both decrease for $t > 2e$. Meanwhile m, l, p increase with t, so c and $P(c)$ decrease. The worst endpoint is therefore $D = 1000$. If $B < 25$, ordinary weighted arithmetic–geometric mean already gives

$$|\det N_X|^2 \le \frac{B^3}{27pD^2} < \frac{25^3}{27(17/500)1000^2} < 1.$$

At $D = 1000$, the following rational bounds certify the first two endpoints. The column headed H bounds $(D-B)(B/D)^3$.

e	V	c	$P(c)$	p	H
$18/5$	$1/1000$	$< 41/200$	$< 1/6$	$> 147/4000$	$< 327/250$
$10/3$	$1/250$	$< 27/100$	$< 7/40$	$> 181/5000$	$< 53/50$

For completeness, $\sqrt{m} > 577/1000$. In the first row, $113 < B < 2277/20$, $l > 307/1000$, and the two numerator terms of c are respectively less than $89/1000$ and $29/1000$; hence $c < 118/577 < 41/200$. In the second row, $105 < B < 211/2$, $l > 281/1000$, and these terms are less than $93/1000$ and $60/1000$; hence $c < 153/577 < 27/100$. The bounds for p and H follow directly by squaring positive rational quantities. The two bounds for P follow from

$$[9 - c(9-5c^2)]^2 - 2(3-c^2)^3 = \frac{3200085294507}{64000000000000} > 0 \quad (c = 41/200),$$
$$[189/20 - c(9-5c^2)]^2 - 2(3-c^2)^3 = \frac{513547573203}{10^{12}} > 0 \qquad (c = 27/100).$$

Equation (10.11) now gives, respectively,

$$|\det N_X|^2 < \frac{436}{441} < 1, \qquad |\det N_X|^2 < \frac{2597}{2896} < 1.$$

In the third variance interval ordinary weighted arithmetic–geometric mean suffices. We have $p > 17/500$, and $B \le (5/2)\sqrt{D}$ implies $(D-B)(B/D)^3 < 12/25$, by the same endpoint monotonicity. Therefore

$$|\det N_X|^2 \le \frac{(D-B)(B/D)^3}{27p} < \frac{80}{153} < 1.$$

The determinant is an integer, so it vanishes in all three cases. □

10.4 The positive-dimension bound

Proposition 10.5. *If a pseudo-unitary fusion category has rank four and all its simple objects are self-dual, then its Frobenius–Perron dimension is less than* 1000.

Proof. Suppose $D \geq 1000$, put $t = \sqrt{D}$, and let X be the largest simple supplied by Lemma 10.3. Since $B < 4\sqrt{D} < D/2$, the largest simple is unique. Write the basis as $1, X, Y, Z$, with dimensions $1, x, y, z$, and put

$$K = \mathbb{Q}(x, y, z), \qquad r = [K : \mathbb{Q}], \qquad B = D - x^2.$$

The rank of L_X is r. The rational semisimple commutative algebra $R \otimes \mathbb{Q}$ is a product of number fields. Its character orbits are the embeddings of those fields, and the Frobenius–Perron orbit corresponds to K. If a character χ lies outside this orbit, all algebraic conjugates of $\chi(X)$ have absolute value less than one because $|\chi_j(X)|^2 < f_j B/D < 20/\sqrt{D} < 1$. As $\chi(X)$ is an algebraic integer, its norm forces $\chi(X) = 0$. On the Frobenius–Perron orbit the values are embeddings of the nonzero element $x \in K$, and are nonzero. Hence

$$\operatorname{rank} L_X = r. \tag{10.12}$$

The principal block of L_X on $1, X$ is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & a \end{smallmatrix}\right)$, so its rank is at least two. Its determinant vanishes by Lemma 10.4. Thus $r = 2$ or 3.

Rank two is impossible. Write

$$L_X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & c \\ 0 & b & e & h \\ 0 & c & h & k \end{pmatrix}.$$

The upper left block is invertible and its Schur complement is exactly the lower right block. Rank two therefore gives $e = h = k = 0$. Consequently $XY = bX$ and $XZ = cX$, so $y = b$ and $z = c$ are positive integers. The second character in the Frobenius–Perron orbit has codegree

$$1 + (x')^2 + b^2 + c^2 < 5.$$

It follows that $b = c = 1$. Thus Y and Z are distinct self-dual invertible objects. Their product is a fourth invertible basis element, necessarily X: it cannot be $1, Y$, or Z by cancellation. This gives $D = 4$, a contradiction.

If $r = 3$, Theorem 9.11 applies. Its codegree polynomial (9.13) has trace 27, so $D < 27$, again a contradiction. This proves the proposition. □

10.5 Pivotal characters outside the FP orbit

A pivotal category whose simples are self-dual is spherical: the left dimension of a simple equals the right dimension of its dual. Its simple dimensions are real, because they form a character of commuting real symmetric fusion matrices.

Lemma 10.6. *Let $\mathcal{C}$ be a self-dual pivotal fusion category of rank four. Either it has a pseudo-unitary Galois conjugate, or* $\operatorname{FPdim}(\mathcal{C}) < 576$.

Proof. Write $D = \operatorname{FPdim}(\mathcal{C})$. If D is rational, it is an integer and the category is pseudo-unitary by [6, Proposition 8.24]. For the rest of the proof, assume that no Galois conjugate is pseudo-unitary. Then D is irrational, and the spherical dimension character is outside the Frobenius–Perron orbit.

The field generated by spherical dimensions contains every formal codegree, including D, by [16, Corollary 2.15]. Let K be the Frobenius–Perron field. Its degree is between two and four. Degree four leaves no other character orbit, and in degree three the other orbit is rational, so neither case is possible. If $[K : \mathbb{Q}] = 2$, the other factor of $R \otimes \mathbb{Q}$ must be a quadratic field containing $\mathbb{Q}(D) = K$. Thus

$$R \otimes \mathbb{Q} \cong K \times K \tag{10.13}$$

for a real quadratic field K, and the spherical dimension character belongs to the other factor.

Denote the four formal codegrees by D, D', H, H', with primes indicating conjugation in K. We allow $H = H'$ and retain both character multiplicities. After Galois conjugation, arrange that the spherical global dimension is $H \geq H'$. Each codegree is positive and greater than one. Also $H \leq D$, because each character value has absolute value at most the corresponding Frobenius–Perron dimension. Equality would make the category pseudo-unitary, so $H < D$ in the exceptional case.

By the central-induction dimension formula, H/D is a simple central dimension and hence an algebraic integer. Put

$$n = DD', \qquad k = \frac{HH'}{DD'} \in \mathbb{Z}_{\geq 1}, \qquad p = \frac{1}{D} + \frac{1}{D'}, \qquad w = \frac{1}{D^2} + \frac{1}{(D')^2} + \frac{1}{H^2} + \frac{1}{(H')^2}.$$

Here k is the field norm of H/D, including when that element is rational. Since $D' > 1$ and $H \geq H'$, we have

$$n > D, \qquad H \geq \sqrt{kn} > \sqrt{kD}. \tag{10.14}$$

The reciprocal-codegree identity gives

$$w - \frac{1}{2} = 2\left(p - \frac{1}{2}\right)^2 - \frac{2(1 + 1/k)}{n}. \tag{10.15}$$

We also need a spherical first-moment bound:

$$w \leq \frac{1}{2} + \frac{1}{2H}. \tag{10.16}$$

Indeed, the dimensions of central simples are real, as they are dimensions of their restrictions to $\mathcal{C}$. Give a central simple A the nonnegative weight $\dim(A)^2$ at its twist θ_A. The total weight is H^2 and its first moment is H. To see both identities directly, multiply the dimension formula for $I(X)$ and the $k = 1$ case of the central indicator formula by $\dim(X)$ and sum over simple X; adjunction identifies the coefficient of $\dim(A)$ with $\dim F(A) = \dim(A)$, while $\nu_1(X) = \delta_{X,1}$. The four constituents of $I(1)$ contribute weight $h = H^2 w$, all at twist one. Since the remaining twists have absolute value one,

$$h - H \leq |H - h| \leq H^2 - h.$$

This proves (10.16) without any positivity assumption on unsquared spherical dimensions.

Set $r = D/H > 1$. By (10.14), $r < \sqrt{D/k} \leq \sqrt{D}$. Moreover, $kr > 2$. This is immediate when $k \geq 2$. When $k = 1$, H/D is a totally positive algebraic-integer unit of norm one. Thus r is also such a unit, with conjugate r^{-1}; the integer $r + r^{-1} > 2$ is at least three, and $r \geq (3 + \sqrt{5})/2 > 2$.

Since $H' = krD'$, the reciprocal-codegree identity gives

$$1 = \frac{1 + r}{D} + \frac{1 + 1/(kr)}{D'}, \qquad p = \frac{kr}{kr + 1} - \frac{kr^2 - 1}{(kr + 1)D} > \frac{2}{3} - \frac{r}{D} > \frac{2}{3} - \frac{1}{\sqrt{D}}.$$

If $D \geq 576$, then $p > 5/8$ and $H > 24$. Equation (10.15) consequently gives

$$w - \frac{1}{2} > \frac{1}{32} - \frac{4}{D} \geq \frac{7}{288} > \frac{1}{48} > \frac{1}{2H},$$

contradicting (10.16). Thus $D < 576$, as required. □

10.6 Non-self-dual categories and conclusion

By Theorem 4.1, a rank-four category with a non-self-dual simple admits a spherical structure. There are exactly two real-valued characters: the complex-linear duality involution has trace two in the distinguished basis, hence fixes exactly two primitive idempotents. These are precisely the real characters. If the FP character is irrational, its Galois orbit consists of these two characters, so the real spherical dimension character belongs to that orbit. If the FP character is rational, weak integrality gives pseudo-unitarity by [6, Proposition 8.24]. Thus a pseudo-unitary Galois conjugate exists in either case. Larson's candidate theorem [11, Theorem 1.1] bounds its simple FP dimensions by the following possibilities, besides the pointed case:

$$1,1,1,\frac{m+\sqrt{m^2+12}}{2}\quad (m=0,2,3,6),\qquad 1,1,c+\sqrt{c^2+1},c+\sqrt{c^2+1}\quad (c=1,2).$$

Their global dimensions are at most $24+12\sqrt{3}<45$ and $20+8\sqrt{5}<38$, respectively. Therefore $D<45$.

For a self-dual pivotal category, Lemma 10.6 gives $D<576$ unless its spherical dimension character belongs to the FP orbit. In the latter case Proposition 10.5 gives $D<1000$. This proves the dimension assertion in Theorem 10.1. Lemma 2.1 gives $N_{XY}^{Z}<\sqrt{D}<32$, and integrality gives the coefficient bound.

11 Uniform bounds for quartic dimension fields

The following bound applies to arbitrary integral orders in the dimension field. Its cutoff 950 and resulting multiplicity 30 are auxiliary refinements; the final finite proof uses the common cutoff 1000 and multiplicity 31.

Theorem 11.1. *Let $\mathcal{C}$ be a self-dual rank-four complex fusion category, and suppose that its Frobenius–Perron dimension D generates a quartic field K. If $D\geq 950$, then $\mathcal{C}$ is spherical.*

We prove the theorem in two field cases. First suppose $K\neq\mathbb{Q}(\sqrt{2},\sqrt{5})$, and assume for a contradiction that $\mathcal{C}$ is nonspherical with $D\geq 950$. We use the sphericalization notation R,T,D,E and the central-column identities. By Corollary 7.6, both rational algebras are the same totally real abelian quartic field. Choose the largest conjugate E of the signed codegree and put

$$\alpha=E/D,\qquad p_\sigma=1/\sigma(D),\qquad q_\sigma=1/\sigma(E),\qquad d=p_0=1/D.$$

Then $\sum p_\sigma=\sum q_\sigma=1$, and $d\leq q_0$ are their respective least entries. Central integrality and the order-discriminant identity show that α is a totally positive integral d-number whose norm h is a positive integer square. In particular, $\prod q_\sigma\leq\prod p_\sigma$. Moreover α_0 is the least conjugate: the signed central constituent labelled by σ has ordinary folded dimension $D/\sigma(D)$ and signed dimension $E/\sigma(E)$, so spectral domination gives

$$E/\sigma(E)\leq D/\sigma(D),\qquad E/D\leq\sigma(E)/\sigma(D).$$

Write $u=\sum p_\sigma^2$ and $v=\sum q_\sigma^2$. Corollary 3.6 gives $u+v\leq 4/5$ if $\sqrt{2}\notin K$, and $2u+v\leq 9/8$ if $\sqrt{5}\notin K$. In the latter case

$$u\geq d^2+\frac{(1-d)^2}{3}>\frac{13}{40}\qquad (D\geq 100)$$

also gives $u+v<4/5$. A quartic field containing both square roots is their compositum. Thus both branches reduce to

$$u+v\leq\frac{4}{5}. \tag{11.1}$$

Schur's inequality and signed trace domination give

$$v \geq \frac{1}{2} - \frac{9}{2h} e_3(p), \qquad e_3(p) \leq \frac{(1-d)^2(1+8d)}{27}. \tag{11.2}$$

If $h \geq 9$, these imply

$$u + v \geq \frac{1}{2} + \frac{(1-d)^2(17-8d)}{54} > \frac{4}{5} \qquad (D \geq 100),$$

a contradiction. Therefore

$$h \in \{1, 4\}. \tag{11.3}$$

11.1 Product and ratio estimates

Lemma 11.2. *Let positive quadruples p, q each have coordinate sum one. Suppose $p_0 = d \leq 1/950$, $q_0 = \min q_i$, $u + v \leq 4/5$, and $\prod q_i \leq \prod p_i$, where $u = \sum p_i^2$, $v = \sum q_i^2$. Then $q_0 < 1/168$.*

Proof. Put

$$d_* = \frac{1}{950}, \quad r = \frac{1}{168}, \quad a = \frac{7}{250}, \quad S = 1 - r, \quad c = \frac{4}{5} - d_*^2 - \frac{(1-d_*)^2}{3}.$$

By monotonicity for $0 < d \leq d_*$,

$$v \leq c, \qquad \prod q_i \leq \prod p_i \leq \frac{d(1-d)^3}{27} \leq \frac{d_*(1-d_*)^3}{27}.$$

Suppose all $q_i \geq r$, and minimize their product subject to this lower bound, their sum being one, and their square sum being at most c. A minimizer has a coordinate equal to r. Indeed, an interior minimizer with inactive square-sum constraint would have all coordinates equal, which is a local maximum. With the constraint active, the multiplier equations for the logarithm of the product give at most two values $a_0 < b_0$. The constrained Hessian is diagonal with entries $1/(a_0 b_0) - 1/q_i^2$. If the smaller value occurs at least twice, the tangent vector increasing one such coordinate and decreasing another has negative second variation. If it occurs only once, the square sum is $a_0^2 + (1-a_0)^2/3 < 1/3 < c$, again impossible. The two constraint gradients are independent at every such two-value point.

At a boundary coordinate r, the other three have sum S and square sum at most $c - r^2 < S^2/2$. Their unconstrained positive product minimum has one small and two equal large coordinates,

$$\left(t, \frac{S-t}{2}, \frac{S-t}{2}\right), \qquad t^2 + \frac{(S-t)^2}{2} = c - r^2, \quad 0 < t < S/3.$$

This is the usual two-value multiplier calculation for three variables; the one-small-coordinate branch has the smaller product. In fact $t > a > r$, because

$$a^2 + \frac{(S-a)^2}{2} - (c - r^2) = \frac{2706003}{35378000000} > 0$$

and the quadratic decreases up to $S/3$. Since $x(S-x)^2$ increases on that interval, every admissible quadruple consequently has product greater than $ra(S-a)^2/4$. But

$$\frac{ra(S-a)^2}{4} - \frac{d_*(1-d_*)^3}{27} = \frac{30295201169}{13793174640000000000} > 0,$$

contradicting the preceding upper bound. □

Lemma 11.3. *Let P, Q be positive triples whose coordinates sum to one, and put $U = \sum P_i^2$, $V = \sum Q_i^2$. Then*

$$U + V \le \frac{201}{250} \implies \sum_i \frac{P_i}{Q_i} < \frac{51}{4}, \qquad U + V \le \frac{8063}{10000} \implies \sum_i \frac{P_i}{Q_i} < \frac{68}{5}.$$

Proof. Put $r = \sum Q_i^{-1}$. Cauchy–Schwarz in the plane of coordinate sum zero gives

$$\sum_i \frac{P_i}{Q_i} \le \frac{r}{3} + \sqrt{\left(B - V - \frac{1}{3}\right)\left(\frac{2r^2}{3} - \frac{4r}{1-V}\right)}.$$

Here B denotes the relevant bound on $U+V$. For fixed $V < 1/2$ the right-hand side increases with $r \ge 9$. The product-minimizing positive triple with prescribed sum and square sum has one small and two equal large coordinates. Consequently it suffices to take

$$Q = \left(t, \frac{1-t}{2}, \frac{1-t}{2}\right), \qquad P = \left(a, \frac{1-a}{2}, \frac{1-a}{2}\right),$$

where

$$a = \frac{1}{3} + \sqrt{\frac{2(B - 2/3)}{3} - (t - 1/3)^2}, \qquad \frac{1 - \sqrt{6B-4}}{3} \le t \le \frac{1}{3}.$$

For the proposed ratio bound R, define

$$N(t) = -3Rt^2 + (3R-3)t - 1, \qquad F(t) = N(t)^2 + (1-3t)^4 - (6B-4)(1-3t)^2.$$

The required inequality follows from $N(t) > 0$ and $F(t) > 0$ by multiplying positive denominators and squaring positive sides; the endpoint $t = 1/3$ gives ratio 3 directly. Indeed,

$$\frac{a}{t} + \frac{2(1-a)}{1-t} < R \quad \Longleftrightarrow \quad a - \frac{1}{3} < \frac{N(t)}{3(1-3t)} \qquad (t < 1/3).$$

For $(B, R) = (201/250, 51/4)$, one has $t > 3/100$, $N(3/100) = 923/40000 > 0$, and N is increasing up to $1/3$. The polynomial $f = (2000/3)F$ is

$$f(t) = 1029375t^4 - 1869750t^3 + 910431t^2 - 51704t + 784.$$

Writing $x = t - 1/32$ gives the exact identity

$$\begin{aligned} f(t) &= H(t)x^2 + Lx + C, \\ H(t) &= 1029375t^2 - \frac{28886625}{16}t + \frac{815705469}{1024}, \\ L &= -\frac{1263121}{8192}, \qquad C = \frac{1325631}{1048576}. \end{aligned}$$

The function H decreases on $[0, 1/3]$, and $H(1/3) = 316577469/1024 > 300000$, $|L| < 155$, $C > 1$. Therefore

$$f(t) > 300000x^2 - 155|x| + 1 > 0,$$

since $155^2 < 1200000$.

For $(B, R) = (8063/10000, 68/5)$, one has $t > 1/36$, $N(1/36) = 1/54 > 0$, and N is again increasing up to $1/3$. Now $f = (5000/3)F$ equals

$$f(t) = 2909400t^4 - 5320800t^3 + 2594833t^2 - 137622t + 1937.$$

Writing $x = t - 29/1000$ gives

$$f(t) = H(t)x^2 + Lx + C,$$
$$H(t) = 2909400t^2 - \frac{25760274}{5}t + \frac{11467835081}{5000},$$
$$L = -\frac{327793717}{1250000}, \qquad C = \frac{2526625707}{5000000000}.$$

Again H decreases on $[0, 1/3]$, and $H(1/3) = 13492231243/15000 > 800000$, $|L| < 263$, $C > 1/2$. Thus

$$f(t) > 800000x^2 - 263|x| + \frac{1}{2} > 0,$$

because $263^2 < 1600000$. □

Lemma 11.4. *Let positive quadruples p, q each have coordinate sum one, write $d = p_0$, $q = q_0$, $u = \sum p_i^2$, $v = \sum q_i^2$, and suppose*

$$0 < d \le \frac{1}{950}, \qquad d \le q < \frac{1}{168}, \qquad u + v \le \frac{4}{5}.$$

Then $\sum_{i=0}^{3} p_i/q_i < 14$.

Proof. Normalize the last three coordinates by $P_i = p_i/(1-d)$, $Q_i = q_i/(1-q)$, and write $U = \sum P_i^2$, $V = \sum Q_i^2$. Since $U \ge 1/3$ and $q \ge d$,

$$U + V \le B(d, q) := \frac{1}{3} + \frac{4/5 - d^2 - (1-d)^2/3 - q^2}{(1-q)^2}.$$

The function B increases in both variables on the indicated domain. If $q \le 1/300$, then

$$B(d, q) \le B(1/950, 1/300) = \frac{77826934}{96821283} < \frac{201}{250}.$$

Lemma 11.3 gives

$$\sum_{i=0}^{3} \frac{p_i}{q_i} < 1 + \frac{300}{299}\frac{51}{4} = 14 - \frac{62}{299} < 14.$$

If $q > 1/300$, then

$$B(d, q) < B(1/950, 1/168) = \frac{15220412926}{18877366875} < \frac{8063}{10000},$$

and the other ratio bound gives

$$\sum_{i=0}^{3} \frac{p_i}{q_i} < \frac{300}{950} + \frac{168}{167}\frac{68}{5} = 14 - \frac{44}{15865} < 14.$$

□

11.2 A finite arithmetic lemma

Lemma 11.5. *Let α be a totally positive integral d-number of degree four, with abelian Galois field, norm $h \in \{1, 4\}$ and trace at most 13. Its minimal polynomial $x^4 - tx^3 + sx^2 - rx + h$*

is one of the following. The column z gives an integer root of the cubic resolvent.

t	s	r	h	z	G	t	s	r	h	z	G
7	14	8	1	3	C_4	12	24	12	1	2	V_4
8	14	7	1	3	C_4	12	28	12	1	2	V_4
8	19	12	1	3	C_4	12	30	24	4	4	V_4
8	20	16	1	4	V_4	12	32	24	4	4	V_4
9	19	9	1	2	C_4	12	34	20	1	6	C_4
9	26	24	1	7	C_4	12	34	23	1	3	C_4
10	26	20	4	4	V_4	12	34	28	1	18	C_4
11	21	11	1	2	C_4	12	45	54	1	9	V_4
11	31	11	1	2	C_4	13	29	17	1	18	C_4
12	19	8	1	3	C_4	13	40	13	1	2	C_4
12	23	12	1	2	V_4	13	49	57	1	18	C_4

If $g(x) = (x+2)^2/(x^2+1)$ and α_0 is the least conjugate, then

$$\sum_{i=1}^{3} g(\alpha_i) < \frac{423}{40}. \tag{11.4}$$

The same inequality holds for a nonrational quadratic d-number whose four-dimensional norm is 1 or 4 and whose four-dimensional trace is at most 13, counting each conjugate twice.

Proof. Here are complete integer tests for the finite table. An abelian quartic group fixes a partition of the four positive roots into pairs. Write their sums as a, a' and products as b, b', and put $z = b + b'$. Then z is an integer and

$$A = (a-a')^2 = t^2 - 4s + 4z, \quad B = (b-b')^2 = z^2 - 4h, \quad AB = (tz - 2r)^2.$$

Positivity and symmetric means give

$$4 \le t \le 13, \quad 2\sqrt{h} \le z \le t^2/4 - 2\sqrt{h}, \quad z < s \le 3t^2/8, \quad 1 \le r \le t^3/16.$$

For $h = 4$ the d-number criterion further requires $2 \mid t, s$ and $4 \mid r$. For each h, t, z, s in these bounds, retain $A \ge 0$ with AB a square and set $r = (tz \pm \sqrt{AB})/2$. Put

$$S = t^2 - 2s - 2z, \qquad J = s^2 + 2sz - 3z^2 - 4tr + 16h.$$

The two pair discriminants have sum S and product J; therefore all four roots are positive precisely when $S, J > 0$, after reducible polynomials are discarded. Reducibility is tested by positive divisors of h for a linear factor, or positive integers b, d with $bd = h$ and $a + c = t$, $ac + b + d = s$, $ad + bc = r$ for quadratic factors. These tests are exhaustive by Gauss's lemma.

Set $d_* = A$ if $A > 0$, and $d_* = B$ otherwise. For an irreducible polynomial this is positive and nonsquare. Its pair field is $\mathbb{Q}(\sqrt{d_*})$. Normality is equivalent to J being a square in this field, hence to J or Jd_* being an integer square. The first case gives V_4, the second C_4. Applying these square and divisibility tests in the displayed bounded ranges gives exactly the table. This is an arithmetic case analysis of bounded integers, independent of fusion-ring data.

For the scalar assertion put $A_0 = 1 - s + h$, $B_0 = t - r$, $C_0 = 3t - r$, and $F_0 = 2s - 4$. Logarithmic differentiation of the polynomial at i gives

$$\operatorname{Tr}(g(\alpha)) = 4 - \frac{(4C_0 + 3F_0)A_0 + (4F_0 - 3C_0)B_0}{A_0^2 + B_0^2}.$$

In every row except four this is at most $583/40$, and $g(\alpha_0) > 4$. The exceptions have the following rational data:

(t, s, r, h)	$\operatorname{Tr} g$	a	$f(a)$	$f(3/10)$
$(7, 14, 8, 1)$	$425/29$	$1/10$	$3331/10000$	$-3209/10000$
$(8, 14, 7, 1)$	$443/29$	$1/4$	$1/256$	$-479/10000$
$(11, 21, 11, 1)$	$278/19$	$1/10$	$991/10000$	$-6989/10000$
$(12, 19, 8, 1)$	$930/61$	$1/4$	$1/256$	$-59/10000$.

In the first and third rows $f' < 0$ on $[0, 1/10]$ follows from $4x^3 + 2sx - r < 0$. In the other two, $f'' > 0$ on $[0, 1/4]$ and $f'(1/4) = -23/16, -11/16$. Thus $a < \alpha_0 < 3/10$. As g increases up to $1/2$, subtracting $441/101$ or $81/17$ from the listed trace proves (11.4). For the other rows the least root is below one: this is immediate at norm one, while the norm-four polynomials are negative at $1/2$.

In degree two the quadratic norm is 1 or 2 and the trace at most six. The large root b exceeds $13/5$, and $g(a) \le 5$. Hence

$$g(a) + 2g(b) < 5 + 2g(13/5) = \frac{1014}{97} < \frac{423}{40}.$$

□

11.3 Proof of the uniform bound

Proof of Theorem 11.1 outside the remaining field. The reductions above give $u + v \le 4/5$ and $h \in \{1, 4\}$. Lemma 11.2 gives $q_0 < 1/168$; hence Lemma 11.4 gives $\operatorname{Tr}(\alpha) < 14$. Its integral trace is at most thirteen. A rational α is a positive integer with $\alpha_0 \le 1$, hence equals one. Suppose $\alpha \ne 1$. For $i = 1, 2, 3$, completing a square and using $q_i = p_i/\alpha_i$ gives

$$p_i^2 + q_i^2 \ge \frac{2}{3} p_i + \frac{4}{3} q_i - \frac{1}{9} \frac{(\alpha_i + 2)^2}{\alpha_i^2 + 1}.$$

Lemma 11.5, including its quadratic case, yields

$$u + v > \frac{33}{40} - \frac{2}{3} d - \frac{4}{3} q_0 > \frac{4}{5} + \frac{3917}{239400} > \frac{4}{5},$$

contradicting (11.1). Consequently $E = D$; spectral saturation gives a spherical section of the sphericalization, proving the theorem. □

11.4 The remaining biquadratic field

It remains to prove Theorem 11.1 for $K = \mathbb{Q}(\sqrt{2}, \sqrt{5})$, under the same assumption $D \ge 950$.

We retain $p_i = 1/D_i$, $q_i = 1/E_i$, $d = p_0 = 1/D$, and $\alpha = E/D$, choosing E to be its largest signed codegree. Thus $d \le q_0$, $q_0 = \min q_i$, and $\alpha_0 = d/q_0 \le 1$ is the least conjugate of α. In this field the averaged indicator inequality is

$$5u + 4v \le \frac{15}{4}, \qquad u = \sum_{i=0}^{3} p_i^2, \quad v = \sum_{i=0}^{3} q_i^2. \tag{11.5}$$

The norm $N(\alpha)$ is a positive integer square. In particular,

$$q_0^3 (1 - 3q_0) \le \prod_i q_i \le \prod_i p_i \le \frac{d(1-d)^3}{27}. \tag{11.6}$$

Lemma 11.6. *If $D \geq 950$, then $q_0 < 1/85$ and $q_i > 1/16$ for $i = 1, 2, 3$. Moreover*

$$E_j + E_k \geq D_j + D_k \qquad (1 \leq j < k \leq 3).$$

Proof. Every conjugate of α belongs to the signed order T:

$$\sigma(\alpha) = (E/\sigma E)^{-1}(E/\sigma D).$$

Both factors on the right belong to T by the central-column identities, and the first denominator is a unit, since its norm is one. For a nonidentity involution σ_i of K, $\alpha\sigma_i(\alpha)$ is therefore totally positive and belongs to T. Its coefficient at the identity is a positive integer, so

$$I_i = dp_i(1/q_0 + 1/q_i) + p_jp_k(1/q_j + 1/q_k) \geq 1, \qquad \{i, j, k\} = \{1, 2, 3\}.$$

Writing $U = dp_i$ and $V = p_jp_k$, one obtains

$$I_i - 1 = U(\operatorname{Tr} E - \operatorname{Tr} D) + (V - U)((E_j + E_k) - (D_j + D_k)). \tag{11.7}$$

We first prove $V > U$. Otherwise $\prod p_l = UV \leq U^2 < d^2$. The first inequality of (11.6) then gives $q_0 < 1/90$, since

$$\frac{1}{90^3}\left(1 - \frac{3}{90}\right) > \frac{1}{950^2}.$$

Consequently (11.5) and $v \geq q_0^2 + (1 - q_0)^2/3$ give $u < 49/100$. But minimizing the square sum at fixed p_jp_k yields

$$p_jp_k \geq \frac{1}{2}\left(\frac{(1-d)^2}{2} - u + d^2\right) > \frac{1}{200} - \frac{d}{2} > d > dp_i,$$

a contradiction. Since $\operatorname{Tr} E \leq \operatorname{Tr} D$, equation (11.7) proves all three pair inequalities.

Relabel so that $q_1 = \min(q_1, q_2, q_3)$ and put $S = q_0 + q_1$. The pair inequality for $2, 3$ implies

$$q_2q_3 \leq \frac{(p_2 + p_3)(q_2 + q_3)}{4}.$$

Writing $A = 1 - d$, $B = 1 - S$, and $x = p_1$, we have

$$u \geq d^2 + x^2 + \frac{(A - x)^2}{2}, \qquad v \geq q_0^2 + q_1^2 + B^2 - \frac{(A - x)B}{2}.$$

Minimization in x gives

$$5u + 4v \geq \frac{21}{5} - \frac{32S}{5} + \frac{118S^2}{15} + \frac{2d}{3}(-2S + 10d - 3) - 8q_0q_1. \tag{11.8}$$

If $S \leq 3/40$, using $q_0q_1 \leq S^2/4$ in this inequality gives

$$5u + 4v \geq \frac{21}{5} - \frac{32S}{5} + \frac{88S^2}{15} - \frac{2d}{3}(2S + 3) \geq \frac{15}{4} + \frac{3}{3800},$$

a contradiction. Thus $S > 3/40$.

Equation (11.6) gives $q_0 < 1/27$. Suppose $q_0 \geq 1/80$. Since $q_i > 3/40 - q_0$ for $i \neq 0$,

$$\prod_i q_i > \psi(q_0), \qquad \psi(x) = x(3/40 - x)^2(17/20 + x).$$

On $[1/80, 1/27]$ one has $\psi''(x) = 12x^2 + 21x/5 - 39/160 < 0$, and $\psi(1/27) > \psi(1/80)$. Hence

$$\prod_i q_i > \frac{69}{1638400} > \frac{1}{27 \cdot 950},$$

contrary to (11.6). Therefore $q_0 < 1/80$ and $q_i > 3/40 - 1/80 = 1/16$ for $i \neq 0$. A final use of the product bound gives

$$q_0 < \frac{256d}{27(1 - 1/80 - 1/8)} = \frac{20480d}{1863} \leq \frac{20480}{1863 \cdot 950} < \frac{1}{85}.$$

□

Lemma 11.7. *Under the hypotheses of Lemma 11.6,* $\operatorname{Tr}_{K/\mathbb{Q}}(\alpha) < 12$.

Proof. Normalize the last three coordinates by $P_i = p_i/(1-d)$ and $Q_i = q_i/(1-q_0)$, and put $U = \sum P_i^2$, $V = \sum Q_i^2$, and $r = \sum Q_i^{-1}$. The preceding lemma and (11.5) give

$$5U + 4V < \frac{15}{4}\left(\frac{85}{84}\right)^2 < \frac{96}{25}, \qquad V < \frac{163}{300}, \qquad Q_i > \frac{1}{16}. \tag{11.9}$$

Cauchy–Schwarz in the plane of coordinate sum zero gives

$$\sum_i \frac{P_i}{Q_i} \leq \frac{r}{3} + \sqrt{\left(U - \frac{1}{3}\right)\left(\frac{2r^2}{3} - \frac{4r}{1-V}\right)}. \tag{11.10}$$

We bound its right-hand side by $163/15$, retaining the tradeoff between U and V in (11.9).

For $V < 1/2$, Schur's inequality gives $Q_1Q_2Q_3 \geq (1-2V)/9$, hence

$$r \leq \frac{9(1-V)}{2(1-2V)}.$$

Thus $r \leq 27/2$ if $V \leq 2/5$, and $r \leq 81/4$ if $V \leq 7/16$. For the remaining interval we use $r < 108/5$. To see this, put $t = \min Q_i > 1/16$. If $t \geq 1/8$, convexity gives $r \leq 2/t + 1/(1-2t) \leq 52/3$. Otherwise,

$$r < \frac{1}{t} + \frac{2(1-t)}{1 - 2t + 2t^2 - 163/300} =: R(t).$$

On $[1/16, 1/8]$ its denominator is at least $571/2400$, so

$$R'(t) \leq -64 + \frac{463/150}{(571/2400)^2} < 0.$$

Therefore $r < R(1/16) = 70144/3259 < 108/5$.

The second factor under the square root in (11.10) increases with r on the domain $9 \leq r$, $1/3 \leq V < 163/300$, and decreases with V. The three intervals just considered therefore give the bounds

$$\sum_i \frac{P_i}{Q_i} < \begin{cases} \frac{9}{2} + \sqrt{\frac{21}{125}\frac{81}{2}}, & V \leq 2/5, \\ \frac{27}{4} + \sqrt{\frac{43}{375}\frac{1107}{8}}, & 2/5 \leq V \leq 7/16, \\ \frac{36}{5} + \sqrt{\frac{127}{1500}\frac{3936}{25}}, & V \geq 7/16. \end{cases}$$

Each displayed bound is less than $163/15$, as follows by squaring. Since $\alpha_0 = d/q_0 \leq 1$, we conclude

$$\operatorname{Tr}\alpha = \frac{d}{q_0} + \frac{1-d}{1-q_0}\sum_i \frac{P_i}{Q_i} < 1 + \frac{85}{84}\frac{163}{15} = \frac{3023}{252} < 12.$$

□

Lemma 11.8. *Let α be a totally positive integral d-number in $K = \mathbb{Q}(\sqrt{2}, \sqrt{5})$, with square norm, trace less than twelve, and least conjugate at most one. Then $\alpha = 1$, or α is a conjugate of $(3 + \sqrt{5})/2$ or $2 + \sqrt{2}$.*

Proof. Put $\phi = (1 + \sqrt{5})/2$. The integer ring has basis $1, \sqrt{2}, \phi, \phi\sqrt{2}$, since the quadratic discriminants 8 and 5 are coprime. In particular its traces are even, so $\operatorname{Tr} \alpha \leq 10$. The arithmetic–geometric mean inequality gives $N(\alpha) < 40$; the possible square norms are $1, 4, 9, 16, 25, 36$.

The principal ideal of a d-number is Galois invariant. At an unramified prime its norm valuation is a multiple of four, since the prime-ideal valuations above it are all equal. Only 2 and 5 ramify in K; this excludes norms 9 and 36. The unique prime $\mathfrak{p}$ above 2 has norm 4 and $\mathfrak{p}^2 = (2)$. Thus norm 16 would imply $\alpha = 2\epsilon$ for a totally positive unit ϵ. Its trace is even and at most five, hence at most four. Equality in the arithmetic–geometric mean inequality forces $\epsilon = 1$, contrary to the least-conjugate hypothesis.

If the norm is 25, the d-number criterion forces the trace to be divisible by five, hence to equal ten. Write

$$2\alpha = 5 + b\sqrt{2} + c\sqrt{5} + e\sqrt{10}, \qquad c \equiv 1 \pmod 2, \quad b \equiv e \pmod 2.$$

Positivity gives $|b|\sqrt{2} < 5$, $|c|\sqrt{5} < 5$ and $|e|\sqrt{10} < 5$. Thus $c = \pm 1$; either $e = 0$ and $b \in \{0, \pm 2\}$, or $e = \pm 1$ and $b \in \{\pm 1, \pm 3\}$. The norm equation is

$$(25 + 5c^2 - 2b^2 - 10e^2)^2 - 5(10c - 4be)^2 = 400.$$

For $e = 0$ it forces $b = 0$. For $|e| = |b| = 1$ its left side is 144 or -656; for $|e| = 1$, $|b| = 3$ it is -16 or -2416. The only remaining elements are $(5 \pm \sqrt{5})/2$, both greater than one. Norm 25 is therefore impossible.

The norm is now 1 or 4. A quartic α must appear in the first seven rows of Lemma 11.5. Their cyclic fields are not K, and their two biquadratic fields are $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $\mathbb{Q}(\sqrt{3}, \sqrt{5})$. Thus α has degree at most two. In degree two its quadratic norm is one or two and its quadratic trace is at most five. At norm one the nonrational possibilities have traces $3, 4, 5$ and quadratic fields $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{21})$, respectively. Only the first lies in K. At norm two the d-number condition requires even trace, leaving $2 \pm \sqrt{2}$. A rational α is a positive integer with least conjugate at most one, so equals one. □

Proof of Theorem 11.1 in the remaining field. The preceding lemmas leave only two nonconstant dimension ratios. Suppose first that the least conjugate is $a = (3 - \sqrt{5})/2$. Since the conjugates a, a^{-1} each occur twice and $\sum p_i = \sum p_i/\alpha_i = 1$, the ordinary reciprocal mass at the two a positions is $a/(1 + a)$, and the signed mass is $1/(1 + a)$. One of these positions has index zero, with $p_0 = d$. Minimizing the square sums in each pair gives

$$5u + 4v \geq \frac{7a^2 + 13/2}{(1 + a)^2} - \frac{10a + 8/a}{1 + a}\, d > \frac{39}{10} - 28d > \frac{15}{4}.$$

Indeed, $a^2 = 3a - 1$ makes the constant $21/5 - 1/(10a) > 39/10$, while $1/3 < a < 2/5$ bounds the coefficient by 28. The last strict inequality holds already for $D \geq 256$.

If the least conjugate is $2 - \sqrt{2}$, the ordinary mass of its two positions is $1/2$, while their signed mass is $(2 + \sqrt{2})/4 > 4/5$. The same pair minimization gives

$$u \geq \frac{3}{8} - d, \qquad v > \frac{16}{25} - 4d, \qquad 5u + 4v > \frac{887}{200} - 21d > \frac{15}{4} \quad (D \geq 256).$$

Both cases contradict (11.5). Therefore $\alpha = E/D = 1$. Spectral saturation gives a spherical section of the sphericalization, proving the theorem. □

Corollary 11.9. *If a nonspherical self-dual rank-four complex fusion category has quartic Frobenius–Perron global dimension, every fusion multiplicity is at most* 30.

Proof. Theorem 11.1 gives $D < 950 < 31^2$. Lemma 2.1 gives $N_{ij}^k < \sqrt{D} < 31$; use integrality. □

12 Proof of the uniform bound

Proof of Theorem 1.2. If $\mathcal{C}$ is spherical, Theorem 10.1 gives $D < 1000$. Suppose therefore that it is not spherical. Theorem 4.1 shows that every simple object is self-dual. If D were rational, it would be an integer, and weak integrality would give a spherical structure by [6, Proposition 8.24]. Thus D is irrational.

The field generated by the Frobenius–Perron character is a number-field factor of the four-dimensional rational Grothendieck algebra. It has degree at most four and contains $\mathbb{Q}(D)$. If D had degree three, that character field would have degree three as well; Theorem 9.11 would make the category spherical. Only degrees two and four remain.

In either case Corollary 7.6 identifies the signed dimension field with $\mathbb{Q}(D)$. In degree two, Theorem 8.3 shows that the signed rational algebra is the product of two copies of this field, with complementary codegree product at least 100. Theorem 8.16 therefore gives $D < 1000$. In degree four, both ordinary and signed rational algebras are the common quartic field. Theorem 11.1 gives $D < 950$. This proves the dimension bound in every case.

By Lemma 2.1,

$$N_{XY}^Z \le d_X < \sqrt{D} < \sqrt{1000} < 32.$$

Integrality gives the bound 31. A multiplication table in a fixed basis of four elements has only finitely many entries, all now bounded; hence only finitely many based isomorphism classes can occur. □

13 Necessary conditions for the finite certificates

The universal dimension bound makes the classification a finite problem. We record the necessary conditions used in its exact certificates. These conditions concern arbitrary complex categorifications; positive unsquared central dimensions are used only after an explicit pseudo-unitary reduction.

13.1 Arithmetic and signed dimension characters

13.1.1 Conventions and standard obstructions

A fusion ring has a distinguished basis $b_0 = 1, b_1, b_2, b_3$, nonnegative integral structure constants, duality $i \mapsto i^*$, and Frobenius reciprocity, with $N_{ij}^0 = \delta_{i,j^*}$. We use the row convention $(N_i)_{jk} = N_{ij}^k$. Commutativity is Lemma 2.1.

Write $d_i = \mathrm{FPdim}(b_i)$ and $D = \sum_i d_i^2$. The four characters χ have formal codegrees

$$f_\chi = \sum_i \chi(b_i)\chi(b_{i^*}) = \sum_i |\chi(b_i)|^2.$$

They are the eigenvalues, with multiplicity, of

$$B = \sum_i N_i N_{i^*} = \sum_i N_i N_i^{\mathsf{T}}. \tag{13.1}$$

The following necessary conditions will be used in their stated scope. The arithmetic inputs and permitted pseudo-unitary reductions are Proposition 2.8. The last assertion concerns a Galois conjugate; it does not claim pseudo-unitarity of every categorification of the ring.

The Galois reduction used here is the one in Section 10: an embedding of the dimension field extends to an automorphism of $\mathbb{C}$ and preserves all tensor and spherical identities. For the two-dimension case we use precisely the published statement in Proposition 2.8(5).

13.1.2 Small arithmetic certificates

For a nonzero algebraic integer u, put $e_u = [\mathbb{Q}(u) : \mathbb{Q}]$ and $n_u = |N_{\mathbb{Q}(u)/\mathbb{Q}}(u)|$.

Lemma 13.1 (Normalized norm obstruction)**.** *If u/v is an algebraic integer, then*

$$\frac{v_p(n_u)}{e_u} \geq \frac{v_p(n_v)}{e_v} \qquad \text{for every rational prime } p. \tag{13.2}$$

Proof. Taking the norm in $L = \mathbb{Q}(u, v)$ shows that $n_u^{[L:\mathbb{Q}]/e_u}/n_v^{[L:\mathbb{Q}]/e_v}$ is an integer. Taking its p-valuation gives the assertion. □

Thus a failed divisibility needs only two irreducible polynomials and one prime. As a preliminary shortcut, a signed Casimir determinant not divisible by a prime required by every possible Δ excludes the whole table: if the minimal polynomial of Δ had norm divisible by that prime, the characteristic polynomial of the Casimir would have constant term divisible by it too.

Lemma 13.2 (Prime certificates for fields)**.** *Let $g \in \mathbb{Z}[t]$ be monic irreducible and $K = \mathbb{Q}(\alpha)$, $g(\alpha) = 0$.*

1. *If K contains a root of a monic irreducible polynomial f, then* $\deg f$ *divides* $\deg g$.
2. *Suppose g splits into distinct linear factors modulo a prime p. If K contains all roots of a monic integral polynomial f, then f splits into linear factors modulo p, with repetitions allowed.*
3. *If the squarefree reduction modulo p of a monic irreducible polynomial has irreducible factors of unequal degrees, its root field cannot be cyclotomic.*

Proof. The first assertion is the tower formula. In the second, p does not divide the discriminant of g, hence not the index of $\mathbb{Z}[\alpha]$ in the ring of integers of K. Every residue field above p is therefore $\mathbb{F}_p$. Reduce the linear factorization of f in the ring of integers of K at any such prime. For the third, Dedekind's factorization theorem identifies the factor degrees with the cycle lengths of a Galois permutation of the roots. A transitive abelian permutation group acts regularly after quotienting its kernel, so all cycles of each element have the same length. Every subfield of a cyclotomic extension is Galois and abelian. □

These are rejection certificates: failure to find a prime certificate proves neither a field inclusion nor cyclotomicity. Positive root lists in a number field are instead verified by multiplying their linear factors and recovering the complete rational polynomial.

13.1.3 Sphericalization and exhaustive signed tables

Lemma 2.2 supplies the signed reduction. For the finite search its coefficient conditions are

$$|T_{ij}^k| \leq N_{ij}^k, \qquad T_{ij}^k \equiv N_{ij}^k \pmod 2. \tag{13.3}$$

All compatible signed tables, and all of their nonzero dimension characters, are retained until a stated necessary condition excludes them. A sign choice does not identify characters with different ordinary or signed provenance.

The signed search is finite by (13.3). In the self-dual case there are ten independent coefficients. The exhaustive cover below retains all zero-denominator branches and all compatible basis-sign transforms. A signed certificate rejects every remaining table and every possible dimension-character factor.

13.2 Central induction and two twist moments

Let $\mathcal{C}$ be pseudo-unitary, with its canonical spherical structure, and let $F : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ have right adjoint I. For simples X_i and A, write $M_{iA} = [F(A) : X_i] = [I(X_i) : A]$. The four simple summands of $I(1)$ occur once, have twist one and dimensions D/f_χ [16, Theorems 2.5 and 2.13]. Let L be their four restriction columns, including the central unit column $(1, 0, 0, 0)^{\mathsf{T}}$. Then

$$MM^{\mathsf{T}} = B, \qquad L\mathbf{1} = B_{\bullet 0}, \qquad B - LL^{\mathsf{T}} \text{ is a nonnegative integral Gram matrix.} \tag{13.4}$$

Indeed $FI(X) = \bigoplus_j X_j \otimes X \otimes X_j^*$, and adjunction gives the first identity. All remaining columns have first entry zero, which gives the other assertions. In particular the remainder is positive semidefinite and entrywise nonnegative.

Lemma 13.3 (Gram saturation)**.** *If $B = \sum_A m_A m_A^{\mathsf{T}}$ and some known columns already satisfy $\sum_{A\ known}(v^{\mathsf{T}} m_A)^2 = v^{\mathsf{T}} B v$, then v is orthogonal to every remaining column.*

Proof. The sum of the remaining nonnegative squares is zero. □

13.2.1 Higher vanishing indicators

The following statement holds at arbitrary rank. For an object V, let $I_{\mathrm{low}}(V)$ be the sum, with its multiplicities in $I(V)$, of those simple constituents that occur in $I(1)$.

Theorem 13.4 (Higher-indicator bound)**.** *Let $\mathcal{C}$ be a pseudo-unitary fusion category over $\mathbb{C}$, with its canonical spherical structure, and let $m \geq 1$. If $\nu_1(V) = \cdots = \nu_m(V) = 0$, then*

$$(m+1)\,\mathrm{FPdim}(I_{\mathrm{low}}(V)) \leq D\,\mathrm{FPdim}(V).$$

More strongly, the same inequality holds with $I_{\mathrm{low}}(V)$ replaced by the full twist-one part of $I(V)$. Equality forces every other constituent to have a nontrivial $(m+1)$st root of unity as twist. In particular, if every simple constituent of V is non-self-dual, then $3\,\mathrm{FPdim}(I_{\mathrm{low}}(V)) \leq D\,\mathrm{FPdim}(V)$.

Proof. Write $w_A = [I(V) : A]\,\mathrm{FPdim}(A) \geq 0$ and $T = D\,\mathrm{FPdim}(V)$. The indicator identity [13, Theorem 4.1] is $\sum_A w_A \theta_A^k = D\nu_k(V)$. Negative moments vanish by complex conjugation. Expanding the nonnegative polynomial

$$|1 + z + \cdots + z^m|^2 = (m+1) + \sum_{k=1}^{m} (m+1-k)(z^k + z^{-k}) \qquad (|z| = 1)$$

gives $\sum_A w_A |1 + \theta_A + \cdots + \theta_A^m|^2 = (m+1)T$. Every twist-one constituent contributes $(m+1)^2 w_A$; discarding the remaining nonnegative terms proves the bound. Equality requires the polynomial to vanish at every remaining twist, proving its stated support. The last assertion follows from additivity and $\nu_1(X) = \nu_2(X) = 0$ for a non-self-dual simple X. □

For $m = 1$ this is the low-part estimate explicit in the proof of [16, Theorem 2.21]. Its higher-indicator proof is included here.

13.2.2 The moment inequality

For a nontrivial simple X, put $d = \mathrm{FPdim}(X)$ and $\epsilon = \nu_2(X)$. The indicator formula [13, Theorem 4.1] gives, with $w_A = [I(X) : A]\,\mathrm{FPdim}(A) \geq 0$,

$$\sum_A w_A = Dd, \qquad \sum_A w_A \theta_A = 0, \qquad \sum_A w_A \theta_A^2 = D\epsilon. \tag{13.5}$$

Here the twists are roots of unity, $\epsilon \in \{-1, 1\}$ for a self-dual simple and $\epsilon = 0$ otherwise.

Proposition 13.5 (Two-moment bound)**.** *If L_X is any known contribution to the twist-one mass of $I(X)$, then*

$$(3d + \epsilon)L_X \leq Dd(d + \epsilon).$$

Consequently the following bounds require no choice of second indicator:

$$(3d + 1)L_X \leq Dd(d + 1) \qquad (X \cong X^*), \tag{13.6}$$

$$3L_X \leq Dd \qquad (X \ncong X^*). \tag{13.7}$$

One may take $L_X = \mathrm{FPdim}(I_{\mathrm{low}}(X)) = (LL^{\mathsf{T}}d)_X$.

Proof. For $P(z) = d + (d + \epsilon)z + dz^2$, expand $|P(z)|^2$ on the unit circle and use (13.5). This gives

$$\sum_A w_A |P(\theta_A)|^2 = Dd(3d + \epsilon)(d + \epsilon).$$

The twist-one terms contribute at least $(3d + \epsilon)^2 L_X$. Since $3d + \epsilon > 0$, division gives the assertion. The function $e \mapsto (d + e)/(3d + e)$ is increasing on $[-1, 1]$; replace ϵ by one in the self-dual case. □

The familiar first-moment bound $2L_X \leq Dd$ and the weaker Fejér bound $9L_X \leq D(3d+2)$ follow as well. The checker uses (13.6)–(13.7); these suffice for all the low-induction inequalities needed here. Positivity is essential: these bounds are not applied to arbitrary spherical categories with signed dimensions.

13.2.3 A finite restriction test

Proposition 13.6 (Exhaustive low-induction certificate)**.** *Fix the positive dimension character d. For each formal codegree f, form the set of vectors $v \in \mathbb{Z}^4_{\geq 0}$ satisfying*

$$v_0 = 1, \qquad \sum_i v_i d_i = D/f, \qquad v_i \leq \min\{B_{0i}, \lfloor \sqrt{B_{ii}} \rfloor\} \quad (i > 0). \tag{13.8}$$

Reserve $(1, 0, 0, 0)^{\mathsf{T}}$ for the central unit. Enumerate all choices of the other three columns, with the codegree multiplicities retained, whose sum together with the unit column is $B_{\bullet 0}$. If every such choice fails the entrywise or positive-semidefinite condition in (13.4), or one of (13.6)–(13.7), then R admits no pseudo-unitary categorification.

Proof. A genuine restriction column satisfies its dimension equation, and $M_{0A} = 1$ on the four constituents of $I(1)$. The first row and the diagonal of (13.4) give the two bounds in (13.8). Hence the actual low part occurs in the finite list and satisfies every tested necessary condition. □

The computation is only linear algebra over a number field. Express the four dimensions and D/f in a rational power basis, row-reduce the dimension equation, and enumerate its bounded free integer coordinates. When the four dimensions are rationally independent, there is at most one possible column for each f; a negative or nonintegral coefficient is already an obstruction. When they are dependent, no uniqueness is presumed. Full induction matrices, modular data and associators are never enumerated.

13.3 Positive central moments and the sign object

We use the canonical spherical structure on the sphericalization $\mathcal{S}$ of a hypothetical fusion category. Its realization as an equivariantization for the canonical involutive pivotal functor gives a central copy of $\mathrm{Rep}(C_2)$; see [1, Section 5.3]. Its central sign object ε has order two, dimension -1, and self-braiding $+1$. Consequently its twist in $\mathcal{Z}(\mathcal{S})$ is -1. The two lifts of each original self-dual simple are self-dual; their dimensions and second Frobenius–Schur indicators have opposite signs. The latter assertion follows directly from the definition of the second indicator: the two lifts have the same evaluation and coevaluation maps after forgetting equivariance, while their pivotal maps differ by -1.

Lemma 13.7 (A positive measure without pseudo-unitarity)**.** *Let $\mathcal{S}$ be a spherical fusion category, and put $d = \dim \mathcal{S}$. For $A \in \operatorname{Irr} \mathcal{Z}(\mathcal{S})$, let $a_A = \dim A$ and θ_A be its twist. Then a_A is real and*

$$\sum_A a_A^2 = d^2, \qquad \sum_A a_A^2 \theta_A = d, \qquad \sum_A a_A^2 \theta_A^2 = d \sum_V \dim(V) \nu_2(V).$$

In particular the coefficients a_A^2 form a positive measure on roots of unity, even when some dimensions of $\mathcal{S}$ are negative.

Proof. Simple dimensions in a spherical fusion category are totally real [16, Section 2.1]. The forgetful functor is spherical, so a_A is an integral linear combination of these real dimensions. Let I be central induction and $m_{V,A} = [I(V) : A]$. Adjunction gives $a_A = \sum_V m_{V,A} \dim V$. The indicator identity of [13, Theorem 4.1]

$$\operatorname{Tr}(\theta^n|_{I(V)}) = d \nu_n(V)$$

therefore gives, after multiplication by $\dim V$ and summation over V,

$$\sum_A a_A^2 \theta_A^n = d \sum_V \dim(V) \nu_n(V).$$

For $n = 1$ only the tensor unit contributes, and for $n = 2$ this is the claimed formula. The zeroth moment is the standard identity $\dim \mathcal{Z}(\mathcal{S}) = d^2$. The twist is a root of unity by the Vafa theorem for the center of a spherical fusion category. □

Proposition 13.8 (A global codegree moment bound)**.** *Let a commutative fusion ring have a spherical categorification, with simple dimensions d_i, global dimension $\Delta = \sum_i d_i^2$ and formal codegrees f_j, counted with multiplicity. Put*

$$w = \sum_j f_j^{-2}, \qquad S = \sum_{i=i^*} |d_i|, \qquad h = \Delta^2 w.$$

At every real embedding of its dimension field one has

$$2(\Delta - h)^2 \le (\Delta^2 - h)(\Delta^2 + \Delta S - 2h). \tag{13.9}$$

Equivalently, $\Delta(1 - 3w) + (1 - w)S + 4w - 2/\Delta \ge 0$. For a self-dual pseudo-unitary category, $S = \sum_i \mathrm{FPdim}(X_i)$.

Proof. The constituents of $I(1)$ are distinct, have twist one and dimensions Δ/f_j. Remove their squared mass h from the positive measure of Lemma 13.7. The remaining mass is $\Delta^2 - h$, its first moment is $\Delta - h$, and its second moment is $\Delta s - h$, where $s = \sum_i d_i \nu_2(X_i) \le S$. The weighted sum of squared real parts of twists is $(\Delta^2 + \Delta s - 2h)/2$. Cauchy–Schwarz gives the displayed inequality with s in place of S; increasing s to S preserves it because the remaining mass is nonnegative. Expanding and dividing by $\Delta^3 > 0$ gives the equivalent expression. The argument applies to every Galois conjugate spherical structure. The rational number w can be computed without finding codegree roots: $w = \operatorname{Tr}(B^{-2})$ for the Casimir matrix B. □

This form of the bound applies to every possible spherical dimension character, including those outside the Frobenius–Perron orbit. A pseudo-unitary obstruction alone would not exclude such branches.

Proposition 13.9 (Determined-sign moment obstruction)**.** *For a self-dual rank-four spherialization use the notation $E = \sum_i x_i^2$, $u = \sum f_i^{-2}$ and $v = \sum g_i^{-2}$ from Section 2. Put $s = 1 + \sum_{i=1}^{3} \nu_i x_i$, where $\nu_i = \nu_2(X_i^+)$. At every real embedding one has*

$$u + v \le 2, \qquad E(u + v - 1) \le s \le E, \tag{13.10}$$

$$(2 - u - v)\left(1 - u - v + \frac{s}{E}\right) \ge \left(u - \frac{1}{E}\right)^2. \tag{13.11}$$

The same indicator assignment must work at every embedding.

Proof. The constituents of $I(1)$ have dimensions $E/f_i, E/g_i$ and twist one. Their disjoint sign translates have opposite dimensions and have twists $+1$ on the ordinary constituents and -1 on the signed ones, by Lemma 2.9. Thus their squared mass and signed first-moment contribution are

$$h = E^2(u + v), \qquad C = E^2(u - v).$$

The first-moment identity for $I(\varepsilon)$ is automatic, since both reciprocal-codegree sums are one. Removing these two families from the positive squared-dimension measure leaves mass $M = 4E^2 - 2h$, first moment $2E - h - C$, and second moment $4Es - 2h$. Lemma 3.1 gives the two mass bounds and

$$(2E - h - C)^2 \le (4E^2 - 2h)(2Es + 2E^2 - 2h).$$

Substituting h, C and dividing by $4E^4$ gives (13.11). The indicators are rational signs and remain fixed under Galois conjugation. □

13.4 Joint arithmetic, parity and moment certificates

Theorem 13.10 (Joint signed certificate)**.** *A proposed signed quotient T and dimension character χ must satisfy all the following conditions. Its codegrees are d-numbers, its character values are cyclotomic integers, and each ordinary or signed codegree belongs to the character field and divides $E = \sum_i \chi(b_i)^2$ integrally. Moreover there must be signs $\nu_i \in \{1, -1\}$ such that*

$$N^i_{j,j} - \nu_i T^i_{j,j} \equiv 0 \pmod 4 \qquad (1 \le i \le 3,\ 0 \le j \le 3), \tag{13.12}$$

and (13.10)–(13.11) hold at every real embedding with $x_i = \chi(b_i)$ and that same indicator assignment. Ordinary and signed occurrences of equal codegrees retain their provenance and multiplicity.

Proof. The arithmetic statements are Lemma 2.2 and the standard codegree and cyclotomicity inputs. The two lifts of b_i occur in an adjoint product with multiplicities $(N^i_{j,j} + T^i_{j,j})/2$ and $(N^i_{j,j} - T^i_{j,j})/2$ and have opposite indicators. By Lemma 2.10, the lift with indicator -1 occurs evenly. This is exactly (13.12). These are the same indicators as in Proposition 13.9; hence the conditions must be imposed jointly, with one assignment at all embeddings. There is no additional twist-sign search. □

For a spherical character the same parity observation improves Proposition 13.8: replace its upper bound S by

$$S_{\mathrm{par}} = 1 + \sum_{i \in J} d_i + \sum_{\substack{i > 0,\ i = i^* \\ i \notin J}} |d_i|, \qquad J = \{i > 0 : i = i^*,\ N^i_{j,j^*} \text{ odd for some } j\}. \tag{13.13}$$

Indeed parity forces $\nu_2(X_i) = 1$ for $i \in J$, so $s \le S_{\mathrm{par}}$. The remaining mass multiplying this upper bound is nonnegative, preserving the inequality at each embedding.

Lemma 13.11 (Parity-restricted signed enumeration)**.** *Write the unsigned and signed tuples as $N = (N_0, \dots, N_9)$ and $p = (p_0, \dots, p_9)$ in the convention (2.1). Let $J_1 = \{0, 3, 5\}$, $J_2 = \{1, 6, 8\}$ and $J_3 = \{2, 7, 9\}$. It suffices to search the union over $\nu \in \{1, -1\}^3$ of*

$$|p_k| \le N_k, \qquad p_k \equiv \nu_i N_k \pmod 4 \ (k \in J_i), \qquad p_4 \equiv N_4 \pmod 2.$$

After identical and empty domains are removed, these are at most eight pairwise disjoint boxes, also after the chosen sign normalization.

Proof. The congruences are (13.12). For a fixed i, the two residue choices agree when all N_k, $k \in J_i$, are even; otherwise an odd coordinate makes them disjoint. Since the three J_i are disjoint, the product boxes have the same property. A basis-sign change multiplies $T^i_{j,j}$ and ν_i by the same sign, so their product is unchanged. The normalization therefore preserves the union and loses no categorical possibility. □

Identical boxes do not identify their indicator assignments: every compatible assignment remains available in the joint moment test. The common bounded signed cover in Section 14.3 applies the universal arithmetic conditions once, then tests all compatible transforms against each ordinary ring.

13.5 Uniform applications of the moment bounds

The following consequences replace separate low-rank exclusion theorems. Throughout, the two-dimension Galois reduction is used only in the form stated in Proposition 2.8(5).

For the twist-sector observations in the next paragraph, work in a pseudo-unitary category with its canonical spherical structure. If g is invertible of order two, its second indicator is $\epsilon \in \{-1, 1\}$. Since $\mathrm{FPdim}(g) = 1$, equality in the triangle inequality for the second indicator forces $\theta_A^2 = \epsilon$ for every constituent A of $I(g)$. More generally, if H_X is any known mass in $I(X)$ supported on squared twist one, taking real parts of the second moment gives

$$2H_X \le D(d_X + \nu_2(X)) \le D(d_X + 1). \tag{13.14}$$

Indeed the remaining real trace is at least the negative of its mass. For a non-self-dual simple, equality in $3W_X \le Dd_X$ forces the complementary twists to be primitive cube roots by Theorem 13.4.

13.5.1 The rank-two subcategory obstruction

Proposition 13.12. *If a fusion ring contains a based fusion subring with basis $1, x$ and $x^2 = 1 + ax$, where $a \ge 2$, it has no complex fusion categorification.*

Proof. The indicated rank-two based subring would be the Grothendieck ring of a fusion subcategory. Ostrik's rank-two classification [18, Theorem 1] allows only $x^2 = 1$ and $x^2 = 1 + x$. □

The criterion applies whenever such a rank-two based subring is present.

13.5.2 A uniform diagonal two-invertible obstruction

Theorem 13.13. *Let R_m have self-dual basis $1, g, X, Y$ and products*

$$g^2 = 1, \quad gX = Y, \quad gY = X, \qquad X^2 = Y^2 = 1 + m(X + Y), \quad XY = g + m(X + Y).$$

For every integer $m \geq 3$, the ring R_m has no fusion categorification over $\mathbb{C}$.

Proof. Its positive dimensions are $(1, 1, d, d)$, where $d = m + \sqrt{m^2 + 1}$ and $d^2 = 1 + 2md$. The two-dimension reduction allows a pseudo-unitary assumption. Put $D = 4 + 4md$ and $V = X \oplus Y$.

The four dimensions in $I(1)$ are $1, d^2, 1 + md, 1 + md$. Moreover

$$FI(1) = 4 \cdot 1 + 2mX + 2mY.$$

In particular the low part of $I(g)$ is empty. Dimension comparison shows that the total multiplicities of X, Y in the last three restrictions are $2m, m, m$. Therefore their known twist-one mass in $I(V)$ is, independently of its allocation between X and Y,

$$W = 2md^2 + 2m(1 + md) = 4m + 6m^2 d. \tag{13.15}$$

The total mass in $I(V)$ is

$$M = 2Dd = 8m + (16m^2 + 8)d.$$

The Gram entry $B_{gg} = 4$ implies that $I(g)$ is either twice one simple or a sum of four distinct simples. The first possibility contradicts its zero first twist trace. Write the four simple dimensions as $\beta_j = 1 + k_j d$: each restriction contains g once, no unit, and k_j copies of X, Y in total. Thus $k_j \in \mathbb{Z}_{\geq 0}$ and $\sum_j k_j = 4m$.

Put $\epsilon = \nu_2(g)$. All four twists have square ϵ. The zero first trace divides these four objects into the two possible twists, with mass $D/2$ at each. Since d is irrational, each side has two objects and the two k_j sum to $2m$. Each side therefore contributes at least

$$2m + 2m^2 d$$

to $I(V)$, by $k_1^2 + k_2^2 \geq (2m)^2/2$.

If $\epsilon = 1$, the new twist-one mass is at least $2m + 2m^2 d$, disjoint from the low part. Adding (13.6) for X, Y yields

$$(3d + 1)(6m + 8m^2 d) \leq 2Dd(d + 1).$$

After reduction by $d^2 = 1 + 2md$, the left side minus the right side is

$$2(4m^2 - m - 4) + 2(8m^3 - 4m^2 - 3m - 4)d > 0$$

for $m \geq 3$. For example, substituting $m = n + 3$ gives coefficients $8n^2 + 46n + 58$ and $16n^3 + 136n^2 + 378n + 334$, respectively.

Suppose instead $\epsilon = -1$. The four new twists are $\pm i$, and their total mass in $I(V)$ is at least

$$H = 4m + 4m^2 d.$$

Remove this mass and the low mass W from the positive measure of $I(V)$. For the remaining measure, the first cosine moment is $-W$, its total mass is at most

$$M - W - H = (6m^2 + 8)d,$$

and its second cosine moment is at most

$$\frac{M + D(\nu_2(X) + \nu_2(Y))}{2} - W \le \frac{M}{2} + D - W = 4 + (2m^2 + 4m + 4)d.$$

Cauchy–Schwarz therefore requires

$$W^2 \le (6m^2 + 8)d(4 + (2m^2 + 4m + 4)d).$$

The difference between the two sides, reduced by the dimension equation, is

$$\begin{aligned} &8(m-2)(3m^3 + 3m^2 + 3m + 2) \\ &\quad + 8(m-2)(6m^4 + 6m^3 + 8m^2 + 5m + 2)d > 0. \end{aligned}$$

This excludes the second indicator choice as well. □

13.5.3 Propagation from an invertible involution

Proposition 13.14 (Residual-involution criterion)**.** *Let $\mathcal{C}$ be pseudo-unitary with commutative Grothendieck ring, and let $g \neq 1$ be an invertible simple with $g^2 = 1$. For its actual low restriction matrix L, put*

$$Q = B - LL^{\mathsf{T}}, \qquad W_i = (LL^{\mathsf{T}}d)_i.$$

Suppose $Q_{gg} = 2$ and $0 < W_g < D/2$. The two constituents of $I(g)$ outside $I(1)$ then have dimensions and twists

$$(D/2 - W_g, 1), \qquad (D/2, -1).$$

If moreover $D/2 - W_g = 1$, their restriction columns are respectively e_g and $Q_{g\bullet}^{\mathsf{T}} - e_g$. Consequently, for every nonunit simple $X_i \neq g$,

$$2W_i + DQ_{gi} \le D(d_i + \nu_2(X_i)) \le D(d_i + 1). \tag{13.16}$$

Proof. The residual Gram entry $Q_{gg} = 2$ is a sum of squares of nonnegative integer multiplicities. Thus exactly two additional simples occur, each once. Equality in the triangle inequality for the second indicator of g forces all twists in $I(g)$ to have square $\nu_2(g)$. The positive low mass has twist one, so $\nu_2(g) = 1$. The vanishing first trace divides the total dimension D equally between twists 1 and -1. Since $0 < W_g < D/2$, both additional sectors have positive mass and each contains one of the two remaining simples. Their dimensions follow.

When the first dimension is one, its restriction contains g and has Frobenius–Perron dimension one; hence its restriction is exactly g. Write v for the other restriction column. In the residual Gram product, only these two columns contribute to row g, so $Q_{gi} = \delta_{gi} + v_i$ for every i. For $i \neq g$, the low constituents and the twist-minus-one constituent therefore give disjoint squared-twist-one mass $W_i + (D/2)Q_{gi}$ in $I(X_i)$. Equation (13.14) proves the required bound. □

The criterion is finite and uniform: enumerate the admissible low parts by Proposition 13.6 and check (13.16) whenever its hypotheses hold. Uniqueness of an admissible low part is sufficient but is not assumed. The automated dispatcher recognizes this obstruction from the complete low-induction proof of any weakly integral input ring; it does not match the tensor against a preselected exceptional ring.

13.5.4 The non-self-dual quadratic family

For an integer $m \geq 0$, let $S(m)$ be the commutative based ring with basis $\{1, g, x, y\}$, involution $g^* = g$, $x^* = y$, and multiplication

$$g^2 = 1, \qquad gx = y, \qquad gy = x, \qquad x^2 = y^2 = g + m(x + y), \qquad xy = 1 + m(x + y).$$

Theorem 13.15. *For every $m \geq 2$, the fusion ring $S(m)$ admits no fusion categorification over $\mathbb{C}$.*

Proof. Put $d = m + \sqrt{m^2 + 1}$. Thus $d^2 = 1 + 2md$, and the Frobenius–Perron dimensions are $(1, 1, d, d)$. By Schopieray's Galois-conjugacy theorem for two-dimension fusion rings [20, Lemma 3.1], any fusion categorification would have a pseudo-unitary Galois conjugate with the same fusion rules. It is consequently sufficient to exclude pseudo-unitary categorifications.

Suppose one exists, and use its canonical spherical structure. Its dimension is $D = 2 + 2d^2 = 4 + 4md$. The four characters of $S(m)$, displayed by columns, are

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ d & -d^{-1} & i & -i \\ d & -d^{-1} & -i & i \end{pmatrix}.$$

Their formal codegrees are D, $2 + 2d^{-2}$, 4, 4, respectively. The formal-codegree description of induction therefore gives

$$I(\mathbf{1}) \cong \mathbf{1} \oplus A \oplus B \oplus C, \qquad \mathrm{FPdim}(A) = d^2 = 1 + 2md, \quad \mathrm{FPdim}(B) = \mathrm{FPdim}(C) = 1 + md,$$

where the four central simples are pairwise nonisomorphic and occur with multiplicity one. Indeed, every irreducible representation of this commutative rank-four ring is one-dimensional. Furthermore,

$$FI(\mathbf{1}) \cong \bigoplus_{T \in \mathrm{Irr}\,\mathcal{C}} T \otimes T^* \cong 4\mathbf{1} \oplus 2mX \oplus 2mY,$$

where $X^* \cong Y$ correspond to x, y. By adjunction each of $F(A)$, $F(B)$, $F(C)$ contains $\mathbf{1}$ once, and the displayed decomposition shows that none contains g. Since d is irrational, comparison of dimensions now forces

$$[F(A) : X] + [F(A) : Y] = 2m, \qquad [F(B) : X] + [F(B) : Y] = [F(C) : X] + [F(C) : Y] = m.$$

These statements determine the sum of the two low-part dimensions without requiring the individual restriction multiplicities:

$$\begin{aligned} L &= \mathrm{FPdim}(I_{\mathrm{low}}(X)) + \mathrm{FPdim}(I_{\mathrm{low}}(Y)) \\ &= 2m(1 + 2md) + 2m(1 + md) = 4m + 6m^2 d. \end{aligned}$$

Both X and Y are non-self-dual. Applying (13.7) to each of them and adding gives $3L \leq 2dD$. But

$$\begin{aligned} 3L - 2dD &= 12m + 18m^2 d - 2d(4 + 4md) \\ &= 12m + 18m^2 d - 8d - 8m(1 + 2md) \\ &= 4m + (2m^2 - 8)d > 0 \end{aligned}$$

for $m \geq 2$, a contradiction. □

13.6 A quadratic root-of-unity obstruction

Let R_n be the ordinary table in (8.11), with $m = n \geq 1$ and its tuple specified in the same calculation. Its positive dimension row is $(1, \delta - 1, \delta, \delta + 1)$, where

$$\delta = \frac{3n + \sqrt{9n^2 + 4}}{2}, \qquad \delta^2 = 3n\delta + 1, \qquad D = 3 + 3\delta^2 = 6 + 9n\delta.$$

Lemma 13.16. *Let $d > 1$ be squarefree and let f be the conductor of $\mathbb{Q}(\sqrt{d})$: $f = d$ if $d \equiv 1 \pmod 4$ and $f = 4d$ otherwise. Put $\kappa = \sqrt{f/d}$ and $\lambda = (\varphi(f) - |\mu(f)|)/\kappa$. Here φ and μ denote Euler's totient and the Möbius function. If a sum of k roots of unity equals $a + b\sqrt{d}$, with $a, b \in \mathbb{Q}$, then $k \geq -a - \lambda b$.*

Proof. Let $K = \mathbb{Q}(\sqrt{d})$, let L be a common cyclotomic overfield, and write $T = \mathrm{Tr}_{L/K}/[L : K]$. Define $\ell(a + b\sqrt{d}) = -a - \lambda b$. We prove $\ell(T(\xi)) \leq 1$ for every root of unity ξ.

If m is the exact order of ξ and $f \nmid m$, then $K \cap \mathbb{Q}(\xi) = \mathbb{Q}$, because the conductor of K is f. Consequently $T(\xi) = \mu(m)/\varphi(m)$, whose absolute value is at most one. If $m = fh$, let χ be the primitive quadratic character modulo f and write $\xi = \zeta_m^r$, with $(r, m) = 1$. The induced quadratic Gauss sum gives

$$T(\xi) = \begin{cases} \dfrac{\mu(f)\mu(h) + \chi(r)\mu(h)\chi(h)\kappa\sqrt{d}}{\varphi(f)\varphi(h)}, & (f, h) = 1, \\ 0, & (f, h) > 1. \end{cases}$$

For clarity, this follows by splitting the units modulo m into the two values of χ: their sum over $\chi = 1$ is half the sum of the Ramanujan sum $\mu(m)$ and the induced Gauss sum. The latter equals $\chi(r)\mu(h)\chi(h)\sqrt{f}$ when $(f, h) = 1$ and vanishes otherwise; an additional power of a prime dividing f cancels upon summing over its last prime-power layer. The same formula includes nonsquarefree h through $\mu(h) = 0$. In the nonzero case, the definition of λ gives

$$|\ell(T(\xi))| \leq \frac{|\mu(f)| + \lambda\kappa}{\varphi(f)\varphi(h)} \leq 1.$$

Applying $\ell \circ T$ to the sum proves the lemma. □

13.6.1 The conductor-forty equality case

Lemma 13.17. *If k roots of unity have sum $a + b\sqrt{10}$, then $k \geq -a - 8b$. Equality forces each root to be either -1 or a primitive fortieth root ζ_{40}^r with $\chi_{40}(r) = -1$. In particular, if eighteen roots have sum $-2 - 2\sqrt{10}$, exactly two equal -1 and the normalized absolute trace of the sum of their squares is 2.*

Proof. In Lemma 13.16, $f = 40$, $\kappa = 2$ and $\lambda = 8$. For a root of order m not divisible by 40, equality in the normalized trace bound requires $-\mu(m)/\varphi(m) = 1$, hence $m = 2$. For $m = 40h$, the nonzero trace formula requires $(h, 40) = 1$, and equality requires $\varphi(h) = 1$. Thus $h = 1$, and equality with the positive sign means $\chi_{40}(r) = -1$. Equality in a sum requires these conditions for every summand. Primitive fortieth roots have normalized absolute trace zero, whereas -1 has trace -1. The trace of $-2 - 2\sqrt{10}$ is -2, so precisely two roots equal -1. Squaring a primitive fortieth root gives a primitive twentieth root, also of trace zero since $\mu(20) = 0$. The other two squares are 1, as asserted. □

Proposition 13.18. *Let $n \geq 2$, write $9n^2 + 4 = t^2 d$ with d squarefree, and define λ as in Lemma 13.16. If $n = 2$, or if*

$$n^2 + nt\lambda > 8n^2 + 4, \tag{13.17}$$

then R_n admits no spherical categorification over $\mathbb{C}$. The condition is an exact scalar arithmetic test for each parameter.

Proof. We first derive a necessary root-of-unity identity for a pseudo-unitary categorification of R_n. Give such a category its canonical spherical structure, and use F, I for forgetful functor and central induction. The four characters, displayed by rows, are

$$\begin{pmatrix} 1 & \delta-1 & \delta & \delta+1 \\ 1 & \delta'-1 & \delta' & \delta'+1 \\ 1 & 0 & 1 & -1 \\ 1 & 1 & -1 & 0 \end{pmatrix}, \qquad \delta' = -\delta^{-1}.$$

The formal codegrees are $D, D' = 3 + 3(\delta')^2, 3, 3$. Accordingly,

$$I(1) = 1 \oplus A \oplus B \oplus C, \qquad d_A = D/D' = 1 + 3n\delta, \quad d_B = d_C = D/3 = 2 + 3n\delta.$$

Every summand has multiplicity one and twist one. Comparing the coefficients of 1 and δ in their restriction dimensions shows that their columns are

$$(1, a, 3n - 2a, a), \quad (1, b, 3n - 1 - 2b, b + 1), \quad (1, c, 3n - 1 - 2c, c + 1).$$

The first row of $MM^{\mathsf{T}} = Q := \sum_U N_U^2$, for the full induction matrix M, gives $a+b+c = 3n-1$.

Here the low part is forced without searching over a, b, c. Put $v = (0, 1, -2, 1)^{\mathsf{T}}$. Multiplication of the displayed matrices gives $v^{\mathsf{T}}Qv = 18$. On the three low columns, the values of v are

$$6(a - n), \quad 6(b - n) + 3, \quad 6(c - n) + 3.$$

Their squares sum to at most 18. The last two are odd multiples of three, so their squares already sum to at least 18. It follows that $a = n$ and $\{b, c\} = \{n - 1, n\}$. Thus

$$F(A) = (1, n, n, n), \quad F(B) = (1, n - 1, n + 1, n), \quad F(C) = (1, n, n - 1, n + 1).$$

Lemma 13.3 now makes v vanish on every remaining column. Such a column is therefore

$$(0, b - t, b, b + t), \qquad b \geq |t|, \quad b > 0,$$

and its central dimension is $3b\delta + 2t$. Subtracting the four low columns from Q and comparing its entries gives

$$\sum b^2 = 6n^2 + 2, \qquad \sum bt = 2n, \qquad \sum t^2 = 2. \tag{13.18}$$

Thus precisely two high columns have nonzero t, which equals 1 or -1. Set $T = 3n\delta + 2$. Subtracting the first twist-trace equation for X from that for Z, then dividing by two, gives

$$\sum t(3b\delta + 2t)\theta = -T. \tag{13.19}$$

If the two signs of t differ, their b values differ by $2n$, by (13.18). Their positive dimensions then differ by $6n\delta + 4 = 2T$, contradicting the reverse triangle inequality in (13.19). Both signs cannot be negative. Hence both are positive; write their b values as p, q, with $p+q = 2n$. Their dimensions $\alpha = 3p\delta + 2$, $\beta = 3q\delta + 2$ satisfy

$$\alpha + \beta = 2T, \qquad \alpha\beta = (9pq + 4)\delta^2, \qquad T^2 = (9n^2 + 4)\delta^2.$$

If their twists are ζ, η, equation (13.19) implies

$$\zeta\eta^{-1} + \eta\zeta^{-1} = 2 - \frac{3(9n^2 + 4)}{9pq + 4}.$$

This rational number is an algebraic integer and lies in $[-2, 2]$. Since $pq \leq n^2$, it is at most -1. The value -2 would require $36pq = 27n^2 - 4$, which is impossible modulo nine. The

value -1 forces $pq = n^2$, hence $p = q = n$. The two twists then sum to -1 and are the primitive cube roots of unity.

All other columns have $t = 0$ and are of the form $(0, b, b, b)$. Their squared multiplicities sum to $4n^2 + 2$. Write $S = \sum b^2\theta$ over these uniform columns; it is a sum of $4n^2 + 2$ roots of unity, with repetitions. The first twist-trace equation for X gives

$$S = n^2 - n\delta.$$

For completeness, the low contribution is $n(1 + 3n\delta) + (2n - 1)T$; the two exceptional high columns contribute $-(n - 1)T$, so the equation is $0 = n(1 + 3n\delta) + nT + 3\delta S$, which is equivalent to the claimed identity using $\delta^{-1} = \delta - 3n$.

Writing $9n^2 + 4 = t^2 d$ with d squarefree, we have

$$S = -\frac{n^2}{2} - \frac{nt}{2}\sqrt{d}.$$

Lemma 13.16 therefore requires

$$4n^2 + 2 \geq \frac{n^2 + nt\lambda}{2}.$$

Condition (13.17) contradicts this inequality. For $n = 2$, the same reduction gives eighteen roots whose sum is $S_1 = -2 - 2\sqrt{10}$. The second indicator of X is $\epsilon \in \{1, -1\}$. The same twist-trace equation with twists squared, using that the two exceptional twists still sum to -1, gives

$$S_2 = S_1 + \epsilon\frac{D}{3\delta} = -2 + (2\epsilon - 2)\sqrt{10}.$$

For $\epsilon = -1$, Lemma 13.17 applied to the eighteen squared roots requires at least 34 roots. For $\epsilon = 1$, the same lemma applied to S_1 shows that the normalized absolute trace of S_2 is 2, whereas $S_2 = -2$. Both cases are impossible.

Finally, a spherical categorification has a dimension character with no zero entries, so the displayed character table restricts it to the two quadratic characters. These are Galois conjugate; conjugating its structure constants if necessary would give a pseudo-unitary categorification. This completes the proof. □

14 Exhaustive finite certification

The argument in this section is computer assisted. The unrestricted bounds proved earlier reduce the categorical problem to finitely many integral tensors. We enumerate every ordinary fusion ring with $D < 1000$ and retain an explicit certificate for each input. Independently, a complete signed cover through multiplicity 31 supplies all possible sphericalization quotients needed by the general-case exclusions. Both programs and both data sets are included; neither uses an external census or a pivotal classification as an oracle.

14.1 Tensor conventions and exact elimination

Every rank-four fusion ring is commutative by Lemma 2.1. For a self-dual basis $1, x, y, z$, reciprocity gives

$$N_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & c \\ 0 & b & d & e \\ 0 & c & e & f \end{pmatrix}, \quad N_y = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & b & d & e \\ 1 & d & g & h \\ 0 & e & h & i \end{pmatrix}, \quad N_z = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & c & e & f \\ 0 & e & h & i \\ 1 & f & i & j \end{pmatrix}.$$

These formulas also define signed tables, with integer coefficients. Associativity is equivalent to the following six equations:

$$ad - b^2 + bg + ch - d^2 - e^2 + 1 = 0, \tag{14.1}$$
$$ae - bc + bh + ci - de - ef = 0, \tag{14.2}$$
$$be - cd + dh - eg + ei - fh = 0, \tag{14.3}$$
$$af + bi - c^2 + cj - e^2 - f^2 + 1 = 0, \tag{14.4}$$
$$bf - ce + di - eh + ej - fi = 0, \tag{14.5}$$
$$df - e^2 + gi - h^2 + hj - i^2 + 1 = 0. \tag{14.6}$$

This follows either by comparing the three pairwise matrix commutators or by direct substitution in the associativity identities. If $b, c > 0$, put

$$H_0 = e(d+f) + bc, \quad G_0 = b^2 + d^2 + e^2 - 1, \quad J_0 = c^2 + e^2 + f^2 - 1,$$
$$A = e(b^2 - c^2) + bc(f - d), \qquad C = e(bf - ce),$$
$$B_0 = b^3 e - b^2 cd + b(d-f)H_0 - ebG_0 + ecH_0.$$

Then

$$Ca + Ai + B_0 = 0, \tag{14.7}$$

with

$$h = (H_0 - ae - ci)/b, \qquad g = (G_0 - ad - ch)/b, \qquad j = (J_0 - af - bi)/c.$$

Divisibility, bounds and all six original equations are checked after this elimination. In particular the linear equation is used only as a necessary condition and does not replace the original equations.

14.2 The complete ordinary census

The ordinary census uses $M = 31$ and the strict dimension condition $D < 1000$. For the displayed matrices, put $B = \sum_i N_i N_i^{\mathsf{T}}$. The positive dimension vector is an eigenvector of this symmetric nonnegative matrix with eigenvalue D. Thus D is its largest eigenvalue, and

$$D < 1000 \quad \Longleftrightarrow \quad 1000I - B \text{ is positive definite.}$$

The generator and checker test its leading principal minors by exact integer arithmetic. The boundary $D = 1000$ is excluded, not rounded. Appendix A gives every elimination branch and proves that the resulting census contains each ring below the bound exactly once. It has 1 692 classes, split into 1 642 self-dual classes and 50 classes with a dual pair.

The local source and data are in `unrestricted/supplementary/census`. The full tensors, their dualities, the ten- or six-parameter presentations, canonical fingerprints, exact dimension certificates, and counts at each multiplicity are supplied. The independently implemented Python tensor checks verify all fusion-ring identities and all unit-preserving basis permutations. Completeness rests on the exhaustive branch proof, rather than only on agreement of totals.

14.3 A shared exhaustive signed cover

The signed reduction permits an arithmetic exclusion to be shared among all unsigned rings compatible with a signed table. The following construction makes that shared search exhaustive.

Proposition 14.1 (A shared exhaustive signed cover)**.** *Fix $M \geq 1$. Consider the integral self-dual rank-four signed tables with the prescribed unit and reciprocity, parameter tuple $(a, b, c, d, e, f, g, h, i, j)$, and all ten coefficients of absolute value at most M. The six associativity equations define a finite set preserved by the group of nonunit basis permutations and basis-sign changes, of order* 48*.*

Every orbit has a representative satisfying

$$b, c, d \geq 0, \qquad b = \min\{|b|, |c|, |d|, |f|, |h|, |i|\}, \qquad e \geq 0 \text{ if } b = c = 0.$$

Enumerating all such representatives and retaining only the lexicographically least preferred tuple in each of their 48 *transforms gives exactly one table per orbit.*

For every self-dual unsigned fusion ring R of multiplicity at most M, all its possible sphericalization quotients are covered by the transforms of these representatives that satisfy its signed coefficient bounds, parities, and adjoint-parity conditions. A representative failing the formal-codegree d-number criterion or character cyclotomicity can be discarded for every R simultaneously.

Proof. The permutation group is transitive on the six mixed positions b, c, d, f, h, i. Place a position of minimum absolute value at b. Under a basis-sign change (s_1, s_2, s_3), the signs multiplying b, c, d are respectively s_2, s_3, s_1. Hence these three entries can be made nonnegative independently, without changing their absolute values. If $b = c = 0$, the remaining choices of s_2, s_3 also make e nonnegative while preserving $d \geq 0$. Thus the preferred set meets every orbit. Its least element in each finite orbit is unique; checking all 48 transforms selects precisely that element.

For completeness, the preferred set is exhausted by the following disjoint branches of the associativity equations. If $b > 0$, then $c, d \geq b$ and $|f|, |h|, |i| \geq b$; the two-coefficient Diophantine relation (14.7), with the polynomial coefficients from the preceding subsection, holds for signed parameters as well. Exhaust the bounded choices of b, c, d, e, f and solve that relation for all bounded integer pairs a, i by exact greatest-common-divisor progressions. If one coefficient vanishes, exhaust its free variable; if both vanish, retain the entire bounded product precisely when the constant term vanishes. The formulas in that subsection then determine h, g, j. Impose the signed bounds and the preferred-representative conditions on all coordinates, and check the original six equations. If $b = 0 < c$ and $e \neq 0$, the equations determine

$$h = \frac{d^2 + e^2 - ad - 1}{c}, \qquad i = \frac{e(d + f - a)}{c},$$
$$j = \frac{c^2 + e^2 + f^2 - af - 1}{c}, \qquad g = \frac{h(d - f) + ei - cd}{e}.$$

The divisibility for i gives $a \equiv d + f \pmod{c/\gcd(c, e)}$; all bounded choices in this progression are checked. If $b = 0 < c$, $e = 0$ and $d > 0$, then

$$\gcd(c, d) = 1, \quad i = 0, \quad h = \frac{d^2 - ad - 1}{c} \neq 0, \quad f = d - \frac{cd}{h},$$

and j is given by the preceding formula, while g is enumerated. The remaining case $b = 0 < c$, $e = d = 0$ forces $(c, h, f, i, j) = (1, -1, 0, 0, 0)$ and leaves a, g free.

If $b = c = 0 < e$, then $a = d + f$ and $df = e^2 - 1$; the equations determine g, j from d, e, f, h, i, with the case $d = 0$ retained. Finally, $b = c = e = 0$ forces $d = 1, a = 0$ and $f \in \{1, -1\}$. For $f = 1$ the remaining equation is

$$gi + hj = h^2 + i^2 - 2,$$

including its $h = 0$ branch. For $f = -1$ one has $h = i = 0$ and g, j are free. Coordinates whose signs are not fixed by the preferred convention retain both signs and the original size bounds, and every emitted tuple is checked against all six equations. This proves exhaustive enumeration of the preferred set, including the zero denominators.

Any sphericalization quotient for R lies in the bounded signed set by Lemma 2.2. One of the 48 transforms of its global representative is the quotient in the basis of R, so testing all compatible transforms loses no categorical possibility. Formal codegrees are unchanged by permutations and basis signs, and character values change only by permutation and sign. Their d-number and cyclotomicity conditions are therefore orbit-invariant. Both conditions are necessary for a signed sphericalization quotient by Theorem 13.10, which proves the common exclusions. □

Remark 14.2 (Certificate reuse). The global signed census is regenerated exhaustively during the verification workflow; its completeness is not inferred from a cached list. Universally excluded orbits receive a checked common arithmetic rejection. Every other orbit is expanded into all distinct transforms compatible with each unsigned ring, and every such candidate receives the corresponding signed-branch check. Sharing these computations changes their allocation among records, not the necessary criteria or the scope of the classification.

14.4 The positive-orbit bridge

Proposition 14.3. *Let $\mathcal{T}_M$ be the complete signed cover after the common d-number and cyclotomicity exclusions. Every self-dual rank-four Grothendieck ring of multiplicity at most M is isomorphic to a nonnegative gauge transform of a table in $\mathcal{T}_M$. Consequently, certificates for all such nonnegative orbits suffice for the self-dual classification. The ordinary census is a separate exact input list for the successive-filter ledger; its completeness is not assumed in this signed-cover argument.*

Proof. The ordinary fusion table itself is an integral associative signed table with the stated coefficient bound. Its orbit occurs in the exhaustive cover, and its formal codegrees and character values satisfy the common necessary arithmetic conditions. Thus its representative belongs to $\mathcal{T}_M$. Applying all eight sign gauges and all six nonunit permutations recovers its ordinary table.

A sign change taking a nonnegative table to another nonnegative table fixes every nonzero coefficient: the sign multiplying such a coefficient must be $+1$. Thus a signed orbit contains at most one nonnegative orbit modulo permutations. Equality between the list of these positive orbits and the list of certificate inputs is an exact finite coverage check. □

Theorem 1.2 reduces the classification to $M = 31$. The fresh exhaustive signed enumeration has 92 795 orbits. The formal-codegree d-number condition rejects 90 510, and cyclotomicity rejects a further 68, leaving 2 217. Exactly 1 182 retained orbits have a nonnegative transform; each has one checked certificate. This signed-cover scope is all multiplicities at most 31, without the ordinary census's additional condition $D < 1000$. Its counts are therefore distinct from the fusion-ring counts in Table 2.

The supplementary certificate bank keeps the broader signed-cover proof intact. The full-census ledger tests rank-two subrings and universal arithmetic directly. Every later self-dual witness carries an explicit basis isomorphism to its certificate input. A table index, digest, or stored status is never an exclusion criterion.

The automatic-sphericality branch checks its structural hypothesis before using either Theorem 5.1 or Corollary 2.7. It then checks the supplied spherical obstruction, including any required Frobenius–Perron Galois-orbit condition. The arithmetic and divisibility inputs are stated in Proposition 2.8. The induction and moment tests are Propositions 13.6, 13.5

and 13.8; the quadratic root-length branch is Proposition 13.18, including its conductor-40 boundary. Signed tests use Lemma 2.2 and Theorem 13.10, together with the norm and field obstructions in Lemmas 13.1 and 13.2. A reduction to pseudo-unitarity is made only after checking weak integrality, the two-dimension hypothesis, or a spherical dimension character in the Frobenius–Perron Galois orbit. No pivotality assumption is imposed on the original input category.

For clarity, two different enumerations are verified. Every compatible signed table is covered, either by the global cover or by the indicated complete bounded local search. For every low-induction certificate, every bounded integral restriction column and every admissible combination of columns is regenerated. In particular, neither a stored classification flag nor a claimed exhaustive list is accepted without checking its witness and its coverage. Character fields are represented by squarefree polynomials over $\mathbb{Q}$; supplied root lists are checked by multiplying linear factors. Algebraic signs use rational isolating intervals.

Theorem 14.4 (Computer-assisted self-dual classification)**.** *The self-dual rank-four complex Grothendieck rings of multiplicity at most* 31 *are exactly the nine self-dual rings in Table 1.*

Proof. Propositions 14.1 and 14.3 reduce the question to the 1 182 positive orbits above. The exact verifier checks a based-ring isomorphism for each of the nine model records and accepts a necessary- condition contradiction for each of the other 1 173 records. The mathematical implications are precisely those listed above. All nine surviving tensors are realized by the cited unitary categories. □

14.5 The non-self-dual case

Theorem 14.5. *There are exactly six non-self-dual rank-four complex Grothendieck rings: the six non-self-dual entries in Table 1. This assertion does not require finite enumeration.*

Proof. Corollary 4.4 permits the pseudo-unitary reduction without any initial pivotality hypothesis. Larson's pseudo-unitary candidate theorem [11, Theorem 1.1] therefore applies. It gives the six stated rings and the one additional candidate $S(2)$. Theorem 13.15 excludes $S(2)$ (indeed every $S(m)$ with $m \geq 2$). The six remaining rings have the unitary realizations in Table 1. □

14.6 Successive filters and complete coverage

The full-census ledger checks all 1 692 input rings and records 1 677 exclusions and 15 models. Its stages are shown in Table 2. Each ring is counted exactly once, at its recorded rejection stage. These are counts for the implemented ordered certificate pipeline; they do not assert that no earlier criterion could also exclude a ring assigned to a later stage.

The rank-two and universal arithmetic stages apply Propositions 13.12 and 2.8. For a non-self-dual input, the verifier checks a based isomorphism with one of Larson's seven candidates or its absence from that list; the extra candidate $S(2)$ is excluded by Theorem 13.15. The reduction to the pseudo-unitary candidate theorem is justified by Corollary 4.4, not imposed on the input. The remaining stages replay precisely the necessary conditions described above, with all reduction hypotheses and basis changes checked. The final signed stage covers all compatible quotients and all required character and restriction-column choices, rather than a preferred lift.

14.7 Models and exact dimensions

Table 1 supplies one unitary realization and a construction reference for every surviving based ring. Here $\mathrm{Ad}(\mathrm{SU}(2)_k)$ denotes the adjoint subcategory, $\mathcal{H}_1$ is the rank-four Haagerup even

Stage	Excluded	Remaining
Complete census, $D < 1000$	—	1 692
Rank-two subring obstruction	375	1 317
Formal-codegree d-numbers	1 084	233
Character cyclotomicity	2	231
Non-self-dual reduction	21	210
Two-dimension and weak-integral reductions	109	101
Subring dimension divisibility	3	98
Involution moment obstructions	10	88
Automatic sphericality and spherical tests	6	82
Signed characters and central induction	67	15

Table 2: Successive exact filters for the complete ordinary fusion-ring census. Each ring is counted once, at its recorded rejection stage. The final stage covers every compatible signed quotient and every character or induction-column choice required by its certificate.

part, and a near-group ring of type (C_3, m) satisfies $\rho^2 = \sum_{g \in C_3} g + m\rho$. The six non-self-dual models are Vec_{C_4}, the Tambara–Yamagami category on C_3, the principal even part of S', $\mathrm{Rep}(A_4)$, and the two near-group categories of parameters three and six.

The supplementary model data contain the fifteen complete multiplication tensors and positive dimension vectors in their actual bases. Exact multiplication identities identify each vector as the Frobenius–Perron character. Independent reconstructions from the group, product, adjoint, near-group, and exceptional fusion rules check all fifteen based-ring identifications. The polynomials for δ_7 and δ_E in the table are checked as factors of the corresponding Casimir characteristic polynomials; positivity identifies their largest roots. The construction references prove existence. Sphericality of *every* categorification is proved separately below.

Corollary 14.6 (Computer-assisted classification and sphericality)**.** *The rings in Table 1 are exactly the Grothendieck rings of rank-four complex fusion categories. Every such category admits a spherical structure. Each listed ring has a unitary realization.*

Proof. The unrestricted bound of Theorem 1.2 places every categorifiable ring within the verified multiplicity range. Apply Theorems 14.4 and 14.5.

For sphericality, the six self-dual multiplicity-one entries are covered by Corollary 2.7. The $2D2$ ring has a nontrivial invertible basis element, so Theorem 5.1 applies. The E_8 ring is the Fibonacci normal form $(a, b) = (1, 1)$, covered by Proposition 6.2. The Haagerup ring is covered by Theorem 7.7, and the six non-self-dual rings by Theorem 4.1. These arguments apply to every categorification, independently of the chosen unitary realizations. □

Corollary 14.7 (Sharp bounds)**.** *Every rank-four complex fusion category satisfies*

$$\mathrm{FPdim}(\mathcal{C}) \leq 24 + 12\sqrt{3}, \qquad \max_{i,j,k} N_{ij}^k \leq 6.$$

Both bounds are attained, and equality in either bound holds precisely for the near-group ring $\mathrm{NG}(C_3, 6)$.

Proof. Read the multiplicities and dimensions from Table 1. Every other row has $\mu \leq 3$ and $D < 44$. The radical comparisons are immediate. For δ_7 and δ_E, substituting $x = 44 + u$ in their defining polynomials gives strictly positive coefficients, so neither has a root at least 44. Since $44 < 24 + 12\sqrt{3}$, the maximal entry is unique. The near-group unitary realization proves sharpness of both bounds. □

14.8 Reproducibility and the scope of the computation

All code and data are supplied locally. Within `supplements/rank4`, the directory `unrestricted/supplementary/census` contains the complete ordinary fusion-ring enumeration and data. The general-case classification source, signed cover, witness bank, full-census ledger, and replay reports are in `unrestricted/supplementary/classification`. No external census, pivotal package, or external classification status is read. From the repository root, run

```
python3 supplements/rank4/unrestricted/supplementary/check.py
```

This regenerates the ordinary census, checks every full tensor and exact dimension bound, regenerates the full-census ledger, independently regenerates the signed cover, and replays all certificates. It then verifies all model dimensions and reconstructs both manuscript tables from the checked data. The requirements are Python, SymPy, and a C++17 compiler with OpenMP. The optional `python-flint` backend accelerates the same exact arithmetic; it is not required.

The `-deep` flag also removes both dimension-pruning steps from the ordinary enumerator, applies a separately implemented exact dimension test, and checks an exhaustive small parameter box. The unpruned search has 16065 classes at multiplicity at most 31; its exact $D < 1000$ filter recovers precisely the same 1 692 inputs. At multiplicity two the full box of $3^{10} + 3^6$ assignments gives the same 27 classes. These comparisons supplement the proof of coverage, with their shared components stated in the supplementary guide.

The delivered fresh replay checks 1 677 exclusions and all fifteen model identifications, with no unresolved records. Its signed subroutine also checks the broader 1 182-record bank. The optional local Python signed-search engine is provided as a separate replay mode. Cached-cover and ledger-only modes are explicitly conditional and are not used by the complete command above. Character signs are decided by rational isolating intervals; numerical approximations never determine an exclusion. These are exact finite checks, not a proof-assistant formalization of the categorical arguments. The finite computation verifies the explicitly bounded arithmetic and coverage assertions. The categorical arguments and the cited category constructions remain mathematical inputs; the research notice on the first page applies to both parts of the proof.

A Completeness of the bounded census

This appendix supplies the mathematical coverage argument for the finite search. Checking that emitted tensors are valid and pairwise nonisomorphic does not by itself prove that every ring occurs. We therefore give the complete presentations, the elimination branches, and the exact acceptance condition. All parameters range over integers in $[0, M]$, where $M = 31$; write $\Lambda = 1000$ for the exclusive dimension bound.

A.1 Complete presentations

By Lemma 2.1, every rank-four fusion ring is commutative. Duality on its three nonunit basis elements is either the identity or a transposition. Reciprocity gives the following two presentations, with $N_0 = I$ and our row convention $(N_u)_{vw} = N_{uv}^w$.

In the self-dual case use the matrices and six associativity equations of Section 14.1, with nonnegative parameters $(a, b, c, d, e, f, g, h, i, j)$.

In the case with duality (0)(1)(23) the parameters are (a, b, c, d, e, f), and

$$N_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & b \\ 0 & b & c & d \\ 0 & b & d & c \end{pmatrix}, \quad N_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & b & c & d \\ 0 & d & e & f \\ 1 & c & e & e \end{pmatrix}, \quad N_3 = N_2^{\mathsf{T}}.$$

Their commutativity is equivalent to

$$\begin{aligned} ac - b^2 + 2be - c^2 - d^2 + 1 &= 0, \\ ad - b^2 + be + bf - 2cd &= 0, \\ b(c - d) + d(e - f) &= 0, \\ -c^2 + d^2 - e^2 + f^2 - 1 &= 0. \end{aligned} \tag{A.1}$$

In both cases commutativity of the displayed matrices suffices for associativity of the proposed multiplication. Indeed, set $E = N_u N_v - \sum_w N_{uv}^w N_w$. Its zeroth row vanishes, and it commutes with every N_t. Its row t is the zeroth row of $N_t E = E N_t$, hence also vanishes. Thus $E = 0$, proving the full regular multiplication identities. The unit, duality, integrality, nonnegativity, and reciprocity axioms are already built into the presentations.

A.2 The self-dual branches

Permutations of the three nonunit elements act transitively on the six mixed coordinates b, c, d, f, h, i. Every isomorphism class therefore has a representative with $b = \min(b, c, d, f, h, i)$. We enumerate all such representatives before imposing the final isomorphism reduction.

Positive minimal mixed coefficient. Suppose $b > 0$. Enumerate b, c, d, e, f, with $c, d, f \geq b$, and use H_0, G_0, J_0, A, C, B_0 from Section 14.1. The first, second, and fourth associativity equations give the displayed formulas there for h, g, j; the third gives $Ca + Ai = -B_0$, namely (14.7). If $A \neq 0$, put $q = \gcd(A, C)$. There are no solutions unless $q \mid B_0$; otherwise a lies in the unique residue class solving

$$(C/q)a \equiv -B_0/q \pmod{|A/q|},$$

and i is determined by (14.7). A modulus of one imposes no restriction. If $A = 0 \neq C$, determine a and retain every bounded i. If $A = C = 0$, require $B_0 = 0$ and retain every bounded pair (a, i). In all cases retain only integral g, h, j, with $a, g, j \in [0, M]$ and $h, i \in [b, M]$, and test all six original equations.

The implementation discards outer tuples only by necessary interval conditions. Besides $b, c, d, f \in [b, M]$ and $e \in [0, M]$, these are

$$\begin{gathered} b(b + c) \leq H_0 \leq M(e + b + c), \qquad bc \leq G_0 \leq M(d + b + c), \\ b^2 \leq J_0 \leq M(f + b + c), \\ \min(0, CM) + \min(Ab, AM) \leq -B_0 \leq \max(0, CM) + \max(Ab, AM). \end{gathered}$$

The first three follow from the displayed equations for h, g, j; the last bounds the left side of (14.7) over its allowed rectangle. The further necessary dimension inequality

$$2 + 2b^2 + 2c^2 + d^2 + 2e^2 + f^2 < \Lambda$$

is the bound on B_{11} after discarding its nonnegative summand a^2. Here $B = \sum_u N_u N_u^{\mathsf{T}}$ is the Casimir matrix.

One zero mixed coefficient. Suppose $b = 0 < c$. The equations imply

$$h = \frac{d^2 + e^2 - ad - 1}{c}, \qquad i = \frac{e(d + f - a)}{c}, \qquad j = \frac{c^2 + e^2 + f^2 - af - 1}{c},$$

and $eg = h(d - f) + ei - cd$. For $e > 0$, enumerate c, d, e, f and the progression $a \equiv d + f \pmod{c/\gcd(c, e)}$. The bounds on h, i, j give necessary bounds on a; exact division determines g. Test all six equations and all parameter bounds.

For $e = 0$, one has $i = 0$, and the first and third equations imply $d, h > 0$ and $\gcd(c, d) = 1$. Indeed $d = 0$ would give $ch = -1$, and $h(d - f) = cd > 0$ forces $h > 0$; divisibility of $d(d - a) - 1$ by c gives coprimality. Enumerate c, d and $a \equiv d - d^{-1} \pmod{c}$, with the congruence unrestricted for $c = 1$. Determine h, put $f = d - cd/h$, and determine j by the displayed formula. Retain every bounded g, testing exact integrality, all bounds, and all six equations.

Two zero mixed coefficients, with $e > 0$. For $b = c = 0 < e$, the first, second, and fourth equations yield $a = d + f$ and $df = e^2 - 1$. Enumerate every bounded factor pair, including the zero-factor possibilities when $e = 1$. For every bounded h, i, determine g, j from

$$eg = h(d - f) + ei, \qquad ej = eh + i(f - d).$$

Retain the integral bounded solutions satisfying all six equations.

The remaining zero branch. For $b = c = e = 0$, the equations force $a = 0$ and $d = f = 1$ and leave

$$gi + hj = h^2 + i^2 - 2.$$

For $h > 0$, enumerate h, i, g and determine j; for $h = 0 < i$, determine g and retain every bounded j. The case $h = i = 0$ is impossible. As before, test all original equations and bounds.

These four branches are exhaustive. In each division step a nonzero denominator has been explicitly assumed; when a coefficient vanishes, every remaining bounded solution is retained. Finally keep the lexicographically least tuple among its permutations preserving the minimal value of b. Each orbit has exactly one such tuple. Thus the self-dual enumeration has neither an omitted zero case nor a duplicate based-ring isomorphism class.

A.3 The branch with a dual pair

Enumerate c, d, e. The last equation in (A.1) determines the unique possible nonnegative integer $f = \sqrt{1 + c^2 - d^2 + e^2}$; test its square and bound exactly. If $c \neq d$, the third equation determines $b = -d(e - f)/(c - d)$. If $c = d$, require $d(e - f) = 0$ and retain every bounded b. Determine a from the first equation when $c > 0$, from the second when $c = 0 < d$, and retain every bounded a when $c = d = 0$. Test all four original equations and bounds. The only unit-preserving relabelling compatible with this duality exchanges the dual pair. It leaves the six parameters unchanged, so these tuples already represent distinct isomorphism classes.

A.4 The dimension test and the coverage conclusion

The positive dimension vector is an eigenvector of the symmetric nonnegative Casimir matrix B, with eigenvalue $D = \mathrm{FPdim}(R)$. It follows that D is its largest eigenvalue. Therefore

$$D < \Lambda \iff \Lambda I - B \text{ is positive definite} \iff \det(\Lambda I - B)_{[1,k]} > 0 \ (1 \le k \le 4). \quad \text{(A.2)}$$

Here $[1, k]$ denotes the leading principal k-by-k submatrix. The final equivalence is Sylvester's criterion. All determinants are integers; a zero determinant is rejected, so equality at the boundary is never admitted. The C++ enumeration uses fraction-free elimination, and the Python tensor checker recomputes the same quantities with arbitrary-precision integers.

Proposition A.1. *The preceding procedure with $M = 31$ and $\Lambda = 1000$ emits exactly one representative of every rank-four fusion ring with $D < 1000$.*

Proof. Every rank-four fusion ring has one of the two complete presentations. All solutions of their associativity equations in the parameter box occur in the enumerated branches, and the isomorphism reduction retains exactly one representative. The acceptance condition (A.2) is equivalent to the desired strict dimension bound. Conversely, every ring below that bound has $\mu < \sqrt{D}$ by Lemma 2.1, hence $\mu \leq 31$, and therefore lies in the searched box. This proves both directions. □

The executable output has 1 642 self-dual classes and 50 classes with a dual pair, for a total of 1 692. Completeness follows from the proposition, not merely from these totals. As an additional implementation check, removing both dimension-pruning steps gives 16065 classes at multiplicity at most 31; an independently implemented strict dimension test retains precisely the same 1 692 parameter tuples. A full small-box test examines all $3^{10} + 3^6 = 59778$ assignments at multiplicity at most two and recovers exactly the same 27 classes. The unpruned experiment shares the Diophantine elimination; the small-box test avoids it but shares the tensor presentations and canonicalization. Neither is asserted to be a second independent proof of the complete enumeration algorithm.

AI assistance and verification

This manuscript and its accompanying code and data were prepared with AI assistance. Independent human verification and Lean formalization are pending. The exact checks certify the finite assertions described in the verification guide; they do not constitute a formal verification of the categorical proofs.

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